Digital Science Research Report

Australian National Persistent Identifier (PID) Benchmarking Toolkit

Final Report

OCTOBER 2025

DIGITAL science

Australian Research Data Commons

Australian Government

NCRIS National Research Infrastructure for Australia

ARDC is enabled by NCRIS

About Digital Science

Digital Science is an AI-focused technology company providing innovative solutions to complex challenges faced by researchers, universities, funders, industry and publishers. We work in partnership to advance global research for the benefit of society. Through our brands – Altmetric, Dimensions, Figshare, IFI CLAIMS Patent Services, metaphacts, OntoChem, Overleaf, ReadCube, Symplectic, and Writefull – we believe when we solve problems together, we drive progress for all. Visit www.digital-science.com

About the Authors

Simon Porter Is the VP Research Futures at Digital Science, and is also a member of the ORCID Board
https://orcid.org/0000-0002-6151-8423

Dr Juergen Wastl is the VP Research Evaluation and Global Challenges
https://orcid.org/0000-0001-7757-8001

Dr Hélène Draux is a Senior Data Scientist at Digital Science
https://orcid.org/0000-0001-8837-168X

Ann Campbell is the Director Research Impact & Comparative Analytics
https://orcid.org/0000-0002-8575-5052

Acknowledgments

This report has been commissioned by the Australian Research Data Commons (ARDC) — Australia's leading research data infrastructure facility. The ARDC is enabled by the Australian Government's National Collaborative Research Infrastructure Strategy (NCRIS). In particular, the authors would like to acknowledge the contributions of:

Natasha Simons, Director, National Coordination, ARDC
https://orcid.org/0000-0003-0635-1998

Lyle Winton, Program Manager, ARDC
https://orcid.org/0000-0002-3049-1221

Matthias Liffers, Product Manager (Persistent Identifier Services), ARDC
https://orcid.org/0000-0002-3639-2080

Linda O'Brien, Strategy Consultant for ARDC
https://orcid.org/0000-0002-1477-8652

This report has been published by Digital Science which is part of Holtzbrinck, a global media company dedicated to science and education.

Digital Science, 6 Briset Street, London EC1M 5NR, UK.
info@digital-science.com

DOI: dx.doi.org/10.6084/m9.figshare.29281667
This report was created in Overleaf

Summary

The Australian National Persistent Identifier (PID) Strategy aims to accelerate Australian research quality, efficiency and impact through universal use of connected persistent identifiers. This strategy promotes better discovery and reuse of research inputs and outputs, improved reproducibility and attribution, and more effective national planning. To realise the vision requires shared action and accountability across the research ecosystem stakeholders.

To support implementation of this vision, Digital Science has been commissioned by the Australian Research Data Commons (ARDC) to create a benchmarking toolkit. The toolkit is designed to help the Australian research ecosystem assess their progress toward PID adoption aligned to the strategy's five objectives: (1) enhancing the FAIRness of research inputs; (2) increasing the discoverability and reuse of research outputs; (3) improving reproducibility and reducing administrative burden; (4) enabling robust impact assessment through linked metadata; and (5) supporting national capability mapping through PID integration. The toolkit, consisting both of analysis, and runnable code provides a mechanism for tracking collective progress, informing stakeholder action plans, and ensuring that Australia's investment in digital research infrastructure delivers measurable and sustainable benefits.

As this is the first time that Australia's National Identifier strategy has been benchmarked, the report also makes a number of recommendations for interventions that could be made to improve Australia's development and utility of its national persistent identifier infrastructure and practice.

At a high level, a comprehensive review of persistent identifier practice within the Australian research system reveals a mixed score card.

- Australia's adoption of ORCID for research publications across the research sector has been extremely successful. This progress sets Australia up well for more comprehensive use of the ORCID infrastructure.

- The level of ORCID adoption across the Australian research community makes it possible to create grant records that connect researchers via their ORCID to award details. These records in turn provide the seed details to create research activity records necessary to join together the different ongoing components of research, from additional project members, access to instruments and research infrastructure, and relationships to research outputs.

- Workflows between publishers, institutional research information systems and ORCID are also sufficiently strong to envisage new modes of trusted comparative research evaluation.

- Review of non-traditional research output shows that some Australian institutions have strong practices around thesis submission and recording of research reports, and taken collectively, only a few countries have stronger practices for thesis deposit using DOIs. This practice has positive implications not only for the promotion of Australian research, but also for the tracking of outputs of early-career researchers, along with graduates beginning their careers with industry. Not every Australian institution has the same level of DOI enabled thesis deposit however, and there is an opportunity here to define new national expectations that capitalise on Australia's advantage in this area.

The report also reveals significant areas for improvement. As exemplified by the National Persistent Identifier strategy, Australia's understanding of the value of persistent identifier infrastructure has reached a level of maturity where it is no longer enough to count the number of things we create, it matters equally how those things are connected. Against these expectations, it is possible to identify areas for which more progress could be made:

- Although Australian researchers are more engaged than the global average in the practice of data citation, they trail significantly behind their European peers. That higher levels of data citation engagement are possible suggests that increased investment in research data citation literacy is warranted.

- ORCID identifiers recorded against research outputs other than journal articles remains the exception rather than the norm. Institutions should now be considered significant publishers of research output (including datasets, thesis, and grey literature), and the same expectations that apply to traditional publishers (that all authors should have an ORCID) should also apply to these outputs as well.

- To date there have been over 5.8 Million IGSN samples registered in Australian IGSN repositories, however most records remain almost completely disconnected from the rest of the persistent identifier graph.

In all, the report makes 23 recommendations specifically tied to the objectives of the National Persistent Identifier Strategy. It is hoped that, in combination with the benchmarking approaches made in this report, these recommendations model a toolkit for assessing progress against strategy going forward.

Contents

1 Summary of Recommendations **4**

2 A Benchmarking Toolkit for the Australian Persistent Identifier Strategy **7**
2.1 A Note on Methodology, and the associated Toolkit 9
2.2 Benchmarks for Research Inputs 11
2.2.1 Research Organisations 12
2.2.2 Research Activities 18
2.2.3 Grants 20
2.2.4 Facilities 24
2.2.5 Research Data 26
2.2.6 Samples 32
2.2.7 Instruments 35
2.3 Benchmarks for Research Outputs. 39
2.3.1 Research Reports 40
2.3.2 Research Methods 44
2.3.3 Non Traditional Research Outputs 47
2.3.4 Dissertations 49
2.3.5 Publications 54
2.4 Benchmarks for research reproducibility, provenance, and reducing academic burden 59
2.5 Benchmarks for enabling the understanding of the impact of research inputs, evaluating research quality, impact and evidence of public benefit 63
2.6 Benchmarks for research capability 65

3 Persistent Identifier Benchmarking in Practice **66**
3.1 Did Vision 2020 make a difference? Benchmarking the impact of the 2017 Australian ORCID strategy. 66
3.1.1 Benchmarking Australia's ORCID adoption 68
3.1.2 ORCID By Funders 71
3.1.3 ORCID By Publisher by Year 73
3.1.4 ORCID By Research Classification 75
3.1.5 ORCID by Institution 77
3.1.6 ORCID 2030? 78

4 Benchmarking Recipes **81**
4.1 Benchmarks for Research Inputs 81
4.2 Benchmarks for Research Outputs 91

5 List of Acronyms used in this report **97**

1 Summary of Recommendations

Table 1: Summary of Recommendations (Inputs)

Reference	Section	Recommendation Text
2.2.1	Inputs – Research Partners (ROR)	Review the representation of commercial Australian research organisations in ROR. With the consent of new commercial organisations identified in Dimensions (or another appropriate resource,) provide details of these institutions to ROR. The additional institutions to be reviewed can be identified using (Toolkit query 4.1.2)
2.2.2	Inputs – Research Partners (ROR)	Engage with international publishing community to ensure ROR affiliations are being directly asserted onto publication records where available. The International publishing community is far less engaged with implementing ROR than would benefit the community. International advocacy will be required to highlight why this should be a priority for publishing systems.
2.2.3	Inputs – Research Partners (ROR)	Engage with repository vendors and review internal workflows to ensure that ROR affiliations are added to all creators. Progress towards implementing ROR affiliations within Datacite is comparatively strong, but remains inconsistent. Engaging with vendors and reassessing current workflows to capitalise on this strength is recommended.
2.2.4	Inputs – Research Projects (RAiD)	The Research Community adopts a RAiD target that initially is roughly equivalent to the number of active grants. By quickly establishing a large number of RAiD records representing a large proportion of active research within Australia, service providers have an incentive to build functionality around these records.
2.2.5	Inputs – Grants	As soon as is practical, Australian funders create persistent identifiers for Grants including complete metadata. Award metadata should include the amount funded, and uniquely identify researchers via their ORCIDS, and research affiliations (ROR)
2.2.6	Inputs – Grants	The creation of a grant should trigger the creation of a RAiD (or the association of an existing RAiD). Creating a RAiD as soon as a project has been funded means that research objects, people and other associated metadata can begin to be associated with it from day one.
2.2.7	Inputs – Facilities	Facilities confirm that they are represented in ROR
2.2.8	Inputs – Research Data	Research Community Data citation literacy is increased in order to meet levels of peer countries. Whilst Australia has data citation rates that compare well to the global average, many European countries as well as Great Britain have consistently higher engagement rates across most fields of research.
2.2.9	Inputs - Research Data	Australian research institutions work with their repository vendors to ensure that their workflows are 'ORCID' first for all data types

(continues next page)

Reference	Section	Recommendation Text
2.2.10	Inputs – Samples & Observations (IGSN)	The Australian Research Community determine new minimum standards for IGSN records. IGSN records should be more firmly integrated into the persistent identifier graph.
2.2.11	Inputs – Instruments	Encourage links between instruments and the rest of the persistent identifier graph. Data and outputs created from Instruments should include a link to the instrument as standard
2.2.12	Inputs - Instruments	Reinforce instrument citation examples as best practice. Following the best practice that has already been established, use of instruments when recorded in research should always cite their persistent identifier

Table 2: Summary of Recommendations (Outputs)

Reference	Section	Recommendation Text
2.3.1	Outputs – Reports	Encourage the practice of publishing reports in institution repositories. By volume, Australia is a global leader in supporting the publication of reports; however, the practice is not evenly distributed across all institutions
2.3.2	Outputs – Reports (ORCID/ROR links)	Continue to increase the practice of linking reports to people and affiliations via ORCID and ROR. As a publisher of reports, institutions adopt the best practices of publishers to ensure that all authors are described by ORCIDs and affiliations via ROR. This may require additional coordination with repository vendors to ensure workflows can be supported.
2.3.3	Outputs – Methods / Protocols	Guidelines are created for the recording of protocols. Without clear guidelines on how to register protocols and methods, benchmarking will be haphazard.
2.3.4	Outputs – Dissertations	Australia considers national policy for publishing dissertations with DOIs in institutional repositories, formalising the use of ORCIDs for authors and their supervisors] Australian institutions are some of the largest publishers of dissertations with DOIs globally. The same expectations on the use of ORCID and ROR identifiers that the community expects of publishers should also apply to thesis deposit.
2.3.5	Outputs – Publications (coverage)	Benchmarking Analysis of persistent identifiers should not ignore closed access material. For Australia, ignoring closed-access publications would result in missing 35% of Australia's output in 2024.
2.3.6	Outputs – Conference Proceedings	ARDC begins outreach to strategic conferences to understand what support is required to ensure that all conference proceedings have a persistent identifier.

Table 3: Summary of Recommendations (ORCID)

Reference	Section	Recommendation Text
3.1.1	ORCID	ORCID adoption for Australian researchers has progressed sufficiently far to make requiring authenticated ORCIDs mandatory. Requiring ORCID ids on grants ensures that grant records can be created that can be integrated into the Australian persistent identifier landscape, facilitating the downstream creation of RAiD records, and making funding information easier to integrate both into Research Information Management systems as well as ORCID itself.
3.1.2	ORCID – International Advocacy	Australia joins international pressure to ensure that all publishers both record ORCID records and push the associated metadata into Crossref. Publishers that do not support ORCID workflows (including the passing of metadata through to Crossref should not be recommended venues for publication)
3.1.3	ORCID – Researcher Practice	All Australian researchers are required to ensure that their ORCID is part of the metadata of their publication. Regardless of whether or not an Australian researcher is the corresponding author, it should be the default that ORCID records are always supplied.
3.1.4	ORCID – System Integration	Australian researchers are encouraged to turn on both Crossref and Datacite integrations to ensure that the population of ORCID records is as automatic as possible.
3.1.5	ORCID – Institutional Policy	Australian Institutions require ORCID registration within their institutional research information management systems. To get the most out of ORCID (for individuals as well as institutions, ORCIDs need to be linked through to Research Information Systems with the authority to push the research objects collected back to ORCID. In the ideal situation, ORCID ids recorded during publication end up being triple verified: at source by the publisher, endorsed by the researchers, and verified by the institutions - all without additional manual effort.

2 A Benchmarking Toolkit for the Australian Persistent Identifier Strategy

The Australian Persistent Identifier Strategy [4] is a critical national initiative to benefit the Australian people by strengthening our digital information ecosystem, the quality of our research and our capacity for effective research engagement, innovation and impact. The strategy is accompanied by an evolving roadmap, which facilitates a stakeholder-driven approach to designing an actionable national collaborative plan and a set of tools to support implementation of the strategy.

To ensure the strategy's effective implementation and sustained success, it is essential to develop benchmarks that can track progress and measure outcomes. These benchmarks should be specific, measurable, achievable, relevant, and time-bound (SMART).

The "National PID Roadmap" functions as an online resource including case studies, other tools and resources for institutions to develop their own action plans, and the collection of actions plans nationally – a PID Benchmarking approach serves as a crucial tool to contextualise stakeholders' efforts within the broader national landscape. By understanding where organisations stand in terms of PID adoption and maturity, organisations can better inform their action plans, identify areas for improvement, and contribute meaningfully to the National PID Strategy's overall objectives. Furthermore, benchmarking can help in measuring the collective progress of the Australian research sector towards the vision of universal and connected PID use outlined in the Australian National PID Strategy.

Collectively these resources form a benchmarking toolkit designed to work with 5 objectives outlined in the strategy:

Improve Research Quality and Efficiency

Objective 1: Increase the Findability, Accessibility, Interoperability and Reuse of inputs to research

Inputs to research include potential research partners, related research projects, grants, facilities, data, samples and observations. By increasing a researcher's ability to discover relevant inputs, we will reduce duplication, increase research productivity and create opportunities for impactful partnerships.

Objective 2: Increase the Findability, Accessibility, Interoperability and Reuse of research outputs.

Outputs from research include data, reports, methods, non-traditional research outputs and publications. By increasing the discoverability of these outputs, we can improve future research, create new kinds of research, reduce duplication and improve the timeliness of translating research into impact.

Objective 3: Improve research reproducibility, provenance and attribution while minimising administrative burden, enabling researchers to spend more time on research

By linking researchers to projects, grants, organisations and equipment, we can improve research reproducibility, provenance and attribution. Through the use of PIDs, researchers can minimise the amount of time they spend on reproducing the same data many times over, releasing valuable time back into research.

Optimise the National Research and Innovation Ecosystem

Objective 4: Improve our ability to understand the impact of research inputs and evaluate research quality, impact and evidence of public benefit

By linking elements across the research and innovation ecosystem through PIDs, we can better understand the relationship between the elements of the ecosystem, including investment, use, outputs, outcomes and impact. We can more efficiently assess our research institutions' performance with respect to research quality thresholds, better trace the impact of the research and through this, more readily explain the public benefit

of the research.

Objective 5 Improve our ability to map Australia's research capability

Australia needs the research capability to address our national priorities. PIDs for people, organisations, research services and infrastructure would enable us to map our research capability against national priorities and invest appropriately to address gaps.

As written, the five objectives of the Australian National PID Strategy work together to support the Australian Research System.

Starting with inputs and outputs, the first two objectives identify the components of the research system. How we talk about, discuss, and act upon research must be built upon instances of these things. To interact with elements is also to be able to identify them - so they must be Findable, Accessible, Interoperable, and Reusable. It is appropriate then that a benchmarking strategy should also be assessed through this lens.

The journey towards the successful adoption of persistent identifiers for research objects is one that takes time, involves a community of stakeholders working together towards a common purpose, and is further supported within a global research information infrastructure. In creating a benchmarking approach for the inputs and outputs of research, then, we feel it is important to also consider the maturity of the identifier in question, the senior stakeholder (champions), the international engagement required, along with an overall purpose for why it is important.

Having established the objects of research, the third objective of the strategy poses us a challenge that cannot be met simply by persistent identifiers alone. If we are to improve research reproducibility, provenance and attribution while minimising administrative burden, enabling researchers to spend more time on research, then it matters how research objects and actors are connected. It is not simply adequate to describe the relationships between research things. The act of research must also provide a means for its description along the way.

Australian benchmarks around ORCID adoption presented in the final section of this document provide us with a way to envision how this can occur. Through the analysis of works assertions in the ORCID records, we can already see that many Australian institutions have managed to reduce academic burden for their researchers by making sure that their research information systems are connected to their ORCID records. Better still, we also see an increase in assertions from Crossref making their way back to ORCID records, significantly reducing the effort involved in publications reporting.

Of course, the relationship between publications and researchers is just one of the connections that we need to establish; efficient processes must be found across all research activities. Each connection between objects will require national focus, collaboration and effort. Knowing which connections to focus on first will be as important as identifying them.

With its fourth objective "Improving our ability to understand the impact of research inputs and evaluate research quality, impact and evidence of public benefit," the strategy provides us with a mechanism to triage which connections matter most to us as a community. If, for instance, it is most important to be able to trace the impact of research facilities on research policies, then we need to focus on efficient workflows that not only link publications to facilities but also tie those activities back to the grants that supported them, as well as the research data outputs that may have arisen. As part of producing the benchmarking toolkit, it was felt that it was important not to preempt what the most important workflows might be, but rather to establish a framework for linking benchmarkable outcomes back to national results. As part of this toolkit, we provide a worked example that breaks down an impact objective into individual benchmarks.

The fifth objective of the National PID strategy - improving our ability to map Australia's research capability, invites us to consider the benchmarking exercise from the perspective of scenarios. For instance, if the next global pandemic were to happen tomorrow how readily could we identify the right collaborations, and what information would we ideally want to use? Here, although beyond the scope of this report, we seek feedback from the community on identifying the right scenarios, both to stress test the utility of our research information infrastructure, and to consider the tools that are available to surface the insights that we need.

Finally, we reflect on the achievements that previous initiatives have made to shape our national information infrastructure. Vision 2020 outlined a national target that, by 2020, all Australian researchers, regardless of institutional affiliation, will be able to easily claim and manage the data regarding their research grants, research and research-related outputs through ORCID and that their details will travel with them as they build their careers. This target shared between institutions, funding bodies, representatives of the research community, libraries, and administration resulted in interventions that can be shown to have made a difference. In the period between 2020 and 2024, Australia has the third highest ORCID adoption rate for countries with estimated researcher pools of greater than 50,000. The average completeness of these records is 55%, ranking 2nd against the same set of countries. The success of interventions at the funder and institution level are also evident. The ORCID adoption and completion rates for researchers that receive funding from the ARC or the NH&MRC are amongst the best in the world. At an institutional level, it is also shown that local interventions also make a difference on the adoption and completeness. The extent to which these advances are also dependant on publisher commitment to shared research information infrastructure is also explored, as it differences in uptake by research classification. Overall, the message is both that

a) Australia's combined interventions to establish ORCID as a key part of the national research ecosystem have been successful so far, and

b) that benchmarking the adoption of persistent identifiers is achievable, and provides a powerful lens through which we can understand progress.

The final version of this report has benefited significantly from community feedback during its creation. We hope that as a result, the final report represents a benchmarking toolkit that meaningfully supports Austrlaia's implementation of its Persistent Identifier Roadmap.

2.1 A Note on Methodology, and the associated Toolkit

In order to benchmark the Australian persistent identifier landscape, the following snapshots of open datasets have been hosted on Google BigQuery:

- Crossref (Snapshot of the Crossref dataset as at September 2025) Note that abstracts have been removed as they have a different copyright licence. [5]
- DataCite (BigQuery representation of the DataCite yearly data file [7])
- ORCID (ORCID [13])
- Research Organisation Registry (v2 August 2025 [15])
- R3Data

With the exception of the R3dataset from r3data.org, these datasets should be considered the primary sources of the research persistent identifier graph. Further details on how to access these datasets are available in the Section 4.

Where possible benchmarking analysis has been carried out with reference only to primary sources. In some cases however, secondary sources are also required to supply additional information that has often been derived algorithmically or extends beyond the current persistent identifier graph. For these use cases, this report uses Dimensions as it both has links back to Crossref, DataCite, ORCID and the Research Organisation Registry, and also contains elements that the primary graph does not (yet) have, such as a global database of grants and funders, links to patents and policy documents, an expanded set of organisations, and an algorithmically derived set of researcher ids, against which ORCID adoption can be compared. As part of this project Dimensions was also used to create a dataset of data citations extracted from the fulltext of publications that Dimensions indexes. In addition, Altmetric data from Altmetric.com was also used to provide attention metrics. Both Altmetric and Dimensions are products of Digital Science, which also employs the authors of this report.

Section 4 provides benchmarking recipes for every table and chart that has been produced in sec-

tions 2.2, and 2.3 of this report[1]. The tables for each recipe are clearly described, with links out to a google colab notebook that allows the analysis to be run. Where alternatives exist to the secondary tables used from Dimensions and Altmetric, it will be up to future exercises to determine whether it is advantageous to swap these sources out for other providers.

Timelines and Consultation leading to this report

The benchmarking project commenced in February 2025 and was established as a core activity to underpin the Australian National Persistent Identifier (PID) Strategy 2024. The project began by utilising a list of prioritised institutional "value proposition" codesigned with the National Institutions Stakeholder Action Group, reflecting the need for shared value and shared accountability across the research ecosystem. The Institutions Stakeholder Action Group is one of several groups convened by the ARDC to drive PID adoption activities and contribute to the National Collaborative Roadmap.

The project team developed an "initial assessment" which focused on analyzing potential measures of impact and reviewing the long-term effects of the Vision 2020 Australia's ORCiD Aspirations released in 2017. This retrospective review draws on Australia's foundational leadership in PID support, such as the Australian ORCID consortium, which has already delivered substantial cost savings and efficiency gains to the academic research sector. On March 5, 2025, the initial assessment findings were presented to universities at the in-person event, "Research Transformation: Navigating change together," hosted by Digital Science at the University of Technology Sydney. Critical feedback was then sought the following day, March 6, through a dedicated workshop involving the National PID Strategy Advisory Group and the Australian ORCID Steering Committee. These advisory bodies provide essential strategic guidance to ensure implementation aligns with national priorities.

A resulting consultation draft was disseminated widely for sector review in early-June 2025, promoted via the Digital Science blog and LinkedIn, and circulated through the ARDC national newsletter. A significant consultation occurred on July 8, 2025, when the consultation draft was presented and workshopped with representatives from the national Stakeholder Action Groups, which include institutions, funders, and NCRIS facilities. The workshop also included senior stakeholders from the National PID Strategy Advisory Group and the Australian ORCID Steering Committee. Stakeholder engagement has been vital in achieving the review and will be vital in the collective action necessary for widespread PID implementation. Following the conclusion of the open feedback period at the end of July, more complete benchmarking recommendations were developed and reviewed in September 2025 by key ARDC personnel, including the National Product Manager for Persistent Identifiers, the National Product Manager for RAiD, and the ARDC Director for National Coordination. This pre-release version of the benchmarking report and recommendations will be announced and presented at the International Data Week conference and workshopped at the eResearch Australasia conference in October 2025, ensuring that the findings are shared across the national and international research communities.

[1] The analysis for ORCID adoption in section 3 is more complicated, and here the source code for the approach has been provided

2.2 Benchmarks for Research Inputs

Inputs to research include potential **research partners**, related **research projects**, **grants**, **facilities**, **data**, **samples and observations**. By increasing a researcher's ability to discover relevant inputs, we will reduce duplication, increase research productivity and create opportunities for impactful partnerships

Objective 1. Australian National PID Strategy

To support a process for benchmarking the Australian National PID Strategy, we start with the foundational elements of the research lifecycle, its inputs. By understanding how the use of persistent identifier infrastructure can be measured, taking into account current practice maturity along with anticipated adoption paths, we establish the building blocks of the PID benchmarking toolkit.

For each research input, we have identified:

- **Motivation for Measurement** What is the nature of the input being measured.
- **Expected Maturity** How recent is the persistent identifier infrastructure that we are trying to measure? Effective persistent identifier infrastructure takes time to filter through the activities of the research community, and this shapes what we should measure in order to help assess its adoption.
- **Senior Stakeholder** Does the input have a senior stakeholder who is able to drive improvements? Although identifying senior stakeholders for each input is beyond the scope of this report (and has been intentionally left blank), identifying which stakeholders to own implementable benchmarks will be an important part of any future benchmarking implementation.
- **International Dependencies**. Almost all persistent identifier infrastructure must have a global component in order to be effective. Persistent identifier must a global as well as a local focus. Understanding how to focus international advocacy activities is an important part of a national pid benchmarking toolkit
- **Benchmarks through the lens of Findability, Accessibility, Interoperability, and Reuse** By benchmarking the findability, accessibility, interoperability, and reuse, we can understand how well our systems are structured to support high-quality, efficient, and collaborative research.
- **An overall perspective of why the benchmarks for the research input in focus matter** Why is this research input important to the Australian research ecosystem? What is the future state that the community should aspire to?

Finally, for each research input the report provides analysis on the measurements that are possible today, and identifies any appropriate recommendations.

Research Inputs

1 Summary of ROR strategic value and benchmarks. 12

2 Summary of RAiD strategic value and benchmarks. 18

3 Summary of grant strategic value and benchmarks. 20

4 Summary of facility identifiers strategic value and benchmarks. 24

5 Summary of research data strategic value and benchmarks. 26

6 Summary of sample data strategic value and benchmarks 32

7 Summary of Instruments strategic value and benchmarks 35

2.2.1 Research Organisations

Research Input 1.

Research partners: Creating and sustaining research partnerships is critical to fostering a collaborative research environment. Identifiers that describe active research institutions allow past activities to inform new collaborations

Strategy Measurement Motivation
Most research partners should be represented in ROR. This includes universities, research centres, and key infrastructure bodies.
International Engagement/Dependencies
The engagement of international publishers is critical. Without their support, persistent identifiers like ROR will not be fully adopted in the scholarly ecosystem.

Expected Maturity	Emerging
Senior Stakeholder	

Benchmarks

Findability	Representation of Australian Research Partners in ROR. Measures: coverage rate, inclusion lag time.
Accessibility	ROR data is openly available with no restrictions. Usability across platforms is in progress.
Interoperability	Use of Australian ROR IDs by publishers, crosswalks with other metadata. *Note: some disciplines underrepresented.*
Reuse	Ongoing measurement of ROR presence in full-text submissions. Potential vs actual use highlights gaps and opportunities.

Why it Matters

Trusted government and institutional reporting is enabled through the use of authoritative identifiers for all Australian research institutions. Use of algorithmically applied identifiers is reduced, increasing accuracy and ownership of Australian research activity.

Analysis

Findability

Using the Dimensions dataset as a reference point, there are 1070 Australian Research Organisation Registry (ROR) ids associated with publications between 2020 and 2024. Although ROR was based on a set of organisations from Dimensions (GRID)[2], and there has historically been a close correspondence with the set of organisations in Dimensions, recently the set of organisations in Dimensions has been expanded from the set of organisations required to reasonably explain global publication output to the set of organisations required to explain global patent affiliations. As there is now a divergence between the two datasets, Dimensions can be used as a reference to assess the coverage of ROR for Australian institutions that are involved in publication. Currently, there are another 531 organisations that have produced publication outputs (toolkit query 4.1.1). As can be seen in figure 1, the majority of the missing organisations are companies or of type other, indicating that a review of whether these additional organisations should be added to ROR may improve the Australian research sector's ability to represent industry engagement. As this process requires the creation of public records about each new organisation that will be used in the international research system, it is recommended to consult these companies before requesting that they be added to the ROR[3].

Interoperability

Although the Research Organisation Registry was established in 2020, publishers have been slow to

[2] https://www.grid.ac/

[3] for this reason, Digital Science is not considering submitting the broader set of institutions to ROR as a bulk dataset

implement direct assertions in their publication workflows. In total, between 2020 and 2024 there were 27,75 affiliation links that had been made (Figure 2), compared to 1180610 affiliation links that could have been made. Of these links, it can also be observed that affiliation links can only be found in a limited number of publishers, indicating that international engagement with most publishers is essential if practice is to change. Unsurprisingly, the usage of direct Australian affiliations compared to the global number of directly asserted affiliations is also proportionate across fields of research (figure 4)

Without links to affiliations being directly asserted in Crossref records, Australia's ability to understand how different organisations contribute to the research landscape remains dependent on companies to determine affiliations via algorithms.

Although support for ROR ids has only begun to be implemented in publication records, ROR ids are much more common in DataCite records, representing datasets, reports, software, and other output types and a range of other resource types (figure 5. These records are concentrated in many institutional repositories with particularly strong contributions from Swinburne and Monash (figure 6).

The number of ROR affiliated records in repositories registered with DataCite offers a unique way of determining separating institutional repositories from generalist repositories. In the table below, we associate an institution with a repository if that institution has the largest number of works affiliated to it, and the percentage of works is greater than 8%. (This technique will be used as part of the benchmarking strategy for other research types in the analysis below.) Although significantly more populated than the records in Crossref, the variability of the ROR affiliated records by repository indicates that more work is required between institutions and repository vendors to ensure that the affiliation identifiers for all contributors become the default practice.

dimensions organisation type	number or orgs	publications
Company	251	1,602
Education	5	150
Facility	7	671
Government	11	138
Healthcare	18	2,443
Nonprofit	9	414
Other	230	2,584

Figure 1: Missing Australian organisations by dimensions organisation type (TK 4.1.1)

Year	affiliation links
2020	115
2021	204
2022	589
2023	668
2024	1,415

Figure 2: Directly asserted ROR affiliations by year (toolkit query 4.1.2)

publisher	affiliations
eLife Sciences Publications, Ltd	874
John Benjamins Publishing Company	357
American Physical Society (APS)	303
Australasian Association of Writing Programs	228
Springer Science and Business Media LLC	195
Bond University	152
Optica Publishing Group	151
International Union of Crystallography (IUCr)	117
Center for Open Science	109
Rockefeller University Press	69

Figure 3: Publishers with Australian ROR Assertions (toolkit query 4.1.3) The number of participating publishers is small

Recommendations

Recommendation 2.2.1: Review the representation of commercial Australian research organisations in ROR. With the consent of new commercial organisations identified in Dimensions (or another appropriate resource,) provide details of these institutions to ROR. The additional institutions to be reviewed can be identified using (toolkit query 4.1.2)

Recommendation 2.2.2: Engage with international publishing community to ensure ROR affiliations are being directly asserted onto publication records where available. The International publishing community is far less engaged with implementing ROR than would benefit the community. International advocacy will be required to highlight why this should be a priority for publishing systems.

Recommendation 2.2.3: Engage with repository vendors and review internal workflows to ensure that ROR affiliations are added to all creators. Progress towards implementing ROR affiliations within DataCite is comparatively strong, but remains inconsistent. Engaging with vendors and reassessing current workflows to capatilise on this strength is recommended.

field of research	pubs with affiliations	pubs with australian affiliations	percentage	reference percentage
Physical Sciences	26,421	478	2%	2%
Engineering	17,348	156	1%	2%
Biomedical and Clinical Sciences	8,695	262	3%	3%
Language, Communication and Culture	7,986	409	5%	3%
Biological Sciences	6,964	235	3%	3%
Mathematical Sciences	5,472	98	2%	2%
Health Sciences	3,891	113	3%	6%
Chemical Sciences	3,636	84	2%	2%
Psychology	3,338	101	3%	5%
Human Society	2,728	44	2%	4%
Information and Computing Sciences	2,311	33	1%	2%
Commerce, Management, Tourism and Services	2,151	24	1%	4%
Education	1,846	122	7%	4%
Creative Arts and Writing	1,763	194	11%	3%
Philosophy and Religious Studies	1,552	26	2%	3%
Law and Legal Studies	1,162	88	8%	4%
History, Heritage and Archaeology	1,122	7	1%	3%
Agricultural, Veterinary and Food Sciences	848	3	0%	3%
Economics	825	4	0%	3%
Built Environment and Design	809	19	2%	4%
Environmental Sciences	553	22	4%	5%
Earth Sciences	497	20	4%	4%

Figure 4: Publications with Australian ROR Assertions by Field of Research as a proportion of all assertions (toolkit query 4.1.4)

resourceTypeGeneral	ROR affiliation	all works	percentage affiliation	Australian affiliations	percentage inc Aus aff
Dataset	10,820,054	28,318,211	38	11,191	0
Text	411,954	13,930,328	3	31,571	8
PhysicalObject	231,694	13,537,117	2	19	0
Image	49,789	4,836,859	1	2,472	5
JournalArticle	43,008	1,426,411	3	238	1
Preprint	5,625	1,344,393	0	67	1
	2	1,327,008	0	0	0
Other	120,452	1,174,156	10	12,257	10
Collection	104,004	788,441	13	5,510	5
Software	20,435	574,602	4	574	3
Audiovisual	15,353	367,691	4	1,930	13
Dissertation	68,070	268,014	25	26,214	39
ConferencePaper	13,286	179,986	7	1,715	13
Report	13,099	150,946	9	2,963	23
StudyRegistration	10,544	134,746	8	235	2
Book	6,429	117,212	5	67	1
InteractiveResource	2,913	106,715	3	32	1
Event	4,476	79,692	6	20	0
BookChapter	7,364	75,852	10	40	1
Sound	1,046	60,327	2	3	0
Journal	3,079	39,473	8	25	1
Model	10,316	22,829	45	48	0
ConferenceProceeding	4,299	16,226	26	56	1
DataPaper	413	16,060	3	8	2
PeerReview	627	14,658	4	10	2
Workflow	778	7,199	11	28	4
OutputManagementPlan	3,643	5,862	62	109	3
Standard	226	4,397	5	10	4
ComputationalNotebook	2,252	4,224	53	67	3
Service	43	474	9	2	5
Instrument	81	191	42	2	2
Project	11	13	85	0	0
Award	5	5	100	0	0

Figure 5: ROR Assertions in DataCite records 2020-2024 (toolkit query 4.1.5)

institution	repository id	count	all works	percentage
Swinburne University of Technology	swin.repo2	12,475	19,144	65%
La Trobe University	latrobe.repo	8,236	17,369	47%
UNSW Sydney	unsw.repo	4,950	4,960	100%
RMIT University	figshare.rmit	4,875	10,040	49%
Monash University	monash.repo	2,224	24,987	9%
The University of Western Australia	uniwa.repo	1,646	1,904	86%
University of Tasmania	ardcx.utas	1,492	10,603	14%
The University of Melbourne	unimelb.repo1	1,397	2,835	49%
Central Queensland University	ardcx.cqu	954	1,508	63%
Macquarie University	ardcx.mqu	724	6,702	11%
The University of Adelaide	adelaide.repo	542	941	58%
Torrens University Australia	ardcx.torrens	441	517	85%
Western Sydney University	ardcx.wsu	424	553	77%
James Cook University	jcu.repo	417	759	55%
University of Southern Queensland	ardcx.usq	280	650	43%
Edith Cowan University	ardcx.ecu	276	299	92%
Griffith University	griffith.figshare	208	239	87%
The University of Sydney	ardcx.usyd	155	221	70%
Murdoch Children's Research Institute	figshare.mcri	147	271	54%
Charles Darwin University	cdu.repo	129	146	88%
Flinders University	figshare.flinders	104	166	63%
Flinders University	ardcx.flinders	88	109	81%
Deakin University	ardcx.deakin	61	254	24%
Federation University	ardcx.federation	48	98	49%
The University of Queensland	uq.repo	38	463	8%
University of Newcastle Australia	ardcx.uon	36	49	73%
Australasian Leukaemia and Lymphoma Group	cctg.allg	32	34	94%
University of Canberra	ardcx.uc	29	228	13%
Monash University	monash.oer	28	29	97%
University of Technology Sydney	ardcx.uts	17	27	63%
Australian Institute of Marine Science	ardcx.aims	17	19	89%
University of Wollongong	ardcx.uow	16	22	73%
James Cook University	ardcx.aims2	15	31	48%
Australian Research Data Commons	datacite.rda	7	68	10%
Charles Sturt University	ardcx.csu	7	15	47%
University of Technology Sydney	uts.repo2	6	8	75%

Additional 21 rows not shown.

Figure 6: ROR Affiliations by Australian Repositories (toolkit query 4.1.6)

2.2.2 Research Activities

Research Input 2.

Research activities: Research activities define collaborators, organise resources, define security and access. Research activities differ from awards or grants in that they are a dynamic representation of a research project (or research program) – including all participants and their roles. Research activities are also the natural aggregation of the accumulation of research metadata, with resulting research items including publications, patents, and datasets inheriting much of the same metadata including participants, classifications, connected research organisations.

Strategy Measurement Motivation
It is expected over time that research activities should be supported by Research Activity identifiers (RAiD,) and institutions support the creation/maintenance of RAiDs
International Engagement/Dependencies
Service providers to research activities are international - concerted engagement is required to ensure the support and uptake of RAiD identifiers.

Expected Maturity	Early Adoption
Senior Stakeholder	

Benchmarks

Findability	The number of RAiDs that can be found in DataCite, via RAiD Registration agencies
Accessibility	In Progress
Interoperability	*number of institutions with RAiD management strategies *presence of RAiD identifiers in Crossref metadata *The number of research service providers that integrate RAiDs into their workflow, or provide interfaces with existing practices
Reuse	*presence of RAiD identifiers in the free text of publications *The number of identified workflows that reuse metadata from a RAiD

Why it Matters

A community of practice is established across the Australian research community, where research activities are organised around common definitions of research projects

Analysis

To measure the number of RAiDs that have been registered, DataCite records were filtered to find records that were contained in dedicated RAiD repositories (toolkit query 4.1.7). From this analysis only 69 RAiD projects were identified, with most registered at The University of Notre Dame Australia (figure 7). The small number of RAiDs detected reflects the recent support for RAiD registration agencies in DataCite[4], following the establishment of RAiD as an ISO Standard in 2022[5] .

As RAiDs are only just beginning to be adopted within the Australian research ecosystem, it is difficult to assess many proposed benchmarks, however it is possible to define what successful adoption of RAiD should look like. Although research activities are not grants, every grant should be supported by an active research activity record. For a successful implementation then, it is a reasonable target to expect that the number of active RAiD records if adopted and endorsed by the community should initially be roughly the same as the number of active research grants.

It may also be strategic to work towards a process where RAiD records are created at the time that a grant is awarded, where ownership of the RAiD record is assigned to the administering organisation assign to the grant record. Perhaps minted by the funding body, responsibility for updating the RAiD record would then shift to the administering organisation once functionality has been established.

[4] https://support.datacite.org/docs/raids
[5] https://ardc.edu.au/article/new-iso-standard-published-for-the-research-activity-identifier-raid/

This process may be necessary as without a large body of established RAiDs to work form, services providers have little incentive to build functionality that incorporates them into their workflows.

id	name	count
ardcx.raid-ygcuhh	The University of Notre Dame Australia	66
ardcx.raid-yschfy	RAiD Australia and New Zealand	2
ardcx.raid-ynhroz	Australian Research Data Commons	1
datacite.services	PID Services Registry	1

Figure 7: Number of RAiDs that have been created as at January 2025

Recommendation 2.2.4: The Research Community adopts a RAiD target that initially is roughly equivalent to the number of active grants. . By quickly establishing a large number of RAiD records representing a large proportion of active research within Australia, service providers have an incentive to build functionality around these records.

2.2.3 Grants

Research Input 3.

Grants provide funding for research activities, typically awarded by a funding body based on specific criteria and objectives. Grants establish financial resources, reporting obligations, compliance requirements, and timelines. Grants can support multiple research projects and can be linked to various institutions and researchers. Metadata associated with a grant includes funding source, award amount, duration, and compliance regulations, and research classifications. Research outputs such as publications and datasets, and research activities can inherit grant-related metadata.

Strategy Measurement Motivation
Grants are identified (within funder) by funding bodies. A further expectation is that they should have a persistent identifier, and that grants should be linked to outputs (by PID or otherwise) PIDs are also to be expected to apply to consultancies - however a registration process is yet to be identified
International Engagement/Dependencies
A growing international body of funders contributing persistently identifiable grant records will make it more likely that workflows that utilise persistently identified grant records will be incorporated into research submission (such as declaring funding acknowledgements in publication submission processes)

Expected Maturity	Emerging
Senior Stakeholder	

Benchmarks

Findability	Percentage of Grants by funder that have a persistent identifier
Accessibility	Percentage of Australian output with funder acknowledgement statements
Interoperability	Assessment of international progress towards linking RAiDs and grant records in either Crossref or DataCite
Reuse	Percentage papers with grant ids in the funder acknowledgements vs simply acknowledging the funder

Why it Matters

Trusted government and institutional reporting is enabled through the use of authoritative identifiers for all Australian research grants. By ensuring that Australian funding is linked by grant DOIs to publications, a culture of tracking funding impact well after the project has completed can be established

Analysis

Findability Although there are 90 grant identifiers registered in Crossref by the ARDC (figure 8), most grants issued by Australian funders have yet to be associated with persistent identifiers[6]. As figure 8 shows, there are nine funders globally with over 1000 grant records registered in Crossref, with 6 creating records with links to ORCID and ROR. Whilst this still represents a minority of global funders, the existence of these records indicates that end to end workflows can be established at scale, and that there are no insurmountable barriers to implementing DOIs for grant records in either Crossref or DataCite.

The number of Australian grants that could be registered in Crossref can be estimated by looking at the number of active grants in 2025 (figure 9)

Interoperability

Despite the lack of globally adopted persistent identifier practice in either DataCite or Crossref for grants and awards, the practice of including grant ids in the acknowledgement section of publica-

[6] An earlier project supported by the ARDC did assign purls to awards by the National Health and Medical Research Council, and the Australian Research Council. https://researchdata.edu.au/grants

publisher	grants	with orcids	with ror	australian aff
Publications Office of the European Union	53,474	0	0	0
Fonds de recherche du Québec	25,054	3,073	23,415	6
Wellcome	19,929	9,693	18,787	179
Austrian Science Fund (FWF)	19,576	0	0	0
Fundacao para a Ciencia e a Tecnologia (unidadeFCCN)	17,422	12,875	0	0
Office of Scientific and Technical Information (OSTI)	14,231	4,145	3,674	13
Japan Science and Technology Agency (JST)	11,866	0	0	0
American Heart Association (AHA)	2,764	2,758	2,658	2
Dutch Research Council	1,351	0	0	0
American Cancer Society	1,162	1,150	1,141	0
American Institute for Cancer Research	806	260	733	2
Children's Tumor Foundation	735	117	114	2
James S. McDonnell Foundation	688	0	0	0
Asia-Pacific Network for Global Change Research	599	0	0	0
The ALS Association	589	160	140	3
Melanoma Research Alliance	532	139	135	2
Templeton World Charity Organization	478	1	119	5
Chan Zuckerberg Initiative (CZI)	399	162	0	0
Human Frontier Science Program	388	150	374	14
Wiener Wissenschafts-, Forschungs- und Technologiefonds (WWTF)	380	0	0	0

Additional 24 rows not shown.

Figure 8: Current Australian Grants registered in Crossref (toolkit query 4.1.8)

tions is well established. The practice has been further formalised in the Crossref metadata schema, with funders receiving DOIs (this practice will now shift to ROR ids [7]). As figure 9 indicates, almost all active grants from major funders in Australia can be associated with at least one publication that acknowledge it (either in the text of the publication or the Crossref metadata). In figure 10, the difference between grant ids recorded in Crossref and those that can be obtained from also extracting grant ids from the acknowledgement text is approximately 6% of publication output, with the difference between funder ids in Crossref, and those that can be extracted from the text of publications at 8% of publication output.

The existence of a wealth of publication to grant linkages significantly increases the value of creating grant records with persistent identifiers. Provided that these records are completely described, linkages between the amount of funding invested and resultant publications can be openly described. If the links between grants and publications are open, then it becomes possible to assess the association of grant funding on emerging research areas, or even policy (via publication citation.)

[7] https://ror.readme.io/docs/funder-registry

granting body	active grants	acknoweldged grants
Australian Research Council	39,229	37,187
National Health and Medical Research Council	4,597	4,541
Medical Research Future Fund	1,813	762
Department of Industry, Science and Resources	755	4
Department of Health and Aged Care	499	0
Fisheries Research and Development Corporation	258	61
Australian Centre for International Agricultural Research	201	132
Australian Renewable Energy Agency	160	0
National Breast Cancer Foundation	154	129
Department of Climate Change, Energy, the Environment and Water	115	0
Department of Agriculture, Fisheries and Forestry	104	1
Cancer Australia	73	0
Department of Foreign Affairs and Trade	63	0
Department of Defence	55	0
Department of Infrastructure, Transport, Regional Development, Communications and the Arts	37	0
Wine Australia	35	1
Department of Social Services	32	0
Office of National Intelligence	32	0
Department of Education	30	0
Australian Trade and Investment Commission	23	0
National Indigenous Australians Agency	15	0
Garnett Passe and Rodney Williams Memorial Foundation	13	4
Department of Employment and Workplace Relations	3	0
Department of the Treasury	3	0
Attorney-General's Department Australia	3	0
National Emergency Management Agency	3	0
National Blood Authority	3	0
Department of Home Affairs	2	0
Hermon Slade Foundation	2	0
Department of the Prime Minister and Cabinet	2	0
Organ and Tissue Authority	2	0
Grains Research and Development Corporation	1	0
Australian Federal Police	1	0

Figure 9: Number of active grants from Australian funders in 2025 as represented in Dimensions (toolkit query 4.1.9) A grant is counted as having been acknowledged if at least one publication uses its grant id in either the text of the document, or its associated Crossref record

Grant records could potentially play a further role in the persistent identifier landscape. If a RAiD record were also to be created at the same time that a grant identifier was created, then it becomes possible for service providers to offer services to the research activities that grants fund from that day that they are awarded. However, for this workflow to be realised, all researchers on grants must have and ORCID, and the grant record must be as completely described as possible including institutional affiliations and research classifications.

Recommendations

Recommendation 2.2.5: As soon as is practical, Australian funders create persistent identifiers for Grants including complete metadata. Award metadata should include the amount funded, and uniquely identify researchers via their ORCIDs, and research affiliations (ROR)

Recommendation 2.2.6: The creation of a grant should trigger the creation of a RAiD (or the association of an existing RAiD). Creating a RAiD as soon as a project has been funded means that research objects, people and other associated metadata can begin to be associated with it from day one.

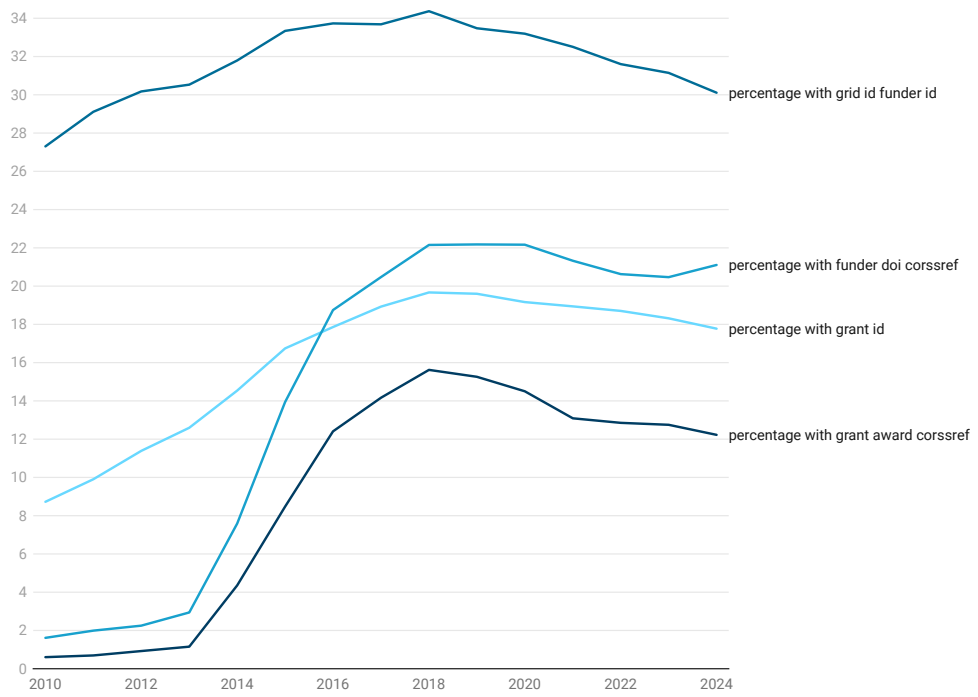


Figure 10: Links between publications and Australian Grants by funder (toolkit query 4.1.10) As of 2024, the difference between grant ids recorded in Crossref and those that can be obtained from also extracting grant ids from the acknowledgement text using Dimensions is approximately 6% of publication output, with the difference between funder ids in Crossref, and those that can be extracted from the text of publications at 8% of publication output.

2.2.4 Facilities

Research Input 4.

A **facility** enables research by offering specialised equipment, laboratories, computing resources, or experimental environments. A research facility is a physical or virtual resource that supports scientific investigation by providing specialised infrastructure, instruments, and services. Facilities may be dedicated to specific disciplines, such as particle accelerators for physics, sequencing labs for genomics, or high-performance computing clusters for data-intensive research. Access to a facility is typically governed by institutional policies, external partnerships, or funding agreements. Facilities serve as key aggregation points for research metadata, linking to associated research projects, grants, and researchers. Metadata related to a facility includes available resources, operational policies, access restrictions, and usage statistics. Research outputs such as datasets and publications often acknowledge facility usage as part of their metadata.

Strategy Measurement Motivation
Representation of Australian Research Facilities should be registered within ROR. Exception: there may be smaller facilities may not meet the threshold, and may be better described as Research Instruments
International Engagement/Dependencies
The scope of facility engagement is largely focused within Australia for this exercise

Expected Maturity	Early Adoption
Senior Stakeholder	

Benchmarks

Findability	Representation of Australian Research Facilities in ROR
Accessibility	–
Interoperability	*Number of institutions that can record links between publications and facilities *Number of facilities that can record links between facility access and research projects.
Reuse	*Number of papers that link to facilities via their identifier (vs full text name)

Why it Matters

Tracking the publication output of facilities becomes as easy as reporting on the output of institutions

Analysis

Findability There are currently 212 Facilities that are identifiable in ROR (toolkit query 4.1.11).

Interoperabilty Facilities as a subset of organisations, do not have a structured way of being associated with research publications beyond author affiliation, accept via free text within publications. Facilities can be associated indirectly with publications via author affiliations (figure 11). As with publications more generally, the number of publications that can be attributed algorithmically far exceeds the number of affiliations that have been directly asserted in Crossref metadata.

Recommendation

Recommendation 2.2.7: Facilities confirm that they are represented in ROR.

ror_id	Facility Name	publications alogrithm asserted	publications directly asserted
https://ror.org/004y8wk30	QIMR Berghofer Medical Research Institute	3,887	9
https://ror.org/0020x6414	Hunter Medical Research Institute	2,995	3
https://ror.org/023331s46	George Institute for Global Health	2,215	1
https://ror.org/01g7s6g79	Neuroscience Research Australia	1,946	2
https://ror.org/03n17ds51	CSIRO Agriculture and Food	1,476	1
https://ror.org/03vk18a84	Australian Synchrotron	1,164	10
https://ror.org/04zj3ra44	Westmead Institute for Medical Research	1,010	1
https://ror.org/03x57gn41	Australian Institute of Marine Science	909	1
https://ror.org/04hy0x592	Woolcock Institute of Medical Research	777	2
https://ror.org/016899r71	Peter Doherty Institute	705	7
https://ror.org/04yn72m09	Perron Institute for Neurological and Translational Science	675	4
https://ror.org/01bsaey45	Children's Medical Research Institute	557	1
https://ror.org/005ynf375	Victorian Infectious Diseases Reference Laboratory	501	2
https://ror.org/046fa4y88	The Heart Research Institute	456	2
https://ror.org/02zp3yd51	ARC Centre of Excellence for Gravitational Wave Discovery	438	7
https://ror.org/02qa5kg76	Australian Regenerative Medicine Institute	342	2
https://ror.org/02aseym49	Australian Centre for Disease Preparedness	327	2
https://ror.org/04ynn1b95	Astronomy and Space	224	1
https://ror.org/03jh4jw93	CSIRO Health and Biosecurity	216	2
https://ror.org/05sh7tb37	ARC Centre of Excellence for Transformative Meta-Optical Systems	176	1

Figure 11: Top 20 Facilities ordered by publications associated with author affiliations 2020-2024 (toolkit query 4.1.12)

2.2.5 Research Data

Research Input 5.

Research data represents the recorded information, observations, or measurements generated or collected during research. It serves as the empirical foundation for scientific inquiry, supporting analysis, validation, and reproducibility. Broadly defined research data can take many forms, including numerical datasets, images, audio recordings, software code, laboratory notebooks, and survey responses. Research data is closely linked to the research life cycle, as it is produced by research projects, often funded by grants, and may be generated using facilities. Metadata associated with research data includes contributors, data formats, methodologies, licences, and storage locations. Research data is often cited in publications, patents, and reports, inheriting project, grant, and facility metadata. As an input to research processes, research data can be reused beyond its original project, supporting new analyses and collaborations.

Strategy Measurement Motivation
Research data are stored in a mixture of institutional and global/national repositories. These repositories can be generalist or subject-specific. Benchmarking of research data sets is primarily achieved via DataCite DOIs; however, there will be other subject-specific solutions that also need to be tracked.
International Engagement/Dependencies
Support for the creation of data availability sections is implemented by international publishers. Advocacy is required to ensure that research data are always properly cited

Expected Maturity	
Senior Stakeholder	

Benchmarks

Findability	
Accessibility	Percentage of papers from Australian research institutions that have data availability sections Percentage of papers that link to a repository Percentage of papers that say "Data available upon request" (analysis by discipline)
Interoperability	*number of Australian institutions with support for research data repositories with persistent identifier infrastructure (general and subject specific)
Reuse	

Why it Matters

Analysis

In this report, the discovery of datasets has been limited to those that have a DOI registered in DataCite.

According to the registry of research data repositories (https://www.re3data.org/), within Australia there are 110 out of 256 repositories (current or previous) that use DOIs as a persistent identifier system (figure 12). This accounts for 42% of identified repository systems, with systems that do not use a persistent identifier system making up 40%.

To identify Australian repositories on DataCite, we use the strategy identified in the Research Partners (Section 2.2.1) above. For each repository, the institution with the highest number of affiliated works is identified, and if the percentage of total works associated with the repository is greater than 8%, then the repository is assigned to the institution. As this approach only identifies a third of repositories on its own, additional country data is recorded against the client (repository record) or in some cases the provider, with a fallback of the most common country associated with registered domains.

Using the approach defined above, it is possible to obtain an overall view of the percentage of

country	ARK	CSTR	DOI	IGSN	PURL	URN	hdl	none	other
USA	60		925	6	22	25	117	1,620	123
DEU	1		627	1	39	70	119	369	67
CAN			243		2		44	373	15
GBR	2		185		10	9	32	281	26
AAA	5		152		2	3	41	157	21
FRA	9		112		6	4	20	140	8
AUS			110		5	6	21	103	11
EEC	3		79		5	7	74	211	26
ESP			62		4	2	48	17	4
NLD	1		52		1	15	31	41	4
POL	1		48			18	14	2	3
CHE	4		47		5		5	65	11
ITA			46		4		14	29	
NOR	1		43			4	11	15	
SWE			35			1	8	24	1
AUT			28			2	11	23	6
ZAF			25				9	1	
CHN	5	1	24				2	42	14
EST			20		1		7	2	1
JPN	4		20					88	5

Figure 12: repositories by PID system as recorded in Re3Data (toolkit query 4.1.15). Note, Re3Data is updated manually, and does not reflect the current status of repositories in DataCite, for instance IGSN repositories in Australia have not been recorded in the statistics above.

works by type that are contained within Australia repositories. Overall, Australia hosts 4% of all repositories hosted in DataCite. Notably, Australia has registered more than 17% of all dissertations in DataCite[8], and 15% of works classified as sounds. At the other end of the scale, Australia is under-represented in works classified as datasets and software.

Further, by removing the largest of the generalist repositories ('cern.zenodo', 'gbif.gbif', 'estdoi.bio', 'figshare.ars', 'arxiv.content', 'fao.itpgrfa', 'zbmed.dsmz', 'rg.rg', 'cos.osf',) it is possible to gain an overall ranking by country of the number of works deposited in Australian repositories. For generalist repositories that store more than one type of work, Australia ranks 6th behind France, the United States, Germany, the United Kingdom, and Switzerland (figure 13). For single-type repositories (figure 13), Japan stands out with 9.9 million works registered in a single-type repository, but only 4,000 works recorded in multi-type repositories.

Extending the same query as in the previous section (toolkit 4.1.17), it is possible to identify 25 institutional repositories, where the greatest number of of outputs is associated with the institution, and this number makes up greater than 8% of total works. Institutional repositories with less than 50 works were not considered (figure 14).

Interoperability

As demonstrated in the previous section on accessibility (figure 14), datasets associated with ROR identifiers have been associated at an overall rate of 43%. ORCID representation over the same records is 9.8% - a rate significantly lower than than ORCID adoption rates for publications (see

[8](this percentage of dissertations drops to 14% when dissertations in crossref are also considered in analysis below)

country code	multi type	single type	works	inst repo ror aff percentage
FR	871,899	529,521	1,401,420	1%
DE	704,663	779,457	1,484,120	41%
US	628,524	2,149,861	2,778,385	4%
GB	305,731	346,047	651,778	19%
CH	201,268	700,894	902,162	0%
AU	184,317	159,967	344,284	31%
CA	94,029	60,306	154,335	2%
NL	84,215	282,865	367,080	1%
ZA	78,724	6,838	85,562	100%
RU	73,395	153,356	226,751	0%
IT	64,739	18,848	83,587	3%
GR	40,378	26,986	67,364	0%
AT	37,111	18,992	56,103	27%
TH	21,472		21,472	
DK	15,699	12,066	27,765	53%
CO	13,611	3,463	17,074	35%
ID	13,321	12	13,333	0%
PT	9,930	1,882	11,812	34%
NZ	9,466	5,531	14,997	47%
CN	8,761	12,828	21,589	

Additional 52 rows not shown.

Figure 13: DataCite deposit statistics by Country 2020-2024 (toolkit query 4.1.16) Use the interactive version of this chart to explore the additional 52 rows, by clicking on the image above

Section 3 below.

Reuse

To track data set reuse through the presence of dataset DOIs in scholarly publications, the Dimensions API is first used to search the full text of articles for DOI prefixes associated with repositories registered in DataCite. This step provides a broad pool of candidate publications that may contain references to datasets.

Once this initial corpus of publications is assembled, the full text of each publication is processed using GROBID (GeneRation Of BIbliographic Data) [10]. GROBID transforms unstructured publication content into machine-readable structured text. This allows for reliable extraction of bibliographic metadata, reference lists, and crucially in-text citation contexts.

Within the structured output, the next step is to locate and extract all DOI strings present in the text. These candidate DOIs are then filtered to retain only those associated with DataCite-registered repositories, using their known DOI prefixes as a reference point. The extracted DOIs are subsequently matched with the DataCite registry to ensure that they correspond to actual datasets.

Using the derived table of dataset citations, it is possible to get an overview of the level of global dataset citation. Dataset citation rates (via DOI) are highest in Earth Sciences at 9.43%, with only 3 other /fields of research having data citation rates above 3% (figure 15.) Against the global average, Australia compares favourably with 18.01% of Australian publications in Earth Sciences citing dataset DOIs, and 5 other fields of research above 3%. However, from figure 16 it can be seen that data citation rates vary greatly by country, with 5 European countries having twice the data citation rate for Earth Sciences. To get an overall picture of relative dataset citation engagement by country, we have chosen to use as a measure of engagement the ratio of the percentage of global papers with data set citations by country over the global percentage of papers for that country. With this measure (figure 17), it can be seen that in general Australia sits in the lower middle of the pack when it comes to researcher dataset citation, with many European countries taking the lead in most fields of research. This result is out of step with Australia's level of engagement with research data practices and indicates an area for improvement.

repository id	repository name	institution	works with institution afilliation	all works	percentage	multi type	works with orcids
swin.repo2	Swinburne University of Technology	Swinburne University of Technology	12,475	35,941	35%	True	2,159
latrobe.repo	La Trobe University	La Trobe University	8,236	29,164	28%	True	1,199
figshare.rmit	RMIT	RMIT University	4,875	14,030	35%	True	1,449
unimelb.repo1	University of Melbourne data repository	The University of Melbourne	1,397	5,722	24%	True	2,137
unsw.repo	University of New South Wales	UNSW Sydney	4,950	4,960	100%	True	2,198
ardcx.cqu	Central Queensland University	Central Queensland University	954	2,334	41%	True	167
uniwa.repo	The University of Western Australia	The University of Western Australia	1,646	1,904	86%	True	59
adelaide.repo	The University of Adelaide	The University of Adelaide	542	1,755	31%	True	866
ardcx.torrens	Torrens University	Torrens University Australia	441	986	45%	True	86
jcu.repo	James Cook University	James Cook University	417	759	55%	True	607
ardcx.usq	University of Southern Queensland research data collection	University of Southern Queensland	280	650	43%	True	323
ardcx.wsu	Western Sydney University ResearchDirect	Western Sydney University	424	553	77%	True	238
uq.repo	The University of Queensland	The University of Queensland	38	463	8%	False	458
ardcx.deakin	Deakin University	Deakin University	61	412	15%	True	313
figshare.flinders	Flinders University	Flinders University	104	351	30%	True	210
ardcx.ecu	Edith Cowan University	Edith Cowan University	276	299	92%	True	247
griffith.figshare	Griffith University Figshare	Griffith University	208	244	85%	True	5
ardcx.uc	University of Canberra Research Portal	University of Canberra	29	228	13%	True	1
ardcx.usyd	University of Sydney Library	The University of Sydney	155	221	70%	True	62
ardcx.federation	Federation University Australia	Federation University	48	174	28%	True	62

Additional 3 rows not shown.

Figure 14: Australian Institutional Repositories. Work counted between 2020-2024 (toolkit query 4.1.17)

field of research	percent	publications with data citation	all research articles
Earth Sciences	9.43%	12,155	128,884
Biological Sciences	4.76%	15,304	321,535
Environmental Sciences	4.24%	5,167	121,892
Physical Sciences	3.09%	5,509	178,416
Psychology	1.92%	2,007	104,708
Economics	1.79%	1,232	68,679
Information and Computing Sciences	1.57%	5,064	322,219
Human Society	1.27%	2,695	212,799
Chemical Sciences	1.16%	3,447	297,936
Agricultural, Veterinary and Food Sciences	1.12%	2,319	206,999
Biomedical and Clinical Sciences	0.98%	7,885	801,971
Built Environment and Design	0.96%	736	76,574
History, Heritage and Archaeology	0.94%	426	45,143
Engineering	0.91%	7,093	777,902
Health Sciences	0.90%	2,584	286,157
Mathematical Sciences	0.63%	779	122,704
Language, Communication and Culture	0.48%	623	129,786
Commerce, Management, Tourism and Services	0.42%	1,095	261,649
Education	0.38%	653	171,108
Creative Arts and Writing	0.30%	163	54,209
Law and Legal Studies	0.28%	234	83,687
Philosophy and Religious Studies	0.18%	177	96,265

)

Figure 15: Global Dataset citation rates 2024 (toolkit query 4.1.18)

Recommendations

Recommendation 2.2.8: Research Community Data citation literacy is increased in order to meet levels of peer countries. Whilst Australia has data citation rates that compare well to the global average, many European countries as well as Great Britain have consistently higher engagement rates across most fields of research.

Recommendation 2.2.9: Australian research institutions work with their repository vendors to ensure that their workflows are 'ORCID' first.

field_of_research	CH	NL	DE	AT	DK	NO	GB	FR	SE	BE	US	CA	AU	ES	IT	KR	JP	BR	CN	IN
Agricultural, Veterinary and Food Sciences	9	7	6	5	6	5	6	8	5	5	4	6	4	3	3	1	2	1	1	0
Biological Sciences	17	14	13	13	13	17	15	13	15	12	11	13	12	10	7	2	5	4	3	1
Biomedical and Clinical Sciences	3	3	3	2	2	2	3	2	2	2	2	2	2	2	1	1	1	1	1	0
Built Environment and Design	4	4	4	6	4	4	3	3	2	4	3	3	2	4	2	1	1	2	1	1
Chemical Sciences	8	5	5	5	6	7	8	4	5	4	3	2	2	3	2	1	1	1	0	0
Commerce, Management, Tourism and Services	2	2	3	3	2	2	1	1	1	2	1	1	1	1	1	1	0	1	1	0
Creative Arts and Writing	3	3	2	2	2	1	1	2	1	2	1	2	1	1	1	2	1	0	1	0
Earth Sciences	33	30	30	24	30	31	26	27	24	27	26	21	18	19	18	12	12	9	8	6
Economics	6	8	8	11	6	8	5	4	9	4	5	5	4	4	4	3	3	2	2	1
Education	3	2	3	1	1	1	2	1	1	2	1	1	1	1	1	2	1	0	1	1
Engineering	7	5	4	4	4	4	4	3	3	3	3	2	2	3	2	1	1	1	0	0
Environmental Sciences	20	15	15	18	14	15	14	15	13	12	14	12	11	10	8	3	5	5	3	2
Health Sciences	2	3	3	2	1	2	3	2	2	2	1	2	2	2	2	1	1	2	1	1
History, Heritage and Archaeology	7	6	7	7	9	4	5	6	9	6	4	3	6	3	3	6	1	1	2	0
Human Society	7	6	8	7	5	4	4	4	4	4	4	3	2	3	4	3	2	1	2	1
Information and Computing Sciences	8	7	7	6	5	5	4	5	4	5	4	3	3	5	4	2	2	3	1	2
Language, Communication and Culture	5	4	4	5	2	2	2	3	3	1	1	1	1	1	2	2	1	0	1	0
Law and Legal Studies	2	2	2	1	3	3	2	4	3	2	3	2	1	1	1	1	1	0	1	1
Mathematical Sciences	2	3	2	2	3	2	2	1	1	1	1	1	2	1	1	1	1	1	0	0
Philosophy and Religious Studies	0	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	0	0	1	0
Physical Sciences	10	12	8	9	12	8	10	8	9	8	9	9	11	7	6	3	4	3	2	2
Psychology	6	5	5	4	4	4	4	5	3	3	3	4	2	3	3	2	3	2	1	1

Figure 16: Data citation percentage by country 2024 (toolkit query 4.1.19)

1 3 10

field of research	DE	NL	CH	GB	FR	US	SE	BE	CA	IT	ES	AU	PT	PL	KR	ZA	CN	BR	JP	MY	IN	TR	SA	PK	ID	IR	UA	RU
Agricultural, Veterinary and Food Sciences	5	6	8	6	7	4	5	5	5	2	3	4	2	2	1	2	1	1	1	1	0	0	0	0	0	0	0	0
Commerce, Management, Tourism and Services	6	6	4	3	2	4	3	4	3	3	3	2	3	2	2	2	2	3	1	1	1	1	2	1	1	1	1	0
Language, Communication and Culture	8	9		3	6	3	6	2	2	4	3	2		1	3	1	2	1	3	1	0	1		1	1		0	0
Education	7	6		5		3	3		3	4	3	2	2	2	4	2	2	1	2	1	1	1	1	1	1	2	0	1
Information and Computing Sciences	5	4	5	3	3	3	3	3	2	3	3	2	3	2	1	2	1	2	1	1	1	1	2	2	1	1	0	1
Law and Legal Studies	9	6		5		10			7	4	5	2		0			4	2			2				1		1	0
Philosophy and Religious Studies	4	6		6	5	3			2	5	3	4		2		1	4	3	2	1	2	1			1		1	1
Human Society	6	5	5	3	3	3	3	3	2	3	3	1	2	1	2	1	1	1	1	0	1	1	1	1	0	1	0	0
Engineering	4	6	7	4	3	3	3	3	2	3	3	2	2	2	1	1	1	1	1	0	1	1	0	0	0	0	1	0
Chemical Sciences	4	5	7	7	4	2	4	4	2	2	2	2	1	2	1	1	0	1	1	0	0	0	0	0	0	0	1	1
Built Environment and Design	4	4	4	3	3	3	2	5	3	3	4	2	2	1	1		1	2	1	1	1	1	0		0	1	0	0
Environmental Sciences	4	4	5	3	3	3	3	3	3	2	2	3	2	1	1	2	1	1	1	1	0	0	0	0	0	0	0	0
History, Heritage and Archaeology	7			5	6	4			3	3	4	6		2			2	1				1			1		1	1
Physical Sciences	3	4	3	3	2	3	3	3	3	2	2	3	1	1	1	2	1	1	1	0	1	1	0	0	0	0	1	0
Mathematical Sciences	4	4	4	3	2	2	2	2	2	1	2	2	1	1	1	1	1	1	1	3	0	1	0	1	1	0	1	0
Economics	4	5	3	3	2	3	5		3	2	2	2		1	1	1	1	1	1	0	0	1	1	0	0		0	0
Biological Sciences	3	3	4	3	3	2	3	3	3	2	2	3	2	1	1	2	1	1	1	1	0	0	0	0	0	0	1	0
Earth Sciences	3	3	4	3	3	3	3	3	2	2	2	2	1	1	1	1	1	1	1	0	1	0	1	0	0	0	0	0
Health Sciences	3	3	2	3	2	2	2	3	3	2	2	2	2	2	1	2	1	2	1	1	1	0	1	1	1	0	0	0
Biomedical and Clinical Sciences	3	3	3	3	2	2	2	2	2	1	2	2	1	1	1	2	1	1	1	1	0	0	1	0	1	0	0	0
Psychology	3	2	3	2	2	2	2	2	2	2	2	1	1	2	1	1	1	1	1	0	1	0			0	0	0	0
Creative Arts and Writing	7			4		4				5	3	3		2			2	0			0	1			1		0	0

Figure 17: Data citation engagement by country 2024 as a ratio of the percentage of global papers with data set citations by country over the global percentage of papers for that country (toolkit query 4.1.19)

2.2.6 Samples

Research Input 6.

Data, samples and observations may be the product output of research, or sometimes the result of other focused activity, but they are also foundational inputs to subsequent research. Samples refer to physical materials collected or prepared for study, such as biological specimens, geological cores, chemical compounds, or archaeological artefacts. They often require specialised handling, storage, and cataloguing to maintain integrity and reproducibility. Observations are recorded measurements, qualitative assessments, or sensory detections made during research. These may include field notes, sensor readings, astronomical records, or behavioural assessments. Observations can be qualitative (e.g., descriptive field notes) or quantitative (e.g., numerical data collected via instruments).

Strategy Measurement Motivation
Samples are physical materials (biological, geological, chemical, archaeological) or observations (field notes, sensor data, astronomical records) that are essential to reproducibility. Increasingly registered in IGSN catalogues.
International Engagement/Dependencies
Global IGSN infrastructure (DataCite + disciplinary registries).

Expected Maturity	Early adoption — strong uptake in Earth Sciences, low elsewhere.
Senior Stakeholder	

Benchmarks

Findability	percentage of samples registered in IGSN catalogues; AU share of global registrations.
Accessibility	Number of AU repositories providing open IGSN metadata.
Interoperability	% of AU IGSNs with links to ORCID, ROR, or datasets.
Reuse	# of publications citing IGSNs (prefix IGSN:XXX).

Why it Matters

Samples underpin reproducibility and provenance. Without persistent identifiers, samples risk being disconnected from publications and datasets.

Analysis

Samples and observations can be registered in DataCite as a physical object deposited in an International Generic Sample Number (IGSN) Catalogue.

Samples with a registered IGSN number have not been universally adopted by the research community, however they have been adopted by the Earth Sciences and Archaeology communities. In total, as at July 2025, 11.6 million IGSN's have been registered globally.

Over the last five years, IGSN samples registered in Australian IGSN catalogues represent 80% of all IGSNs registered globally, however Australian registration of samples has occurred in batches . In 2024 the percentage of samples registered in Australian repositories was 6% (table 18).

Interoperability

Despite there being over 5 Million IGSN records registered within Australia, Australian IGSN samples are only loosely connected to the rest of the persistent identifier graph. There are no Australian IGSN records with links to ORCID or ROR (figure 19), and almost no IGSN records have any links to any other objects. Although most IGSN records follow the same pattern globally, there are a significant number of IGSN records that demonstrate how IGSN samples can be integrated into the rest of the pid graph (figure 20) . These more connected records follow new community expectations on how samples should be referenced and integrated into research [6].

Reuse

year	all samples	australian samples	global percentage
2024	282,605	17,557	6%
2023	1,454,008	1,256,212	86%
2022	232,107	74,369	32%
2021	1,169,861	1,013,148	87%
2020	1,719,387	1,524,449	89%

Figure 18: IGSN Samples by publication year (toolkit query 4.1.21)

location	samples	creators	name identifiers	affiliation identifiers
Australia	5,814,951	5,814,951	0	0
Rest of World	1,787,055	1,936,802	424,663	18,988

Figure 19: Comparison of affiliation and person linkages between Australian IGSN records and the rest of the world (toolkit 4.1.22)

IGSN identifiers can be referenced in papers using the prefix IGSN:XXX. Using a full-text search search in Dimensions, and removing other terms that abbreviate to IGSN as well as articles in Biomedical and Clinical science, we can see that approximately 1,168 have mentioned IGSN's since 2016.

https://app.dimensions.ai/discover/publication?search_mode=content&search_text=igsn%20AND%20NOT%20(%22Image-guided%20surgical%20navigation%22)%20AND%20NOT%20(%22%20integrated%20gene%20similarity%20network%22)%20AND%20NOT%20(%22Service%20Node%22)&search_type=kws&search_field=full_search&not_facet_for=80003&viz-st:aggr=mean&or_facet_year=2025&or_facet_year=2024&or_facet_year=2023&or_facet_year=2022&or_facet_year=2021&or_facet_year=2020&or_facet_year=2016&or_facet_year=2017&or_facet_year=2018&or_facet_year=2019

Australian authors are represented on these publications 7% of the time (50 publications) compared to 3% of the time for Environmental Sciences more generally, indicating that the publication practice of referring to IGSNs has begun, although most samples are yet to be referred to in publications. This engagement rate is roughly inline for the data citation engagement rates observed in the section above.

Recommendations

Recommendation 2.2.10: The Australian Research Community determine new minimum standards for IGSN records.. IGSN records should be more firmly integrated to other persistent identifiers.

location	direction	relationType	resourceTypeGeneral	links
Australia	inbound	IsPartOf	PhysicalObject	166
Australia	inbound	HasPart	PhysicalObject	170
Australia	outbound	HasPart	PhysicalObject	166
Australia	outbound	IsPartOf	PhysicalObject	170
Rest of world	inbound	IsVariantFormOf	PhysicalObject	91
Rest of world	inbound	IsDerivedFrom	Dataset	234
Rest of world	inbound	HasPart	PhysicalObject	18
Rest of world	inbound	Cites	Dataset	1
Rest of world	inbound	References	Dataset	4,501
Rest of world	inbound	IsMetadataFor	Other	186
Rest of world	inbound	IsDerivedFrom	Collection	1
Rest of world	inbound	References	ConferencePaper	2
Rest of world	inbound	IsPartOf	PhysicalObject	55,823
Rest of world	outbound	HasPart	PhysicalObject	18
Rest of world	outbound	IsDescribedBy	Text	8
Rest of world	outbound	IsDescribedBy	Report	60
Rest of world	outbound	IsCitedBy	Other	41,340
Rest of world	outbound	IsVariantFormOf	PhysicalObject	91
Rest of world	outbound	IsPartOf	PhysicalObject	55,823

Figure 20: Comparison of object relationships between Australian IGSN records and the rest of the world (toolkit 4.1.23) Datasets that reference samples are an emerging way that a large number of samples can be referenced at once.

2.2.7 Instruments

Research Input 7.

Strategy Measurement Motivation
Research instruments (from microscopes to HPC systems) are critical infrastructure. Assigning PIDs and tracking their use enables credit for facilities, links instruments to datasets and publications, and improves reproducibility.
International Engagement/Dependencies
DataCite support for instrument registration (since 2020); DOI resolution/landing-page infrastructure.

Expected Maturity	Early Adoption
Senior Stakeholder	

Benchmarks

Findability	Percentage and count of instruments registered with a PID (DataCite)
Accessibility	
Interoperability	Presence of links from instrument records to organisations (ROR), people (ORCID), and related research objects (datasets, projects).
Reuse	Evidence of instrument citation in publications (counts and where they appear in the paper).

Why it Matters

Linking instruments into the persistent identifier types enables measurable credit for facilities, clearer provenance from methods → instruments → data → publications, and straightforward tracking of instrument-enabled outputs. The recommendations in the report explicitly call for strengthening those links and normalising instrument citation.

Analysis

Findability Instruments have been formally categorised in DataCite since 2020 [9]. Since then it has been possible to find Instruments within DataCite via their search interface, although registration numbers have been low. By mapping prior resourceType descriptions to the new resourceTypeGeneral of Instrument, it is possible to find 484 Instruments that have been registered in DataCite including those registered before the change to schema in 2020, and 191 with a resourceTypeGeneral equal to 'Instrument'. (A small nuber of these records refer to survey instruments withing the social sciences.)

Of the larger cohort, Australia has registered 69 instruments (16%) (figure 21). As instruments are registered rather than published, it did not make sense to restrict the reported figures to the publication year window of 2020 to 2024

Interoperability

Of the 60 Instruments registered at the University of Western Australia, 52 records had links back to the institution via a ROR id, along with all 7 records from the Pawsey Super Computing Centre (toolkit query 4.1.26). Links between datasets and instruments were not present in the DataCite records (toolkit query 4.1.27).

Reuse

Although only a small number of instruments have been registered, it is still possible to detect instrument citations. Globally, there have been 27 citations to instruments using dataCite DOIs, with just under half (13) made to instruments registered in Australian repositories. As there are only 13 citations, it is worth displaying them in full, along with the document section that they were found

[9] https://DataCite.org/blog/pids-for-instruments-a-way-forward/

repository name	repository id	institution	country code	Instruments
The University of Western Australia	uniwa.repo	The University of Western Australia	AU	60
Pawsey Supercomputing Centre	pawsey.repo		AU	7
University of New South Wales	unsw.repo	UNSW Sydney	AU	2
The University of Queensland	uq.repo	The University of Queensland	AU	1
Curtin University Research Data Collection	ardcx.curtin	Curtin University	AU	1
Open Collections	ubc.oc	University of British Columbia	CA	1
TUHH Open Research	tib.tuhh		DE	258
Helmholtz-Zentrum Berlin für Materialien und Energie GmbH	tib.hzb	Helmholtz-Zentrum Berlin für Materialien und Energie	DE	24
HZDR-DEVICES	pqip.devices	Helmholtz Association of German Research Centres	DE	3
Research Data Center for Higher Education Research and Science Studies (FDZ-DZHW)	dzhw.fdz-dzhw	German Centre for Higher Education Research and Science Studies	DE	2
DTU EIS	todn.hcsvci	Technical University of Denmark	DK	33
2DCC - DMR-FIRST	psu.dmr-first	Pennsylvania State University	US	61
Open Science Framework	cos.osf	Center for Open Science	US	16
DOE Pacific Northwest National Laboratory (PNNL) Repository	doe.pnnl		US	4
DOE Office of Scientific and Technical Information (OSTI) Repository	doe.osti		US	4
Univ. of Pennsylvania Repository	upenn.repo	University of Pennsylvania	US	2
Smithsonian Research Infrastructure	si.ccrcn	Smithsonian Institution	US	2
Gateway to Global Aging Data	usc.gateway		US	1
Indiana University Bloomington Libraries	iu.bl	Indiana University	US	1
DOE Oak Ridge National Laboratory (ORNL) Repository	doe.ornl		US	1

Figure 21: Instruments by repository (toolkit query 4.1.25)

in. For Australian instrument citations, the practice so far has been to reference instrumentation in the acknowledgement section, often along with funding acknowledgements (figure 22). Interestingly, citations to instruments outside of Australia have occurred in the body of the article within the description of the methodology (**??**. Both citation practices are valid and will make sense for different types of instrument. What is encouraging from a benchmarking perspective is that both approaches can also lead to measurable increases in instrument citation.

Recommendations

Recommendation 2.2.11: Encourage links between instruments and the rest of the pid graph. As data and outputs are created from Instruments, they should include a link to the instrument as standard

Recommendation 2.2.12: Reinforce instrument citation examples as best practice. Following the best practice that has already been established, use of instruments in research should always cite their persistent identifier

name	repository id	location	text	AutralianInstrument
Pawsey Supercomputing Centre	pawsey.repo	references_titles	10.48569/18sb-8s43	True
Pawsey Supercomputing Centre	pawsey.repo	acknowledgement_senteces	This work was supported by resources provided by the Pawsey Supercomputing Research Centre's Setonix Supercomputer (https://doi.org/10.48569/18sb-8s43),	True
Pawsey Supercomputing Centre	pawsey.repo	acknowledgement_senteces	The simulations suite QED and the mock data discussed in this paper were generated with the assistance of resources from the National Computational Infrastructure (NCI Australia), an NCRIS enabled capability supported by the Australian Government, and from the Pawsey Supercomputing Research Centre's Setonix Supercomputer (https://doi.org/10.48569/18sb-8s43),	True
Pawsey Supercomputing Centre	pawsey.repo	funding_sentences	The high-performance computing aspect of this work was supported by the Adaptor grants scheme, with computational resources provided by NCI Australia, an NCRIS-enabled capability supported by the Australian Government, and by the Pawsey Supercomputing Research Centre's Setonix Supercomputer (https://doi.org/10.48569/18sb-8s43),	True
The University of Queensland	uq.repo	acknowledgement_senteces	The Monte Carlo simulation was conducted using resources provided by The University of Queensland Research Computing Centre's Bunya supercomputer (https://doi.org/10.48610/ wf6c-qy55), with funding from The University of Queensland, Brisbane, Australia.	True
Pawsey Supercomputing Centre	pawsey.repo	acknowledgement_senteces	This work was supported by resources provided by the Pawsey Supercomputing Research Centre's Setonix Supercomputer (https://doi.org/10.48569/18sb-8s43),	True
Pawsey Supercomputing Centre	pawsey.repo	acknowledgement_senteces	This work was supported by resources provided by the Pawsey Supercomputing Research Centre's Setonix Supercomputer (https://doi.org/10.48569/18sb-8s43)	True
Pawsey Supercomputing Centre	pawsey.repo	acknowledgement_senteces	org/ 10. 48569/ 18sb-8s43), with funding from the Australian Government and the Government of Western Australia.	True
Pawsey Supercomputing Centre	pawsey.repo	acknowledgement_senteces	This work was supported by resources provided by the Pawsey Supercomputing Research Centre's Setonix Supercomputer (https://doi.org/10.48569/18sb-8s43).	True
Pawsey Supercomputing Centre	pawsey.repo	body_sentences	This work was supported by resources provided by the Pawsey Supercomputing Research Centre's Setonix Supercomputer (https:// doi.org/ 10.48569/18sb-8s43), with funding from the Australian Go v ernment and the Go v ernment of Western Australia.	True
Pawsey Supercomputing Centre	pawsey.repo	acknowledgement_senteces	This work was supported by resources provided by the Pawsey Supercomputing Research Centre's Setonix Supercomputer (https://doi.org/10.48569/18sb-8s43),	True
Pawsey Supercomputing Centre	pawsey.repo	acknowledgement_senteces	This research was supported by the Australian Government's National Collaborative Research Infrastructure Strategy (NCRIS), with access to computational resources provided by the Setonix supercomputer (https://doi.org/10.48569/18sb-8s43)	True
Pawsey Supercomputing Centre	pawsey.repo	acknowledgement_senteces	and Nimbus research cloud (https://doi.org/10.48569/v0j3-qd51),	True

Figure 22: Australian instrument citations (toolkit query 4.1.28)

name	repository id	location	text	AutralianInstrument
2DCC - DMR-FIRST	psu.dmr-first	body_sentences	The MoS 2 and WS 2 films were grown by MOCVD using a multimodule system (doi:10.60551/a0g5-8n60)	False
2DCC - DMR-FIRST	psu.dmr-first	body_sentences	The growth of WSe 2 thin films on c-plane sapphire substrates was carried out in a metal-organic chemical vapor deposition (MOCVD) system (https://doi.org/10.60551/znh3-mj13)	False
2DCC - DMR-FIRST	psu.dmr-first	body_sentences	The growth of a multilayer WSe 2 on two-inch-diameter c-plane sapphire was carried out in an MOCVD system (https://doi.org/10.60551/ znh3-mj13) in the 2D Crystal Consortium Materials Innovation Platform (2DCC-MIP) facility at Penn State.	False
2DCC - DMR-FIRST	psu.dmr-first	body_sentences	Room temperature Raman and photoluminescence were carried out using a Witec Apyron confocal microscope (doi:10.60551/a534-xs21)	False
Helmholtz-Zentrum Berlin für Materialien und Energie GmbH	tib.hzb	body_sentences	relating the HZB instrument "E2 -Flat-Cone Diffractometer" (https://doi.org/10.5442/NI000001)	False
Helmholtz-Zentrum Berlin für Materialien und Energie GmbH	tib.hzb	body_sentences	Listing 2: Snippet of DataCite metadata (https://api.datacite.org/dois/10.5442/NI000001)	False
Leibniz-Institut fuer Ostseeforschung Warnemuende	tib.iow	availability_sentences	The data used to reproduce the results presented here are archived at http://doi.io-warnemuende.de/10.12754/data-2024-0014	False
Helmholtz-Zentrum Berlin für Materialien und Energie GmbH	tib.hzb	body_sentences	Listing 2: Snippet of DataCite metadata (https://api.datacite.org/dois/10.5442/NI000001	False
Open Science Framework	cos.osf	annex_sentences	The Open Science Framework (OSF) repository (https://doi.org/10.17605/OSF.IO/WQEM8) includes the instruments in Word format for download in both Spanish and English, as well as a report on the design process of each of the three rubrics used in this study.	False
Helmholtz-Zentrum Berlin für Materialien und Energie GmbH	tib.hzb	body_sentences	) relating the HZB instrument "E2 -Flat-Cone Diffractometer" (https://doi.org/10.5442/NI000001) with a journal article describing the instrument (https://doi.org/10.17815/jlsrf-4-110	False
2DCC - DMR-FIRST	psu.dmr-first	body_sentences	The growth of MoS 2 film on 2" a-plane sapphire substrates was carried out in a metal-organic chemical vapor deposition (MOCVD) system (DOI: 10.60551/znh3-mj13) equipped with a cold-wall horizontal reactor with an inductively heated graphite susceptor with gas-foil wafer rotation.	False
2DCC - DMR-FIRST	psu.dmr-first	body_sentences	The growth of monolayer WS2 on 2" diameter c-plane sapphire was carried out in a metalorganic chemical vapor deposition (MOCVD) system equipped with a cold-wall horizontal reactor with an inductively heated graphite susceptor with gas-foil wafer rotation (DOI: 10.60551/znh3-mj13).	False
Open Science Framework	cos.osf	body_sentences	The interview questionnaire is available online in the Open Science Framework repository at DOI: 10.17605/OSF.IO/9EC6V. 16	False
2DCC - DMR-FIRST	psu.dmr-first	body_sentences	An inductively heated graphite susceptor equipped with wafer rotation in a cold-wall horizontal reactor (https://doi.org/10.60551/ znh3-mj13) was used to achieve uniform monolayer deposition.	False

Figure 23: International instrument citations (toolkit query4.1.28)

2.3 Benchmarks for Research Outputs.

> Outputs from research include data, reports, methods, non-traditional research outputs and publications. By increasing the discoverability of these outputs, we can improve future research, create new kinds of research, reduce duplication and improve the timeliness of translating research into impact.
>
> **Objective 2. Australian National PID Strategy**

In this section, we turn our attention to the results, the tangible results of research activity. Outputs such as data sets, publications, methods, and reports are essential not only to disseminate knowledge but also to enable its reuse and extension. Research outputs are not only entities to be described and identified, but they also carry with them the expectation that it is possible to describe how they were created (their provenance). In doing so, they provide focal points for the integration of persistent identifier infrastructure, making them essential components of any PID benchmarking approach.

Using the same format as the research inputs section above, for each research output we have identified:

- **Motivation for Measurement**
- **Expected Maturity**
- **International Dependencies**
- **Benchmarks through the lens of Findability, Accessibility, Interoperability, and Reuse**
- **An overall perspective of why the benchmarks for the research output in focus matter**

Each section also includes an analysis of the benchmarks that are assessable today, along with associated recommendations.

Research Outputs

1 Summary of research report strategic value and benchmarks 40

2 Summary of research methods strategic value and benchmarks 44

3 Summary of NTRO benchmarks 47

4 Dissertation benchmarks 49

5 Research Publication Benchmarking Approach 54

2.3.1 Research Reports

Research Output 1.

Research reports are structured documents that formally communicate the progress, findings, or outcomes of a research project. They serve as a key mechanism for documenting and disseminating research activities, often fulfilling compliance, funding, or institutional requirements. Reports may be internal (e.g., project updates, technical reports) or external (e.g., public reports, government submissions, white papers). Research reports are typically generated within research projects, often as required deliverables for contracts. They may summarise findings derived from research data, samples, or observations and may acknowledge the use of facilities where relevant. Metadata associated with research reports include authorship, affiliated institutions, funding sources, and publication date. Reports may be stored in institutional repositories, funder archives, or open-access platforms.

Strategy Measurement Motivation
Research reports can be created by a multitude of external bodies. Reports share many of the same properties as publications in the they need to be easily citable, and cite other literature
International Engagement/Dependencies

Expected Maturity	Established
Senior Stakeholder	

Benchmarks

Findability	Number of reports that can be found across national repository infrastructure
Accessibility	(similar to Findability - reports that are not not hosted within repositories may not be hosted permanently)
Interoperability	integration of report records with other persistent identifiers (including ROR and ORCID)
Reuse	Reuse of policy persistent identifiers in the research and policy literature

Why it Matters

Reports are often produced in response to a specifically funded piece of work by government or industry, where a research publication is not appropriate due to the confidentiality or type of material. Due to this, research reports can often demonstrate engagement and impact, particularly for the funder. Citing of reports in policy and other document further demonstrates research impact.

Analysis

Findability

Reports within Australian repositories represent just over 6% of global output. The percentage rises to 10% if reports in ResearchGate and Zenodo are not included. Notably Australian institutions host many more reports in their institutional repositories than other nations, with 2 repositories in the top 20 by number of reports published between 2020 and 2024 (figure 24)

Interoperability As with many object types registered within Australian repositories, links to ORCID and affiliations ROR, are present but highly variable by repository. Given the effort that institutions have placed on appropriately depositing reports into repositories, this is an area for improvement.

Reuse As the audience for reports may not be academic, being able to trace references in broader discussions is important to determine the context in which a report has been received. Figure 25 uses Almetric as well as data citations to explore the volume of these additional connections.

Recommendation 2.3.1: Encourage the practice of publishing reports in institution repositories. By volume, Australia is a global leader in supporting the publication of reports, however the practice

repository name	repository id	institution	country code	Reports
ResearchGate	rg.rg		DE	40,489
Zenodo	cern.zenodo		CH	39,853
National Bureau of Economic Research	crossref	National Bureau of Economic Research	US	30,199
TIB Hannover	tib.tib		DE	21,820
Archaeology Data Service	bl.ads		GB	13,477
National Bureau of Standards	crossref	National Bureau of Standards	US	11,113
Reserve Bank of Australia	crossref	Reserve Bank of Australia	AU	8,631
National Institute of Standards and Technology	crossref	National Institute of Standards and Technology	US	8,325
International Council for the Exploration of the Sea (ICES)	dk.ices	International Council for the Exploration of the Sea	DK	7,028
Defense Technical Information Center	crossref			4,193
Office of Scientific and Technical Information (OSTI)	crossref	Sandia National Laboratories	US	2,546
Office of Scientific and Technical Information (OSTI)	crossref	National Energy Technology Laboratory	US	2,064
AARP Research	crossref	AARP	US	1,864
Washington State University	wsu.wsu		US	1,438
Office of Scientific and Technical Information (OSTI)	crossref	Idaho National Laboratory	US	1,404
Asian Development Bank	crossref	Asian Development Bank	PH	1,387
KITopen	tib.kit		DE	1,367
Federal Reserve Bank of St. Louis	crossref	Federal Reserve Bank of St. Louis	US	1,278
RMIT	figshare.rmit	RMIT University	AU	1,095
Robert-Koch-Institut (RKI)	tib.hu-rki		DE	1,039

Figure 24: Top repositories with reports published between 2020 and 2024. Both reports placed in Crossref and DataCite are considered. 2 Australian repositories feature in the top 20, with the first of them being the Reserve Bank of Australia historical archives (Unreserved) . (toolkit query 4.2.1).

is not evenly distributed across all institutions

Recommendation 2.3.2: Continue to increase the practice of linking reports to people and affiliations via ORCID and ROR. As a publisher of reports, institutions adopt the best practices of publishers to ensure all authors are described by ORCIDs and affiliations via ROR. This may require additional coordination with repository vendors to ensure workflows can be supported.

repository id	repository name	reports	citations	policy	wikipedia	twitter	bluesky
crossref	Reserve Bank of Australia	8,631		0	92	0	0
figshare.rmit	RMIT	1,095	2				
latrobe.repo	La Trobe University	1,007	20	1	0	0	0
unsw.repo	University of New South Wales	862	47	0	3	0	0
swin.repo2	Swinburne University of Technology	849	8	0	0	1,496	0
monash.repo	Monash University	657	15	27	0	1,993	3
ardcx.wsu	Western Sydney University ResearchDirect	223	19	1	16	101	1
smf.repo	Analysis & Policy Observatory	205	55	13	4	1,177	0
unimelb.repo1	University of Melbourne data repository	150	8	0	0	0	0
ardcx.cqu	Central Queensland University	139	1				
crossref	The University of Queensland	113		0	0	3	0
ardcx.griffith	Griffith University Research Data Repository	90	2				
crossref	Australian Council for Educational Research	81		0	0	133	0
ardcx.usyd	University of Sydney Library	63	4	0	0	8	0
crossref	Queensland University of Technology	61		1	5	131	0
ardcx.vu	Victoria University	59	4				
figshare.mcri	Murdoch Childrens Research Institute (MCRI)	56	4	3	0	0	0
ardcx.scu	Southern Cross University	55	0				
ardcx.aodn	Australian Ocean Data Network	46	6				
crossref	VicHealth	42		0	1	0	0

Additional 35 rows not shown.

Figure 25: Australian repositories citations and attention measures for reports published between 2020 and 2024 (4.2.3)

repository id	all reports	with ORCID	with ROR	percent ORCID	percent ROR
figshare.rmit	1,095	333	912	30	83
latrobe.repo	979	85	310	9	32
unsw.repo	828	770	0	93	0
swin.repo2	828	53	495	6	60
monash.repo	631	241	187	38	30
ardcx.wsu	196	123	172	63	88
smf.repo	162	1	5	1	3
unimelb.repo1	146	61	106	42	73
ardcx.cqu	139	20	118	14	85

Figure 26: Australian repositories with reports - ORCID and ROR usage (4.2.2)

2.3.2 Research Methods

Research Output 2.

Research methods define the structured approaches, techniques, and procedures used to conduct scientific investigations. They provide the framework for collecting, analysing, and interpreting research data, samples, and observations. Methods may be qualitative (e.g. interviews, case studies), quantitative (e.g., statistical analysis, computational modelling), or experimental (e.g., laboratory protocols, fieldwork techniques). Research methods are fundamental to research projects, guiding how investigations are carried out. They are often influenced by the requirements of grants, which may specify methodological approaches to ensure rigour and reproducibility. Some methods require access to specialised facilities, such as laboratories, computing clusters, or field stations. Metadata associated with research methods include protocols, instrumentation, software tools, standard operating procedures (SOP), and methodological references. Research outputs such as research reports and publications frequently describe the methods used to ensure transparency and reproducibility.

Strategy Measurement Motivation
(more domain expertise required)
International Engagement/Dependencies
International publishers and protocol repositories dominate current practice

Expected Maturity	Early Adoption
Senior Stakeholder	

Benchmarks

Findability	The number of methods/protocols with persistent identifiers
Accessibility	
Interoperability	
Reuse	Citations of protocols in publications.

Why it Matters

Separately registered methods and protocols greatly assists efforts to make research more reproducible

Analysis

Currently, there are a variety of approaches to uniquely identify research methods. One approach is to separately publish protocols and methods in dedicated journals. Globally, protocols published in methods and protocol journals have increased to just over 2000 articles per year since 2021 (figure 27), (figure 28). Approximately 20,000 protocols are also registered as preprints in Crossref as part of protocols.io. Just over 68,000 works in DataCite were published with the word "protocol" in their title, although there is little agreement as to how to describe them (figure 29). Without a firmer common understanding on how protocols and methods (even if there are multiple definitions,) it is difficult to proceed with further benchmarks.

Recommendation 2.3.3: Guidelines are created for the recording of protocols. Without clear guidelines on how to register protocols and methods, benchmarking will be haphazard.

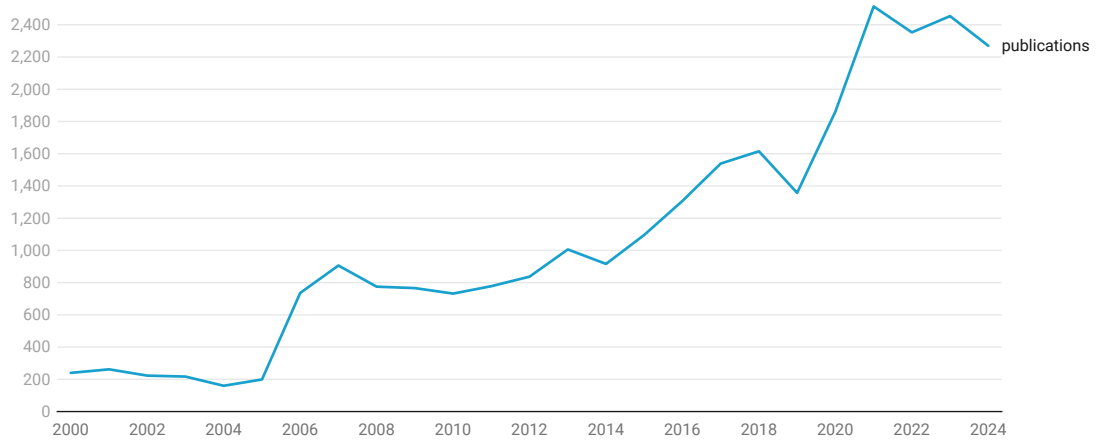


Figure 27: Articles published in Protocol journals by year (toolkit query 4.2.4)

journal	name	protocols articles
STAR Protocols	Elsevier	3,428
JMIR Research Protocols	JMIR Publications	2,736
Bio-protocol	Bio-Protocol	1,713
Current Protocols	Wiley	1,044
Cold Spring Harbor Protocols	Cold Spring Harbor Laboratory	559
Methods and Protocols	MDPI	459
Nature Protocols	Springer Nature	342
Medicine Case Reports and Study Protocols	Wolters Kluwer	274
Biology Methods and Protocols	Oxford University Press (OUP)	194
International Journal of Internet Protocol Technology	Inderscience Publishers	143
International Journal of Surgery Protocols	Wolters Kluwer	92
Procedimientos y Protocolos con Competencias Específicas para Enfermería Nefrológica	Sociedad Espanola de Enfermeria Nefrologica	60
Journal of Surgical Protocols and Research Methodologies	Oxford University Press (OUP)	46
Social Science Protocols	University of Edinburgh	41
Current Protocols in Microbiology	Wiley	37
Current Protocols in Immunology	Wiley	23
Current Protocols in Stem Cell Biology	Wiley	23
Current Protocols in Nucleic Acid Chemistry	Wiley	21
Current Protocols in Neuroscience	Wiley	21
Current Protocols in Molecular Biology	Wiley	19

Figure 28: Protocol journals (toolkit query 4.2.5)

repository	resourceTypeGeneral	resourceType	works	example doi
figshare Academic Research System	Text	Journal contribution	17,611	10.6084/m9.figshare.9759257.v3
figshare Academic Research System	Collection	Collection	8,246	10.6084/m9.figshare.c.7604458.v1
Open Science Framework	StudyRegistration	Pre-registration	6,272	10.17605/osf.io/zyxpd
Open Science Framework	Text	Project	3,713	10.17605/osf.io/zyj7r
Greater South Information System	Text		2,750	10.60692/zzxcj-gv394
Open Science Framework	Text	Pre-registration	2,531	10.17605/osf.io/zyvbj
Faculty of 1000 Research Ltd	Text		2,331	10.5256/f1000research.82451.r125654
figshare Academic Research System	Dataset	Dataset	1,875	10.6084/m9.figshare.8063105.v2
arXiv	Preprint	Article	1,279	10.48550/arxiv.2412.20472
Sage Journals Data	Text	Journal contribution	1,078	10.25384/sage.25975696.v1
figshare Academic Research System	Other	Online resource	827	10.6084/m9.figshare.5244571.v2
Zenodo	JournalArticle		707	10.5281/zenodo.8433552
Sage Journals Data	Collection	Collection	633	10.25384/sage.c.7270426.v1
Global Biodiversity Information Facility	Dataset	OCCURRENCE	543	10.15468/zzs6mk
arXiv	Text	Article	529	10.48550/arxiv.2412.14544
figshare Academic Research System	Image	Figure	514	10.6084/m9.figshare.27991940.v1
figshare Academic Research System	Text	Preprint	495	10.6084/m9.figshare.6199325.v4
La Trobe University	Text	Journal contribution	485	10.26181/7859945.v3
Zenodo	JournalArticle		472	10.5281/zenodo.8432648
Mendeley Data	Dataset	Dataset	420	10.17632/zvfnvxfyw6.1

Figure 29: DataCite works with ”protocol” in the title (4.2.6)

2.3.3 Non Traditional Research Outputs

Research Output 3.

Non-traditional research outputs (NTROs) are research-driven works that extend beyond conventional scholarly publications, such as journal articles and books. They include creative works, software, technical designs, performances, exhibitions, datasets, and industry reports. These outputs are often designed for diverse audiences, including industry, practitioners, educators, policymakers, and the public, rather than solely for academic communities. NTROs emerge from research projects and may be supported by grants that recognise the impact of research beyond traditional publishing. Many NTROs rely on specialised facilities, such as studios, laboratories, or computing environments, for their creation and dissemination. NTROs often incorporate or visualise research data, samples, and observations, transforming them into accessible formats. Metadata associated with NTROs includes contributors, medium, dissemination platforms, licensing, associated projects and grants, and audience engagement metrics.

Strategy Measurement Motivation
Non traditional research objects require specific metadata requirement, as well as tailored support
International Engagement/Dependencies

Expected Maturity	Evolving
Senior Stakeholder	

Benchmarks

Findability	Number of protocols and methods with a DOI
Accessibility	Number deposited in institutional/national repositories.
Interoperability	Presence of ORCID/ROR metadata. (not assessed)
Reuse	Citations and altmetrics of NTROs. (Assessed for specific NTRO in other sections)

Why it Matters

Clearly recording NTROs supports investigation of research impact and engagement across and increased number of disciplines, especially humanities, arts, and applied fields.

Analysis

Findability

Using the resource types within DataCite, 19 additional types or research output can be found (figure 30), whilst many NTRO data types have been specifically highlighted in this report, it is worth highlighting that different NTROs will require different reporting strategies. It has already been noted that the number of Software works in Australian repositories is low. However, in the same period, there are just under 2000 software works associated with researchers Australian institutions (By ORCID) within the generalist repository Zenodo. (toolkit query 4.2.8). Not all benchmarks can be answered with reference to National repositories; however, this is likely to remain the dominant mode on enquiry until creating records with ROR ids and ORCID records becomes the norm. For other query types, there will be other modes of query that are just as important. An international example is the Local Contexts initiative that supports participating indigenous communities with tools that can reassert cultural authority in heritage collections and data[10]. Within DataCite, works can be associated with Local Contexts labels and projects via that dataCite rights relationship, and the yearly dataCite 2024 snapshot contains 14 works that can be identified as having local contexts labels and projects **??**

[10] https://localcontextshub.org/

resourceTypeGeneral	Australian Works	Australian citations	all works	all citations	citation ratio	works ratio
Dataset	16,164	1,634	21,555,982	170,271	1	0
Report	5,633	207	155,150	6,725	3	4
DataPaper	367	189	6,262	2,480	8	6
Collection	799	91	292,477	9,278	1	0
Software	364	78	464,087	51,304	0	0
Other	5,854	59	534,328	13,894	0	1
Dissertation	57,299	48	330,225	1,286	4	17
InteractiveResource	1,946	47	99,777	1,198	4	2
Image	2,606	23	1,459,045	3,567	1	0
Instrument	16	12	415	21	57	4
ConferencePaper	5,959	12	340,156	9,964	0	2
Audiovisual	2,742	11	196,901	1,503	1	1
Service	16	5	176	423	1	9
PhysicalObject	901	3	1,136,593	104	3	0
ConferenceProceeding	10	3	13,421	223	1	0
Workflow	37	3	5,657	548	1	1
Event	498	2	23,091	51	4	2
Model	779	2	15,393	823	0	5
StudyRegistration	6	1	112,621	2,583	0	0
OutputManagementPlan	14	1	5,693	78	1	0

Additional 3 rows not shown.

Figure 30: Non Traditional Research Outputs in DataCite (toolkit query 4.2.7)

2.3.4 Dissertations

Research Output 4.

Strategy Measurement Motivation
International Engagement/Dependencies

Expected Maturity	Early Adoption
Senior Stakeholder	

Benchmarks

Findability	
Accessibility	*Accessibility*: % openly available via institutional repositories.
Interoperability	*Interoperability*: % with ORCID/ROR metadata.
Reuse	*Reuse*: Citations of dissertations in scholarly literature.

Why it Matters

Dissertations are critical to mapping researcher career pathways. ORCID/ROR links ensure visibility and attribution.

Analysis

14% of all dissertations published between 2020 and 2024 with DOIS from DataCite or Crossref are from Australian institutions, making it the third largest publisher of dissertations with DOIS behind only the United States and Brazil (Figure 31). Four Australian repositories rank in the top 20 repositories (figure 32). With the exception of the University of Queensland and the Queensland University of Technology, dissertations have been registered in DataCite, rather than Crossref (figure 33). DOIs are not the only identifiers for Thesis used. The Global Open Access Thesis and Dissertation aggregator service oatd.org, identifies many Australian Institutions that offer openly discoverable thesis records identified by handles (https://oatd.org/oatd/search?q=cc_s:au&pubdate.facet=2020), however there is a significant national trend towards the use of DOIs to describe thesis within institutional repositories.

Findability

Interoperability

With so many dissertations published in Australian repositories, if properly recorded with ORCID identifiers and institutional affiliations, the aggregate dataset has the potential to be invaluable. These records are particularly important when it comes to identifying the progress of early-career researchers. Without an ORCID, and importantly, work associated with it, early-career researchers are poorly served by author disambiguation algorithms as there is not enough information in the system yet to make confident assertions. Ensuring that graduates who leave academia to become part of industry can identify themselves in public reports opens up further possibilities for tracking commercial engagement.

To date however, Australian repositories still mostly record dissertations without links to ORCID identifiers and institutional affiliations, leaving the role that these records could play unrealised. The University of Queensland and University of New South Wales both have relatively high percentages of dissertations with ORCID assertions (25%), indicating that systematic approaches to increasing the number of dissertations with DOIs and ORCIDS is possible.

Just as there is a community expectation on publishers that they should do their upmost to work with their submission system vendors to ensure that most authors supply their ORCID, there is perhaps a similar obligation on institutions to work with repository vendors to ensure that the ORCIDs for dissertations also become the norm.

country_code	Dissertations
BR	128,058
US	123,360
AU	75,210
DE	40,918
GB	37,935
NL	16,315
CO	13,800
AT	12,134
ES	11,639
CH	11,425

Figure 31: Dissertations with a DOI by Country 2020-2024 top 10 (toolkit query 4.2.10)

Going beyond the information that is already possible to record in DataCite and Crossref, there is also an opportunity to record the supervisors of the thesis, allowing the impact of mentorship to more accurately be measured [19]. This information is frequently recorded on the source record, but not easily recorded in either DataCite of Crossref. Establishing the preferred way of recording supervisor relationships will require community consultation.

Recommendation 2.3.4: Australia considers national policy for publishing dissertations with DOIs in institutional repositories, formalising the use of ORCIDs for authors and their supervisors. Australian institutions are some of the largest publishers of dissertations with DOIs globally. The same expectations on the use of ORCID and ROR identifiers that the community expects of publishers should also apply to thesis deposit.

repository name	repository id	institution	country code	Dissertations
Universidade Estadual de Campinas	crossref	Universidade Estadual de Campinas	BR	62,319
ResearchGate	rg.rg		DE	45,176
Universidade de Sao Paulo, Agencia USP de Gestao da Informacao Academica (AGUIA)	crossref	Universidade de São Paulo	BR	25,319
Monash University	monash.repo	Monash University	AU	19,886
Zenodo	cern.zenodo		CH	15,972
University of Queensland Library	crossref	The University of Queensland	AU	15,888
Universidade de São Paulo. Agência de Bibliotecas e Coleções Digitais	crossref	Universidade de São Paulo	BR	14,477
Louisiana State University Libraries	crossref	Louisiana State University and Agricultural and Mechanical College	US	12,290
University of Tasmania	ardcx.utas	University of Tasmania	AU	10,291
University of Vienna	univie.univie		AT	10,254
American University Research Archive	au.ir		US	9,941
EDUFU - Editora da Universidade Federal de Uberlandia	crossref	Universidade Federal de Uberlândia	BR	8,592
Repositorio Académico	vwys.repositorio		CL	7,873
Purdue	figshare.purdue	Purdue University System	US	7,279
Macquarie University	ardcx.mqu	Macquarie University	AU	6,602
University of Liverpool	bl.lpool		GB	6,362
Libra	uva.libra	University of Virginia	US	6,233
Editorial Pontificia Universidad Javeriana	crossref	Pontificia Universidad Javeriana	CO	6,150
Northeastern University Library	crossref			6,093
Faculdades Catolicas	crossref	Pontifícia Universidade Católica do Rio de Janeiro	BR	6,080

Figure 32: Top 20 global repositories with DOI identified dissertations (toolkit query 4.2.10)

repository name	repository id	institution	country code	Dissertations
Monash University	monash.repo	Monash University	AU	19,886
University of Queensland Library	crossref	The University of Queensland	AU	15,888
University of Tasmania	ardcx.utas	University of Tasmania	AU	10,291
Macquarie University	ardcx.mqu	Macquarie University	AU	6,602
RMIT	figshare.rmit	RMIT University	AU	5,603
University of New South Wales	unsw.repo	UNSW Sydney	AU	3,568
La Trobe University	latrobe.repo	La Trobe University	AU	3,498
Swinburne University of Technology	swin.repo2	Swinburne University of Technology	AU	3,196
Queensland University of Technology	crossref	Queensland University of Technology	AU	2,023
The University of Western Australia	uniwa.repo	The University of Western Australia	AU	1,621
Central Queensland University	ardcx.cqu	Central Queensland University	AU	1,050
University of Southern Queensland research data collection	ardcx.usq	University of Southern Queensland	AU	562
James Cook University	jcu.repo	James Cook University	AU	421
Edith Cowan University	ardcx.ecu	Edith Cowan University	AU	230
Western Sydney University ResearchDirect	ardcx.wsu	Western Sydney University	AU	220
University of Canberra Research Portal	ardcx.uc	University of Canberra	AU	219
Australian Catholic University	ardcx.acu		AU	158
Charles Darwin University	cdu.repo	Charles Darwin University	AU	106
Torrens University	ardcx.torrens	Torrens University Australia	AU	39
University of Sydney Library	ardcx.usyd	The University of Sydney	AU	10

Additional 7 rows not shown.

Figure 33: Australian repositories with DOI identified dissertations (toolkit query 4.2.10)

id	all dissertations	with ORCID	with ROR	percent ORCID	percent ROR
monash.repo	19,886	23	51	0%	0%
University of Queensland Library	15,888	3,930	0	25%	0%
ardcx.utas	10,291	1,056	1,441	10%	14%
ardcx.mqu	6,602	3	657	0%	10%
figshare.rmit	5,603	91	1,206	2%	22%
unsw.repo	3,568	860	0	24%	0%
latrobe.repo	3,498	9	852	0%	24%
swin.repo2	3,196	1	362	0%	11%
Queensland University of Technology	2,023	209	0	10%	0%
uniwa.repo	1,621	0	1,467	0%	91%
ardcx.cqu	1,050	31	582	3%	55%
ardcx.usq	562	82	81	15%	14%
jcu.repo	421	296	406	70%	96%
ardcx.ecu	230	191	221	83%	96%
ardcx.wsu	220	0	217	0%	99%
ardcx.uc	219	0	24	0%	11%
ardcx.acu	158	19	0	12%	0%
cdu.repo	106	1	98	1%	92%

Figure 34: Australian Dissertations interoperability with ROR and ORCID (toolkit query 4.2.11)

2.3.5 Publications

Research Output 5.

Research publications are formal written outputs that communicate findings, methodologies and methods, and analyses from research activities. They are the primary means of disseminating scientific knowledge within the academic community and often undergo peer review. Common types of research publications include journal articles, conference papers, books, book chapters, and review articles. Publications are produced as part of research projects and frequently acknowledge grants that funded the work. They often analyse and interpret research data, samples, and observations, presenting them in a structured format. Some publications rely on facilities for experimental work, computing, or analysis. Metadata associated with research publications includes authorship, institutional affiliations, project and funding acknowledgements, journal or publisher details, DOI (Digital Object Identifier), and citation metrics.

Strategy Measurement Motivation
Publication outputs cover journal articles as well as books an book chapters (perhaps even blogs) A number of identifiers exist for publications including ISSNs for journals, ISBNs for books, and DOIs for journal articles For publications in particular, the form of a publication provides the structure that provides the context that links all of the identifiers together Publications also have multiple representations preprint/published version, pdf/JATS XML/html . Each of these formats can encode persistent identifiers at different levels of specificity. Because of the organising role that publications play, many measures involving other parts of the research ecosystem also involve publications
International Engagement/Dependencies

Expected Maturity	Established/Evolving
Senior Stakeholder	

Benchmarks

Findability	*percentage of publications with persistent identifiers that are represented in major databases (such as Dimensions, Scopus, Web of Science) [sampling survey required]
Accessibility	Percentage of Australian research output that is OA, by publisher, institution, and field of research percentage of publications with an associated preprint. Percentage of papers with the correct structure for provenance: Conflict of interest statement, Data Availability, Funding Statement, Ethics Approval Statement, Author Contributions...
Interoperability	Percentage publications represented in at least one ORCID record, by source (institution vs Crossref for instance) Percentage of publications that have ORCIDs represented in Crossref but are not reflected back in ORCID Identification of publishers that are used by Australian researchers that collect ORCIDs but do not pass them on.
Reuse	Reuse of publication metadata, particularly for government reporting is covered below

Why it Matters

Analysis

Findability

The use of persistent identifiers for publications is now more or less ubiquitous. The established practice is that DOIS should be used for journal articles and book chapters. Books are less likely to use DOIS, using the ISBN standard instead.

Throughout this report, the interaction between publications and other identifiers has been a key benchmark. (ORCIDs, grants, datasets) The emergence of new publication categories has been covered including reports and dissertations.

Some areas of research still operate outside of the scope of persistent identifiers. Of particular interest is the role of large conferences and conference proceedings in AI. For example, the (Conference on Neural Information Processing Systems (NeurIPS)) does not use persistent identifiers for its papers despite producing over 4000 publications per year (figure 35)

conference	inproceedings	dois
Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024.	4,494	0
Thirty-Eighth AAAI Conference on Artificial Intelligence, AAAI 2024, Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence, IAAI 2024, Fourteenth Symposium on Educational Advances in Artificial Intelligence, EAAI 2014, February 20-27, 2024, Vancouver, Canada	2,866	2,866
IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2024, Seattle, WA, USA, June 16-22, 2024	2,715	2,715
IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2024, Seoul, Republic of Korea, April 14-19, 2024	2,679	2,679
Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024	2,610	0
IGARSS 2024 - 2024 IEEE International Geoscience and Remote Sensing Symposium, Athens, Greece, July 7-12, 2024	2,589	2,589
15th International Conference on Computing Communication and Networking Technologies, ICCCNT 2024, Kamand, India, June 24-28, 2024	2,463	2,463
The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024	2,260	0
IEEE International Conference on Robotics and Automation, ICRA 2024, Yokohama, Japan, May 13-17, 2024	1,760	1,760
International Joint Conference on Neural Networks, IJCNN 2024, Yokohama, Japan, June 30 - July 5, 2024	1,645	1,645

Additional 3082 rows not shown.

Figure 35: Top ten conferences in 2024 by volume of conference papers (source dblp). Three conferences in the top 10 do not publish their papers with DOIs. (toolkit query 4.2.14)

https://dblp.org/db/conf/nips/index.html.

To get a sense of how much Australian researchers are participating in venues without persistent identifiers, the volume of works recorded without a persistent identifier can be at least partially tracked. Although this is an incomplete measure as works without identifiers could be due to manual data entry, it can be observed that a) , the largest number of works without identifiers are conferences (figure 36), and that b) there has been an overall decline in works without identifiers over time (figures 37, and 37).

Interoperability

As many of the benchmarks for persistent identifier adoption still rely on inspecting the full text of an article for mentions (data citations/ funding mentions), the accessibility to the full text of papers is a key determinant in constructing comprehensive benchmarks. In 2024, the percentage of publications considered openly accessible for publications with Australian authors remained stable at 65% (Figure 39). 2024 is the first time the open access percentage has not increased in the previous decade. This means that for Australian research, analysis that is made solely on open access publications in 2024 would miss assessing practices encoded within 28,352 closed publications.

Recommendation 2.3.5: Benchmarking Analysis of persistent identifiers should not ignore closed access matarial. For Australia, ignoring closed access publications would result is missing 35% of Australia's output in 2024.

field_of_research	journal-article	conference-paper	conference-poster	working-paper	conference-abstract
Engineering	224	368	20	2	43
Information and Computing Sciences	339	340	13	4	21
Commerce, Management, Tourism and Services	120	256	2	6	48
Health Sciences	285	189	243	4	357
Education	215	181	13	2	68
Biomedical and Clinical Sciences	429	174	196	3	218
Built Environment and Design	112	87	5		5
Human Society	227	81	5	27	20
Language, Communication and Culture	117	75	13	8	4
Creative Arts and Writing	86	66	2		3
Psychology	79	65	7		25
Environmental Sciences	69	55	4	10	12
Biological Sciences	143	48	25	3	32
Earth Sciences	25	46	8		46
Agricultural, Veterinary and Food Sciences	49	41	7		15
Law and Legal Studies	213	35		3	
History, Heritage and Archaeology	88	29	2	1	3
Physical Sciences	75	25	1		7
Chemical Sciences	68	18	1		5
Mathematical Sciences	16	17		2	
Economics	47	11		12	2
Philosophy and Religious Studies	43	10	1		3

Figure 36: Works reported in 2024 with no persistent identifier by Field of Research (4.2.15)

Recommendation 2.3.6: ARDC begins outreach to strategic conferences to understand what support is required to ensure that all conference proceedings have a persistent identifier.

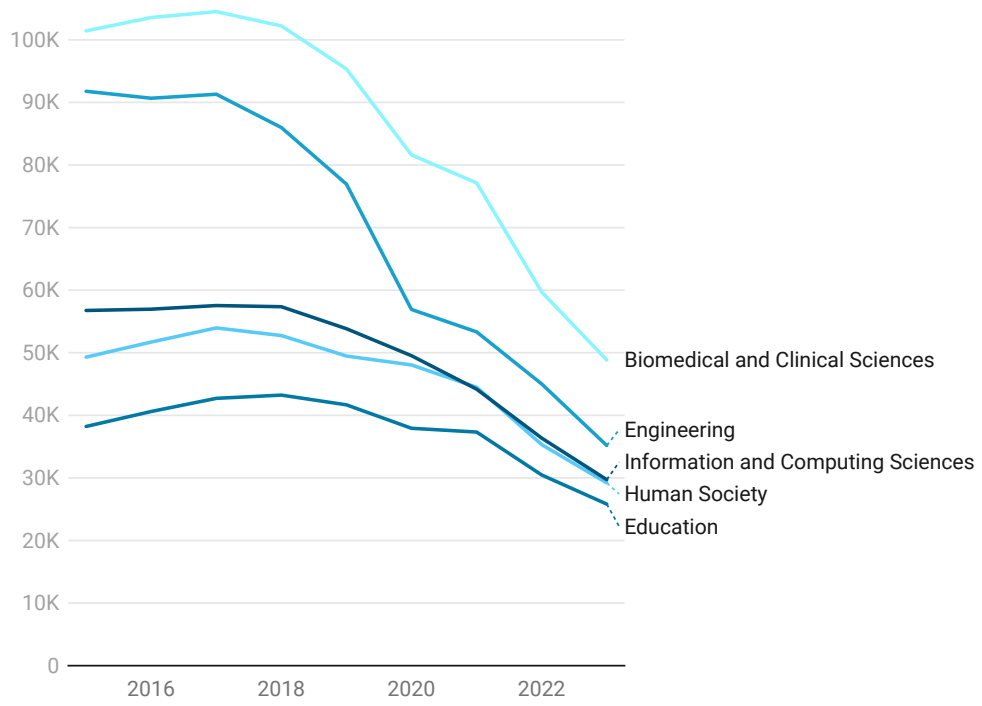


Figure 37: Global works with no pid (toolkit query 4.2.12)

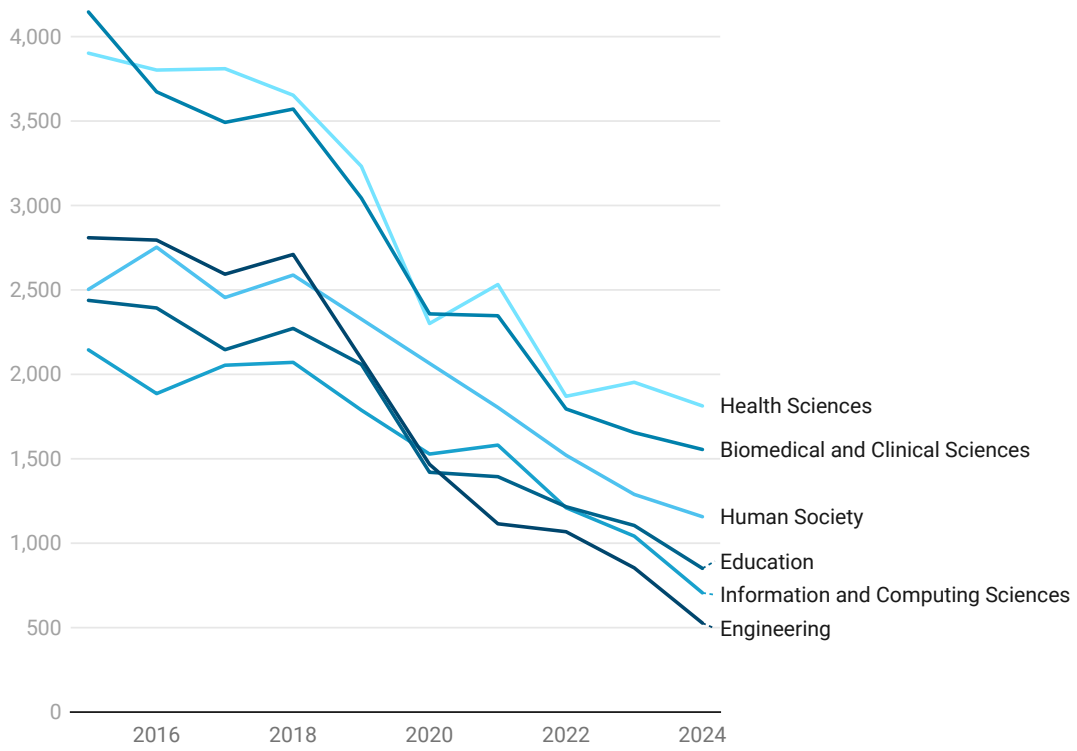


Figure 38: Australian works with no pid (toolkit query 4.2.13)

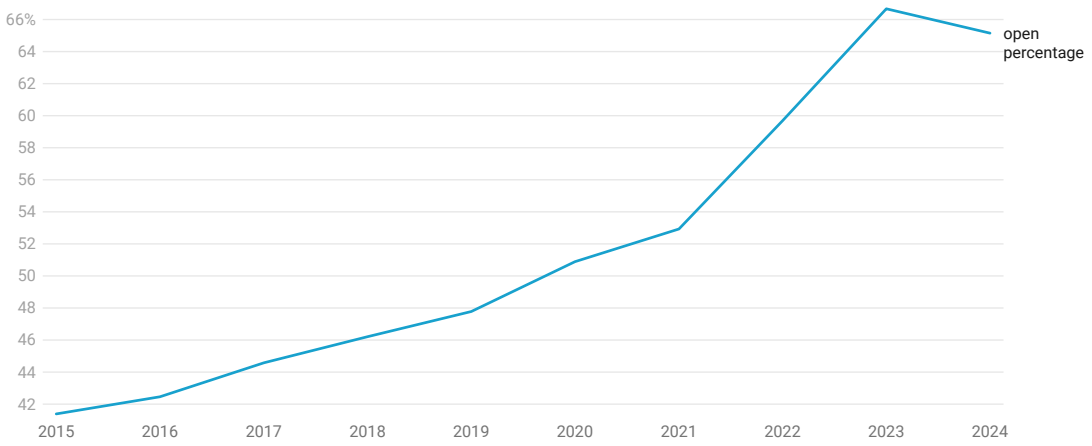


Figure 39: Open Access percentage of Australian research output over time. For this analysis, all access types accept closed are considered open. (toolkit query 4.2.16)

2.4 Benchmarks for research reproducibility, provenance, and reducing academic burden

By linking researchers to projects, grants, organisations and equipment, we can improve research reproducibility, provenance and attribution. Through the use of PIDs, researchers can minimise the amount of time they spend reproducing the same data many times over, releasing valuable time back into research.

Objective 3. Australian National Persistent Identifier Strategy

Beyond inputs and outputs, the maturity research system lies in its ability to link people, projects, institutions, and infrastructure in a way that enhances reproducibility and reduces administrative load. Persistent identifiers (PIDs) provide this connective tissue. This section focuses on how we can benchmark and improve the quality of these connections, ensuring metadata integrity, minimising academic burden [1], and making the act of research itself more traceable, efficient, and trusted.

Within the rubric of reducing academic burden, the recently convened PIDs Value Proposition group [11] identified the following scenarios.

What PIDs enable	Related PIDs
Automatically populates publication and employment records for grant applications.	ORCID, DOI (Publications and grants)
Global unique identification and metadata aiding in reporting, project management, and compliance.	DOI, ORCID, RAID
The automatic population and maintenance of RIMS with accurate and up-to-date information on researchers, publications, grants, and projects.	ORCID, DOI (Publications, Grants, Equipment), RAiD
Integration of priority PIDs with Current Research Information System providers (like Symplectic Elements, Elsevier Pure, ResearchMaster, ReDBox, VIVO) enables wholesale adoption of PIDs, making PID-enabled workflows more accessible and reducing implementation costs for individual institutions.	DOI (Publications, Grants, Equipment), ORCID, RAiD
Researchers and institutions to pass accurate and linked data about outputs, organisations, people, and projects to funders' reporting systems.	DOI, ORCID, RAiD, ROR
Automatic updates of new publications from PID registries to institutions provides more timely data, and improves accuracy by minimising manual data entry.	ORCID, DOI

Table 4: Scenarios: Reducing Academic Burden

These scenarios involve a range of persistent identifier infrastructures, which as we have seen in the previous seen in the previous sections, differ in maturity and adoption. Tables 5 and 6 summarise how the recommendations in sections 2.2 and 2.3 help support these scenarios. Although ORCID integration was measured across all inputs and outputs, Recommendations specific to ORCID adoption and integration will be handled seperately in section 3.1.

Perhaps new to the national conversation is the idea that research management systems might not only be consumers of persistent identifiers, but might also serve as a source for records of authority around Research Projects (RAiD), and equipment. To support these practices, new processes and, in some cases, new functionality will need to be created. For these instances, it is less useful to focus on metadata metrics, but rather audit the development of the system as a whole.

By 2030:

- What percentage of CRIS systems do we expect to have RAiD integrations that also function

[11] The PIDS Value Proposition group was recently convened by the ARDC between Janurary and March 2025

as research activity registries?

- How many equipment registries should have been established?

As the benchmarks associated with the adoption of ORCID in the second section of this document will demonstrate, different fields of research adopt persistent identifier infrastructure at different rates. Beyond prior system integrations, it may also be useful to target priority areas of research. Establishing the right partners to begin building the right processes, even though all the pieces may not yet be in place, will be critical in laying the foundations for broader adoption.

In the next three years, what areas of research should be prioritised? What are the qualitative changes that we should be looking for that will then help shape broader adoption?

A commitment to reducing administrative burden around identifiers such as research activities and equipment also requires a shift in focus from the administration and communication of research to the research process itself. Which integrations between research administrative systems and research platforms should be prioritised?

In addition to the processes and systems that need to be established, there is also the challenge of optimising the use of research information infrastructure that should already be in place. Today, these processes are largely centred around encouraging the optimal use of ORCID.

Within the context of reducing academic burden:

- What is the average level of academic completeness of ORCID records that would indicate an optimum level of metadata reuse?
- What percentage of the Australian research community have turned on Crossref and DataCite integrations for their own ORCID ids?
- What percentage of Australian CRIS systems are connected to ORCID, increasing overall trust in ORCID assertions?

Table 5: How *Inputs* recommendations support the objective "Benchmarks for Research Reproducibility, Provenance, and Reducing Academic Burden".

Input	Recommendation (cross-ref)	Reasoning (reproducibility, provenance, reduced burden)
Research partners (ROR)	Review representation of commercial AU orgs in ROR (2.2.1)	Improves organisational disambiguation and trusted affiliation chains; reduces manual reconciliation in reporting workflows.
	Engage publishers so ROR affiliations are asserted to publications (2.2.2)	Pushes authoritative affiliations into publication metadata at source, strengthening provenance and reducing after-the-fact curation.
	Engage repositories / workflows to add ROR to all creators (2.2.3)	Normalises affiliation capture at deposit time, ensuring automated propagation across the PID graph.
Research projects (RAiD)	Community adopts an initial RAiD target (2.2.4)	Ensures project-level identifiers exist at scale so inputs and outputs inherit consistent context, making work easier to reproduce and audit while reducing duplicative data entry.
Grants	Funders create persistent grant identifiers (2.2.5)	Links funding → projects → outputs via authoritative IDs; enables automated attribution and downstream reporting.
	Grant award triggers RAiD creation/association (2.2.6)	RAiD exists "day one," so people and outputs can attach immediately. Provenance starts at award and admin effort is minimised.
Facilities	Facilities confirm representation in ROR (2.2.7)	Facility use can be credited and traced across outputs; benchmarking and reporting become query-based instead of manual tallying.
Data	Increase data-citation literacy (2.2.8)	More dataset DOIs in the record strengthens input–output links, improving reproducibility and simplifying evidence gathering for impact.
Repositories	Make repository workflows "ORCID-first" (*no label*)	ORCID-by-default reduces rekeying, improving person–output linkage quality for audits and replication.
Samples (IGSN)	Set new national minimum standards for IGSN records (2.2.10)	Embeds samples in the PID graph (links to ORCID/ROR/datasets), tightening provenance from sample → method → publication.
Instruments	Encourage links between instruments and the PID graph; normalise instrument citation (2.2.11)	Connects methods → instruments → data → papers, enabling credit/traceability and lowering curation burden through standard practice.

Table 6: How *Outputs* recommendations support the objective "Benchmarks for Research Reproducibility, Provenance, and Reducing Academic Burden".

Output	**Recommendation (cross-ref)**	**Reasoning (reproducibility, provenance, reduced burden)**
Reports	Encourage deposit of reports in institutional repositories (2.3.1) Link reports to ORCID/ROR (2.3.2)	Preserves grey literature, makes it citable and harvestable; simplifies evidence-of-impact with consistent metadata. Tightens person/organisation attribution; enables automated roll-ups for assessments and audits.
Methods	Create guidelines for recording/registering protocols (2.3.3)	Standardises method disclosure, improving reproducibility and comparability across studies with less bespoke admin.
Dissertations	Consider national policy for publishing dissertations; require ORCIDs for authors/supervisors (2.3.4)	Increases visibility and author disambiguation; reduces ad-hoc local processes and improves provenance for early-career outputs.
Publications (coverage)	Include closed-access material in benchmarking (2.3.5)	Prevents selection bias; ensures system-level metrics reflect the full corpus without manual exceptions handling.
Conference proceedings	Outreach so proceedings have PIDs (*no label*)	Ensures traceable citations and consistent inclusion in output tracking; reduces the need for one-off data corrections.

2.5 Benchmarks for enabling the understanding of the impact of research inputs, evaluating research quality, impact and evidence of public benefit

> By linking elements across the research and innovation ecosystem through PIDs, we can better understand the relationship between the elements of the ecosystem, including investment, use, outputs, outcomes and impact. We can more efficiently assess our research institutions' performance with respect to research quality thresholds, better trace the impact of the research and through this, more readily explain the public benefit of the research
>
> **Objective 4. Australian National Persistent Identifier Strategy**

With robust data on inputs, outputs, and processes we can begin to understand and evidence research impact by further linking these elements across the ecosystem - connecting grants, projects, people, outputs, and institutions to outcomes and public benefit. Persistent identifiers (PIDs) provide the infrastructure to enable this, but the true benchmark for success is whether they allow us to trace research activity into demonstrable impact: both within societal, environmental, cultural, and economic.

National impact reporting should ultimately be supported by our national research information infrastructure—rather than relying on resource-intensive reporting exercises. The key question is: when will there be enough shared trust in the system to make this shift possible, not just for publications, but for all types of research output?

Articulated as a goal, for which we can benchmark against, we might express the objective as follows:

Trusted Government and institutional reporting is enabled through the use of authoritative identifiers for all Australian research institutions. Use of algorithmic identifiers is reduced increasing metadata data sovereignty, accuracy and ownership of Australian research activity:

- All Australian Research Institutions should have an ROR id (recommendations 2.2.1, 3.1.1)
- All Australian Researchers should have an ORCID (recommendation 3.1.5)

In addition, to facilitate accurate national reporting, the following community practices must also be adopted:

- Institutions publishing trusted affiliation assertions to ORCID, enabling easy identification of researchers. (recommendations 3.1.2)
- All Australian Researchers keep up to date ORCIDs records, as well as enabling trusted assertions from CRIS systems and Crossref/DataCite integrations (recommendation 3.1.5)
- All Australian Researchers supply their ORCID when publishing - always (recommendation 3.1.3)

These goals must in turn be supported by international norms:

- Publishers adopting ROR identifiers for institutional affiliations of all (or at least most) authors (recommendation 2.2.2)
- Publishers communicate all information to ORCID via Crossref (recommendation 3.1.2)
- Publishers record ORCIDs for all authors/creators (recommendation 3.1.2)

As identified in sections 2.2, and 2.3 many Australian universities also have strong practices around the recording of research outputs other than publications. Australia in particular has a number of institutions that systematically deposit research reports into their repositories (as well as many other non traditional research outputs), and many universities have established processes for recording dissertations. In these cases institutions are the primary publishing point for these outputs. As

publishers of these outputs, institutions also take on the same metadata expectations and responsibilities as publishers.

Equivalent norms for institutions publishing works in repositories might be expressed as:

- Institutions adopting ROR identifiers for institutional affiliations associated with authors/creators (recommendation 2.2.3)
- Institutions record ORCIDs for all authors/creators
- Institutions communicate all information about works to ORCID via Crossref and DataCite

These behaviours are by no means an exhaustive list of what is required to establish a trusted national reporting environment, but they do represent a starting point that allow us to benchmark progress.

Behaviour	**Benchmark**
All Australian Research Institutions should have an ROR id	Adoption vs potential adoption (using algorithmic grid/ROR assignments as a benchmark)
All Australian Researchers should have an ORCID	Comparison of the completeness of ORCID records belonging to Australian researchers when compared to profiles approximated by Dimensions
Institutions publishing trusted affiliation assertions to ORCID, enabling easy identification of researchers.	Analysis of ORCID records for trusted affiliations
All Australian Researchers keep up to date ORCIDs, as well as enabling trusted assertions from CRIS systems and Crossref/DataCite integrations	Proportion of Australian ORCID records with assertions from Crossref, and local institutions
All Australian Researchers supply their ORCID when publishing - always	Proportion of Australian papers that have ORCIDs for all Australian authors
Publishers adopting ROR identifiers for institutional affiliations	Identification of publishers that are using ROR, and the percentage of time that they are using them
Publishers communicate all information to ORCID via Crossref	Proportion of publishers that communicated ORCID data to Crossref
Publishers record ORCIDs for all authors/creators	Analysis of publishers that support ORCIDs for all authors

Table 7: Behaviour and Benchmarks

Many of these benchmarks have been covered in sections 2.2, and 2.3. Detailed benchmarks exploring ORCID connectivity will be covered in section 3.1.

2.6 Benchmarks for research capability

Australia needs the research capability to address our national priorities. PIDs for people, organisations, research services and infrastructure would enable us to map our research capability against national priorities and invest appropriately to address gaps.

Objective 4. Australian National Persistent Identifier Strategy

Improving our ability to map Australia's research capability invites us to consider the benchmarking exercise from the perspective of scenarios.

For a given national research priority, how do we understand who is investing in what? Today, we might begin to answer this question by conducting keyword searches over national grants and publication outputs. With the appropriate investment in research information infrastructure however, we should expect a more complete picture.

Beyond obvious inputs and outputs, what research facilities and equipment have been critical in helping to achieve our priorities? How is data from previous initiatives scaffolding our understanding? How well do we understand the research community that has self-organized to meet our national challenges?

In contrast to national reporting, which by definition takes a broad view of Australian research capability, questions expressed within the context of national priorities reach deeper into how our research activities are organised. Then by specifically identifying national priorities, along with the necessary information requirements, we can stress test both our current ability to retrieve the necessary information, along with validating that our shared vision for national research information infrastructure is fit for purpose.

For instance, if the next global pandemic were to happen tomorrow, how readily could we identify the right collaborations, and what information would we ideally want to use? As the national PID roadmap is developed, and national priorities identified, in a similar fashion to the benchmarking approach outlined in the previous section, benchmarks arising from objectives that support the identified priorities can be identified.

Based on the observations in this report, it is clear that Australia still has a lot of progress to make before exploring the nations research capability based on the national persitent identifier graph becomes a reality. Sustainable processes for establishing many of the connecting objects such as research activities and instruments, and even grants have yet to be established, and there is still a lot of work to do to connect Australia's research to its researchers and institutions.

Achieving these goals will require both national and international collaboration, however the challenges are not insurmountable. Many of the collaborations required to make this vision a reality have already been created as part of Australia's efforts to implement ORCID. In the final section of this report, we will focus on benchmarking ORCID adoption within Australia, but also the national coordination that has been required to make progress to date such a success. Should these collaboration patterns continue, then the objectives outlined in the National Persistent Identifier Strategy will likely be achieved.

3 Persistent Identifier Benchmarking in Practice

In order to explore persistent identifier benchmarking in practice, we reflect on the achievements that previous initiatives have made to shape our national information infrastructure. Released in 2017, Vision 2020 outlined a national target that, by 2020, all Australian researchers, regardless of institutional affiliation, will be able to easily claim and manage the data regarding their research grants, research and research-related outputs through ORCID and that their details will travel with them as they build their careers.

In the context of the proposed toolkit, these goals would now be articulated both as part of the broader objectives of improving research reproducibility, provenance, reducing academic burden (Benchmarks **??**,) as well as providing much of the underlying capability that supports the communication of research impact (Benchmarks 2.5).

Overall, the message of the following section is both that

a) Australia's combined interventions to establish ORCID as a key part of the national research ecosystem have been successful so far, and

b) that benchmarking the adoption of persistent identifiers is achievable, and provides a powerful lens through which we can understand progress.

3.1 Did Vision 2020 make a difference? Benchmarking the impact of the 2017 Australian ORCID strategy.

> Our vision is that, by 2020, all Australian researchers, regardless of institutional affiliation, will be able to easily claim and manage the data regarding their research grants, research and research-related outputs through ORCID and that their details will travel with them as they build their careers. – VISION 2020 (2017) Australian ORCID Governance Committee [2]

ORCID AUSTRALIA

ORCID

VISION 2020

AUSTRALIA'S ORCID ASPIRATIONS

Our aspiration is that, by 2020:

- All active researchers in Australia have an ORCID identifier
- All Australian researchers actively use their ORCID throughout the research lifecycle
- ORCID is seen as the best way in which to identify international research experts
- The ARC/NHMRC will integrate ORCID with their grant management systems
- Government agencies draw upon ORCID data, where possible, for institutional research performance reporting and assessment
- The Australian ORCID Consortium is a thriving, sustainable and valued organisation which has steadily grown its membership base.

As a matter of principle, we recognise the value of unique researcher identifiers in reducing red tape, increasing efficiency, improving data quality, integrating disparate data, promoting the reuse of data, and enhancing the online presence of Australian research to the global market and commit to support the use of ORCID as the preferred common researcher identifier.

Our vision is that, by 2020, all Australian researchers, regardless of institutional affiliation, will be able to easily claim and manage the data regarding their research grants, research and research-related outputs through ORCID and that their details will travel with them as they build their careers.

Further to that, all Australian research organisations and agencies will be able to ingest relevant details into grant applications, research review processes and create profiles of their researchers with improved data integrity.

Australia will benefit from reducing the administrative burden on researchers and research managers/administrators, and increase Australia's international research profile via increased access to Australia's research outputs and associated activities.

Australian ORCID Governance Committee

Supported by the Australian ORCID Advisory Group

October 2017

Visit orcid.org

aaf.edu.au/orcid to learn more.

Join ORCID today and become part of the solution.

Figure 40: ORCID Australia Vision 2020

Like the Australian National Persistent Identifier Strategy, the Vision 2020 statement by the Australia ORCID Governance Committee was created by senior leadership representation across the research community including [3]:

1. the Chair of the Australian ORCID Advisory Group, ex officio
2. the Chair of the Board of the Consortium Lead, or nominee
3. the Chair of the CAUL, or nominee
4. the Chair of ARMS, or nominee
5. the Chair of Universities Australia Deputy Vice-Chancellors, Research Committee, or nominee

6. A representative from the NHMRC, nominated by the CEO

7. A representative from the ARC, nominated by the CEO

8. An Australian ORCID Board Member (if elected), ex officio

The Governance committee was supported by an equally diverse Advisory Board (Table 8). It is significant to note not only the persistence of vision that has underpinned Australia's commitment to ORCID in the following 8 years, but also the persistent involvement of a community of national organisations that have supported and led these activities.

Name	**Organisation (Abbreviation)**
Michelle Duryea	Australasian Research Management Society (ARMS)
Simon Huggard	Council of Australian University Librarians (CAUL)
Natasha Simons	Australian National Data Service (ANDS)
Adrian Burton	Australian National Data Service (ANDS)
Dr Liz Eedle	Universities Australia (UA)
Steve Hannan	Australasian Research Management Society (ARMS)
Wee-Ming Boon	National Health & Medical Research Council (NHMRC)
Justin Withers	Australian Research Council (ARC)
Cherie Atkinson	Australian Research Council (ARC)
Heath Marks	Australian Access Federation (AAF)
Elleina Filippi	Australian Access Federation (AAF)
Melroy Almeida	Australian Access Federation (AAF)
Anne Kealley	Council of Australian University Directors of Information Technology (CAUDIT)
Jens Klump	CSIRO

Table 8: Members of the Australian ORCID Advisory Group in 2017: ADVISORY GROUP The Australian ORCID Advisory Group acts in the interests of the nation to provide expertise and advice on leveraging maximum benefits through national adoption and use of ORCID.

This coalition of interests built around a common goal emerged as a result of at least 3 years work establishing a common understanding of research information citizenship across the research community [18, 17]. Each stakeholder understood the importance of ORCID, but also the role that they needed to play - both in terms of advocacy, systems development and workflow support. Some of the initiatives emerging from this coalition included:

1. University led advocacy for ORCID adoption

2. inclusion of ORCID workflows within research information management workflows

3. incorporation of ORCID into funding application workflows

4. recognition of the role of ORCID within national research strategy.

Now a decade after these efforts began, what effect did these efforts have on the Australian research information infrastructure? How do these efforts compare to international efforts? To answer these questions, this report compares ORCID representation of the Australian and international research landscape to a reference dataset. A reference dataset must be chosen for consistent comparison and should approximate a more complete dataset. For the purposes of this initial benchmarking project Dimensions has been used as the reference dataset. The ARDC will determine the most suitable reference dataset(s) once this project is completed.

Using this methodology, we can ask:

- Approximately what percentage of the research community has an ORCID, and how complete are these records on average?

- How does Australian ORCID adoption and completeness compare internationally for those researchers that apply for funding?

- How well are Australia's aims supported by publisher support for ORCID?

- How have different disciplines adopted ORCID?
- How much do institutional culture and initiatives affect ORCID adoption?

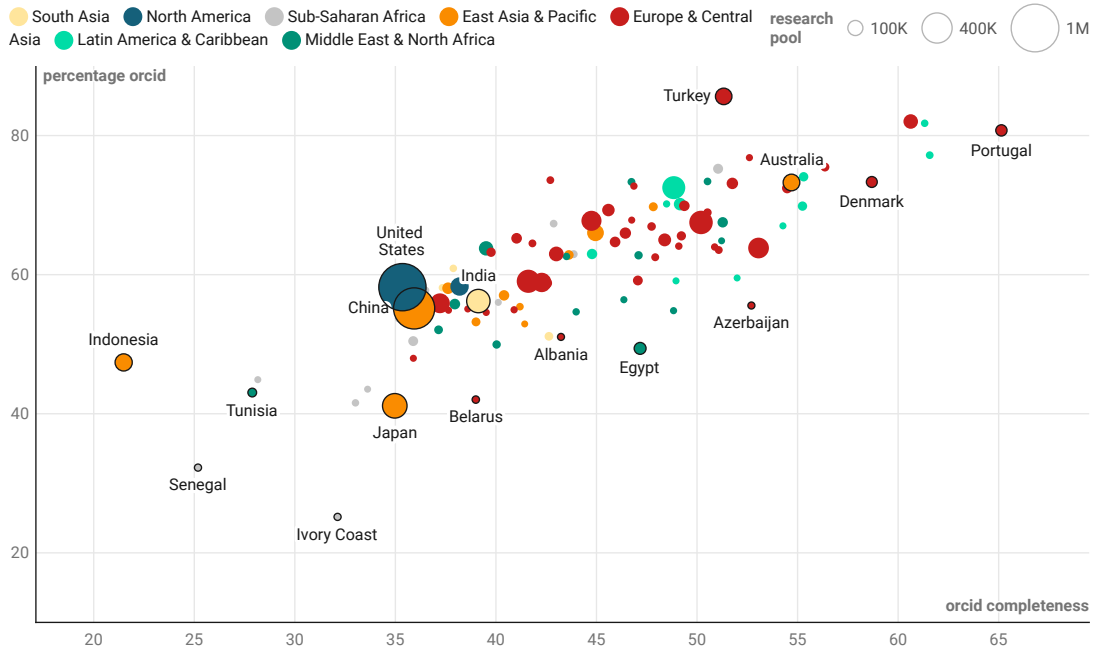
3.1.1 Benchmarking Australia's ORCID adoption

To benchmark Australia's ORCID adoption at a country level, we compare the total number of identifiable researchers within Dimensions [9] that have claimed an affiliation to an Institution within a country in a five-year period. It should be noted that Dimensions is not the only reference dataset that could be used to benchmark ORCID adoption; however, it is particularly useful for this project as Dimensions covers publications, as well as datasets, grants, patents, and policy documents [9]. Dimensions also integrates well with the persistent identifier ecosystem. This is particularly so for institutional identifiers, with Dimensions GRID identifiers serving as the seed dataset for ROR ids [16]. Dimensions also categorises research outputs by ANZSRC fields of research, making benchmarks easy to contextualise within an Australian research reporting framework.

As Dimensions creates profiles algorithmically, we further restrict the researcher pool to those researchers that have been publishing for more than five years, as the fewer years that a researcher has been publishing, the fewer data points there are to accurately identify a new researcher. These profiles are then matched to an ORCID identity, allowing the percentage of researchers by country that have an ORCID to be measured. To assess the average completeness of each record, for each researcher, we then count the number of works assertions with DOIs that have been recorded in ORCID with the number of publications that have been recorded against the equivalent researcher_id within Dimensions. This technique is covered in detail in "Measuring Research Information Citizenship Across ORCID Practice" [14]

Within the period 2000-2024 (figure 41), Australia has an estimated adoption rate of 73%. This is the third highest ORCID adoption rate for countries with estimated researcher pools of greater than 50,000. The average completeness of these records is 55%, ranking 2nd against the same set of institutions.

To get a sense of progress, we also calculated the same set of benchmarks for the period 2015-2019. Within this period, an estimated 48% of Australian researchers had created an ORCID for themselves. In the subsequent 5 year period then, ORCID adoption increased by 25%. Completeness in the period 2015-2019 was 50%, increasing by 5% in the following 5 years. As can be seen in figure 42, these increases are in line with global trends.



country	research pool	orcid researchers	percentage orcid
Turkey	101,457	86,881	86%
Poland	74,274	60,922	82%
Australia	103,734	75,990	73%
Brazil	216,874	157,260	73%
Italy	160,126	108,476	68%
United Kingdom	226,735	153,065	68%
South Korea	105,126	69,407	66%
Switzerland	54,039	35,126	65%
Spain	172,153	109,890	64%
Iran	72,452	46,211	64%
Netherlands	76,723	48,320	63%
Germany	220,842	130,409	59%
Russia	148,136	87,222	59%
Canada	130,786	76,242	58%
United States	983,050	572,158	58%
India	224,978	126,452	56%

Additional 4 rows not shown.

country	orcid completeness
Poland	61%
Australia	55%
Spain	53%
Turkey	51%
United Kingdom	50%
Brazil	49%
Switzerland	48%
South Korea	45%
Italy	45%
Netherlands	43%
Russia	42%
Germany	42%
Iran	40%
India	39%
Canada	38%
France	37%
China	36%
United States	35%
Japan	35%
Indonesia	21%

Figure 41: ORCID Adoption and Completeness by Country 2000-2024. The chart above plots the approximate percentage of researchers (y axis) by country, against the average completeness of those profiles (x axis). The tables below the chart rank countries by Adoption (left) and Completeness (right) (Click on the figures for interactive versions)

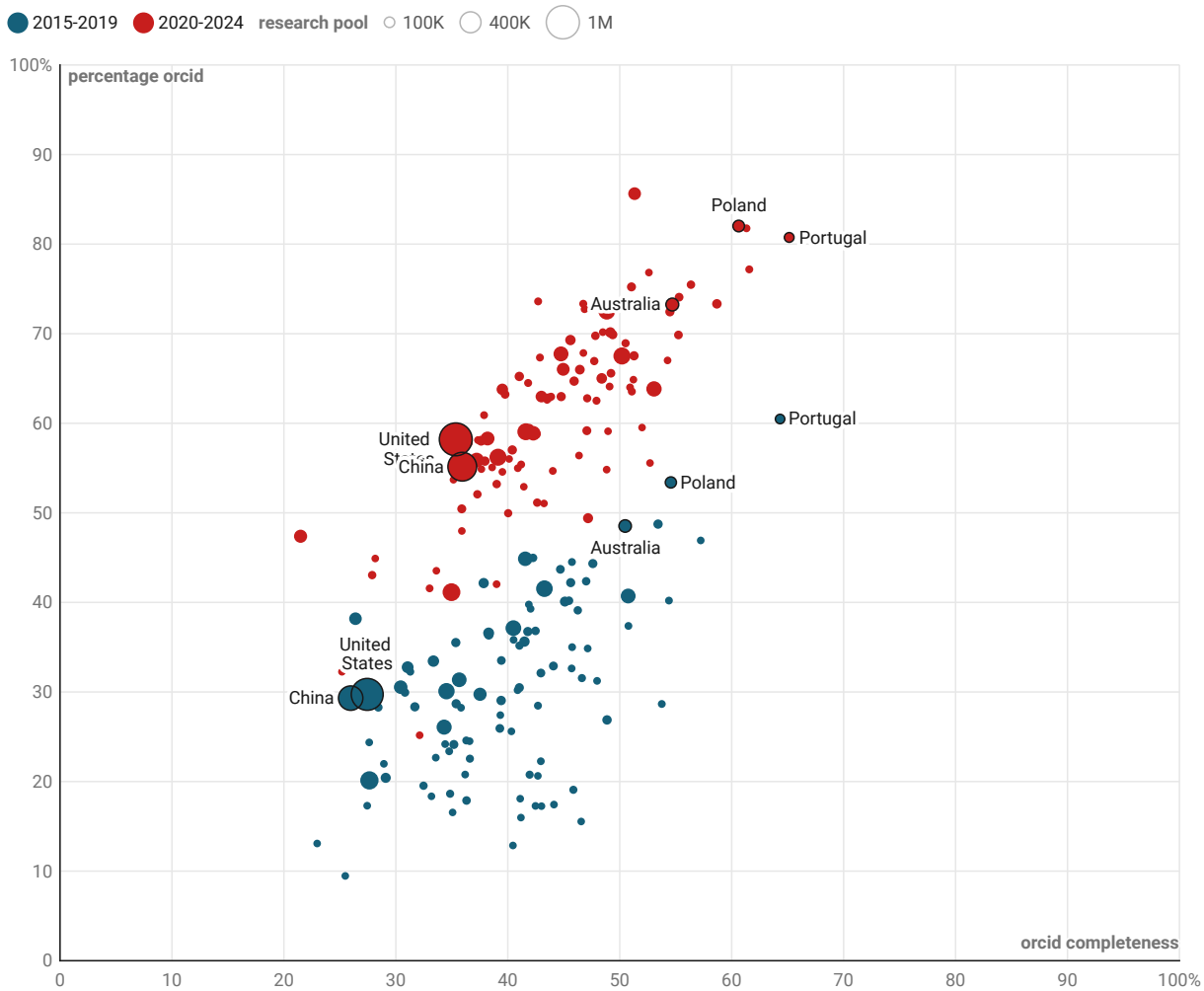


Figure 42: ORCID adoption and completeness by country compared across two time periods, 2015-2019 and 2020-2024. The percentage of researchers by country that have an ORCID has risen significantly between the two time periods, as has the average completeness of these records. (Click on the figure for an interactive version)

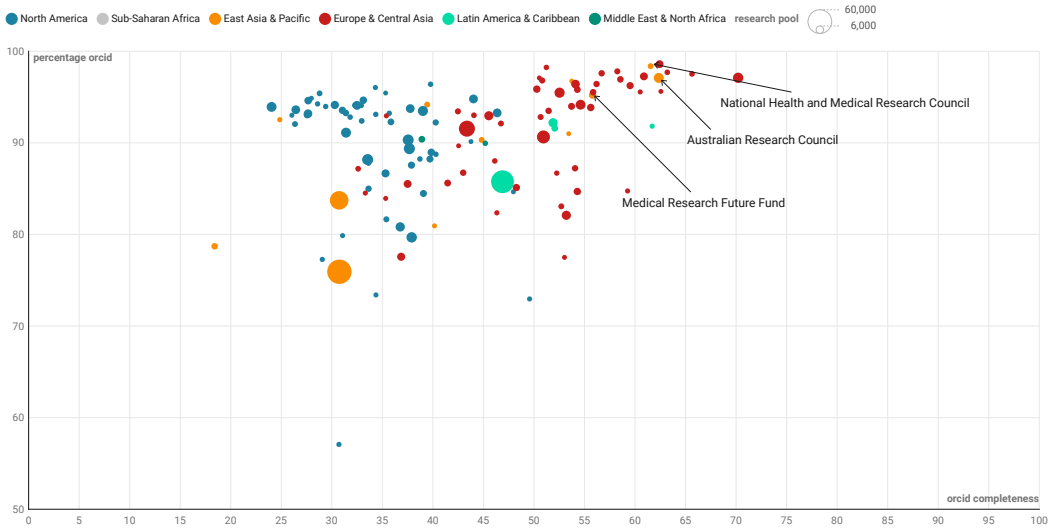


Figure 43: ORCID adoption and completeness by funder 2020-2024
. The average completeness of Australian and most European funders is much greater than those of North American funders. (Click on the figure for an interactive version)

3.1.2 ORCID By Funders

For researchers who apply for and receive funding, the adoption levels of ORCID are close to 100% for many funders worldwide, with most achieving adoption rates greater than 90%. For the period 2020-2024, the National Health and Medical Council has the second highest ORCID adoption rate globally at 98.5%. ORCID completion rates are also significantly higher for funded researchers rather than non-funded researchers and have dropped only marginally when compared across the two five-year periods. In the period 2000-2024, researchers funded by the Australia Research Council had the sixth highest average ORCID completeness rate, behind only funders from Portugal, Denmark, Ireland, and Poland44.

This difference in funded and non-funded researchers in terms of ORCID adoption and completeness is at least partly due to the strategic priority that the international ORCID research community has placed on supporting funding workflows [8].

When viewed through a country lens (figure 43) another significant trend emerges. Although the level of adoption is not significantly different across most funders, researchers that have received funding from North American institutions have less complete ORCID records on average. One hypothesis for this difference is that might be that the North American research community has approached ORCID from a compliance perspective. Researchers funded from European and Australian institutions have significantly more complete records, and are more likely to benefit from the advantages that follow from this, both in terms of the reduction in academic burden, as well as control of their academic identity. It also follows that funded researchers from European and Australian institutions are more likely to positively promote ORCID among their peers than their North American counterparts.

Recommendation

Recommendation 3.1.1: ORCID adoption for Australian researchers has progressed sufficiently far to make requiring authenticated ORCIDs mandatory.. Requiring ORCID ids on grants ensures that grant records can be created that can be integrated into the Australian persistent identifier landscape, facilitating the downstream creation of RAiD records, and making funding information easier to integrate both into Research Information Management systems as well as ORCID itself.

funder	country	2015-2019	2020-2024
Fundação para a Ciência e Tecnologia	Portugal	73%	70.8%
Danish Agency for Science and Higher Education	Denmark	69%	65.7%
Novo Nordisk Foundation	Denmark	66%	63.3%
Science Foundation Ireland	Ireland	64%	62.5%
National Science Center	Poland	71%	62.4%
Australian Research Council	Australia	64%	62.4%
Agencia Nacional de Investigación y Desarrollo	Chile	63%	61.5%
National Health and Medical Research Council	Australia	59%	61.5%
Biotechnology and Biological Sciences Research Council	United Kingdom	62%	60.9%
Ministry of Education Youth and Sports	Czechia	65%	60.5%
Technology Agency of the Czech Republic	Czechia	66%	59.3%
Natural Environment Research Council	United Kingdom	61%	59.2%
National Institute for Health and Care Research	United Kingdom	61%	58.5%
Czech Science Foundation	Czechia	63%	58.3%
Research Foundation - Flanders	Belgium	61%	56.7%
Academy of Finland	Finland	61%	56.2%
FWF Austrian Science Fund	Austria	61%	56.0%
Medical Research Future Fund	Australia	62%	55.7%
Economic and Social Research Council	United Kingdom	58%	55.6%
UK Research and Innovation	United Kingdom	58%	54.7%

funder	country	2015-2019	2020-2024
National Science Center	Poland	86%	98.6%
National Health and Medical Research Council	Australia	85%	98.5%
Italian Association for Cancer Research	Italy	89%	98.2%
Czech Science Foundation	Czechia	84%	97.8%
Novo Nordisk Foundation	Denmark	90%	97.8%
Research Foundation - Flanders	Belgium	75%	97.6%
Danish Agency for Science and Higher Education	Denmark	86%	97.5%
Biotechnology and Biological Sciences Research Council	United Kingdom	86%	97.3%
Fundação para a Ciência e Tecnologia	Portugal	83%	97.3%
Cancer Research UK	United Kingdom	86%	97.1%
Australian Research Council	Australia	87%	97.1%
National Institute for Health and Care Research	United Kingdom	85%	97.0%
Royal Society of New Zealand	New Zealand	84%	96.9%
European Research Council	Belgium	87%	96.8%
Office of Basic Energy Sciences	United States	87%	96.4%
Academy of Finland	Finland	82%	96.4%
Medical Research Council	United Kingdom	82%	96.4%
Natural Environment Research Council	United Kingdom	80%	96.2%
National Human Genome Research Institute	United States	78%	96.0%
Agence Nationale de la Recherche	France	80%	95.9%

Figure 44: ORCID Adoption and Completeness by Funder patterns 2015-2019 compared to 2000-2024. The chart above plots the approximate percentage of researchers (y axis) by funder, against the average completeness of those profiles (x axis). The tables below the chart rank countries by Adoption (left) and Completeness (right) (Click on the figures for interactive versions)

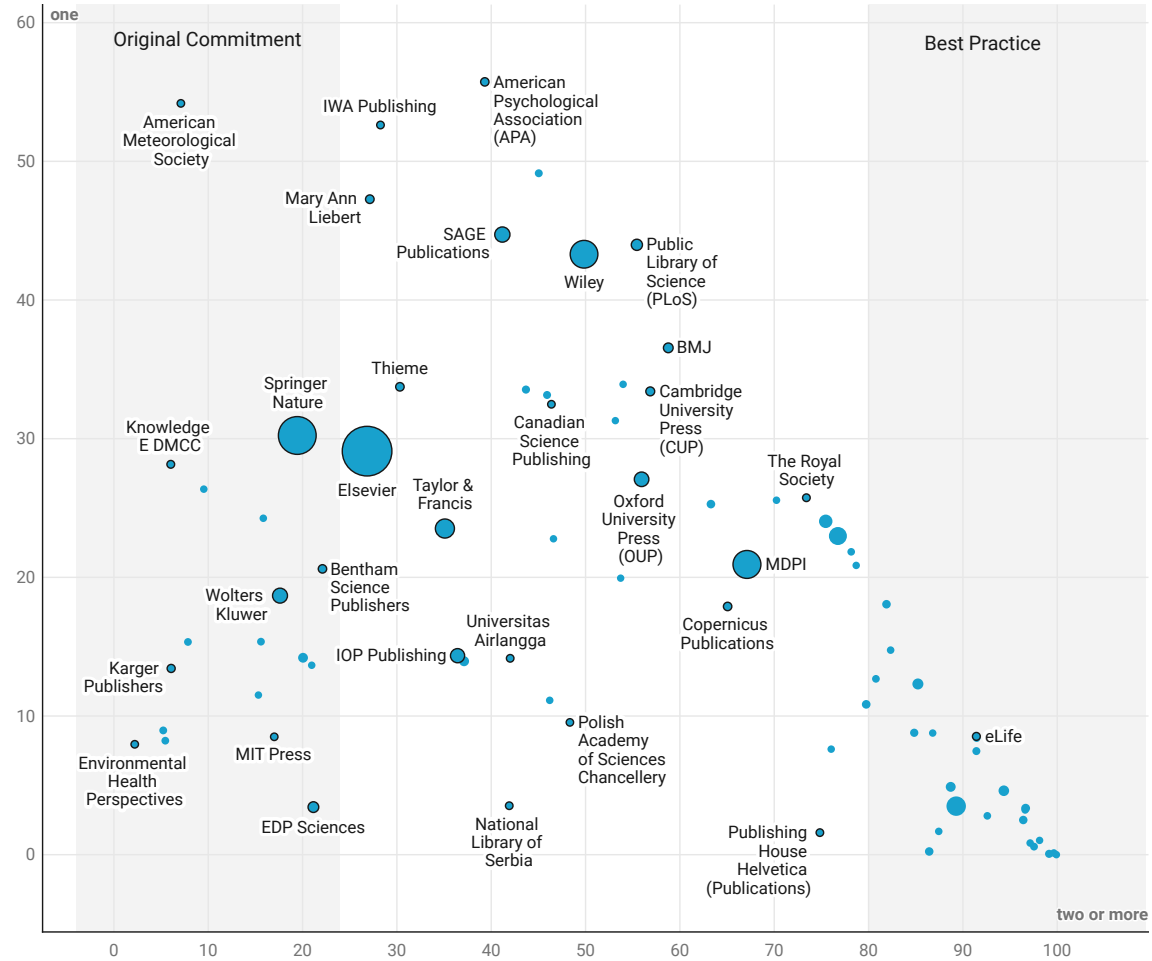


Figure 45: The journey from initial to best practice. The chart above looks at the percentage of journal articles with more than one author by publisher in 2024. On the y axis is the percentage of articles that have only one ORCID, and on the x axis is the percentage of articles that have two or more

3.1.3 ORCID By Publisher by Year

Adoption and use of ORCID is also influenced by international behaviour outside of interventions that can be made in the national sphere. In 2016, many publishers signed on to the commitment to require at least the corresponding author to connect their ORCID iD, understanding that all authors should have the option to assert their relationship to the article [12]. Most publishers began by implementing the first requirement with support for additional authors proceeding at different paces [11]. Like the difference between funded and non funded researchers, without support to strongly encourage the collection of ORCIDs from all authors, the role that a researcher plays in research process also impacts their opportunities to benefit from ORCID supported workflows. Figure 45 looks at the percentage of multi-author articles by publishers in 2024 that have more than one ORCID vs the percentage that have two or more. Even some 9 years on from the original declaration, publisher workflows for Elsevier and Springer Nature still marginally favour the collection of a single ORCID for the corresponding author, leaving most authors the additional work of asserting their relationship to a publication after the fact. Newer publishers such as PLoS, MDPI, and eLife take a more progressive approach to the collection of ORCIDs.

Encouragingly, there is no evidence to suggest that the collection of more than one ORCID on an article is being systematically prevented. This suggests that within a national strategy that maximises the utility of persistent identifiers, it is up to Australian researchers to insist that their ORCID is associated with their authorship regardless of the role that they played on the paper.

Beyond the collecting of ORCIDs during the article submission process, publishers also have a re-

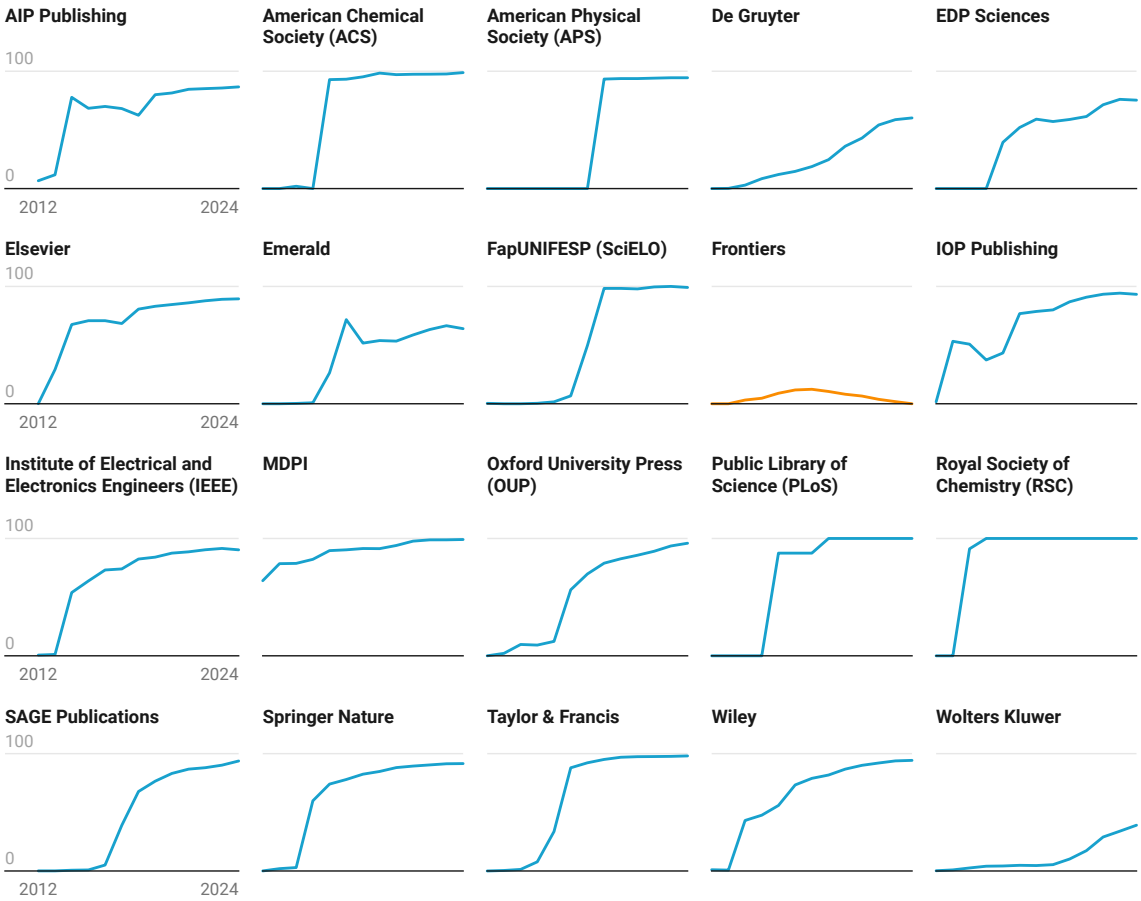


Figure 46: The journey from initial to best practice. The yearly percentage of journals by publisher for which at least one record can be found in Crossref containing ORCID records. (Click on the figure for an interactive version)

sponsibility to pass this metadata onto Crossref via the article metadata, as it is by this mechanism that publisher asserted metadata can make its way back to researcher's ORCID records, reducing academic burden. As figure 46 makes clear, by 2024 most publishers are passing on ORCID assertions for most of their journals; however, Frontiers stands out as one journal that fails on this measure of research information citizenship. In 2024, articles in Frontiers made up 1.4% of Australian publications output; however, it is an example of why national strategy also requires international advocacy to ensure that Australian researchers get the maximum benefit from research information workflows.

Recommendation 3.1.2: Australia joins international pressure to ensure that all publishers both record ORCID records and push the associated metadata into Crossref. Publishers that do not support ORCID workflows (including the passing of metadata through to Crossref should not be recommended venues for publication)

Recommendation 3.1.3: All Australian researchers are required to ensure that their ORCID is part of the metadata of their publication . Regardless of whether or not an Australian researcher is the corresponding author, it should be the default that ORCID records are always supplied.

3.1.4 ORCID By Research Classification

Up to this point, this report has been discussing researchers as if they were a single undifferentiated community. As figure 47 describes, different research disciplines have different attitudes toward ORCID:

- Biomedical and Clinical Sciences are the least engaged on the whole in terms of both adoption and average completeness (50%, 34%)
- Psychology has the highest ORCID adoption rate (67%, 47%)
- Chemical Sciences has good relative adoption and completion rates at (61%, 53%)

Compared with the two five-year periods (2015-2019 and 2000-2024)(50) , it can be seen that all research areas have increased in adoption, with Information and Computing Sciences increasing the most at 34%, and Chemical Sciences the least at 24%. Environmental Sciences stands out as the only research category that did not record an increase in completion rate.

When broken down by research classification, and compared to the global average (figure 51,) Australia's achievements in establishing higher adoption rates for ORCID can be seen across all research classifications. This is particularly so for the Humanities, along with Education and Built Environment and Design. In terms of the average completeness of ORCID records of Australian researchers, the biggest difference from global averages is in Education (16%,) and Law and Legal Studies (14%.)

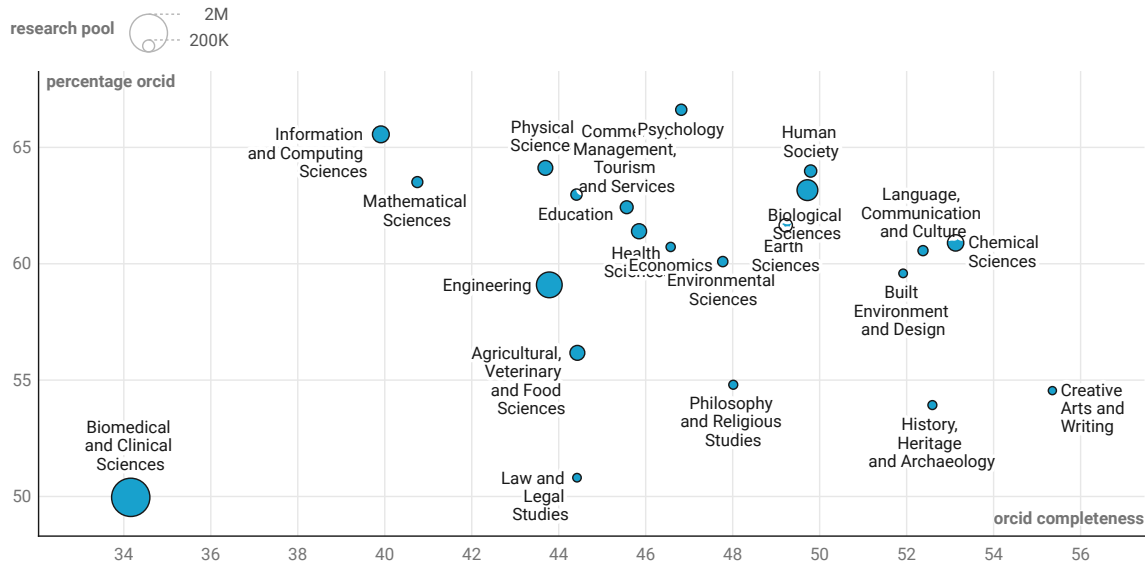


Figure 47: ORCID adoption and completeness by Field of Research 2000-2024. (Click on the figure for an interactive version)

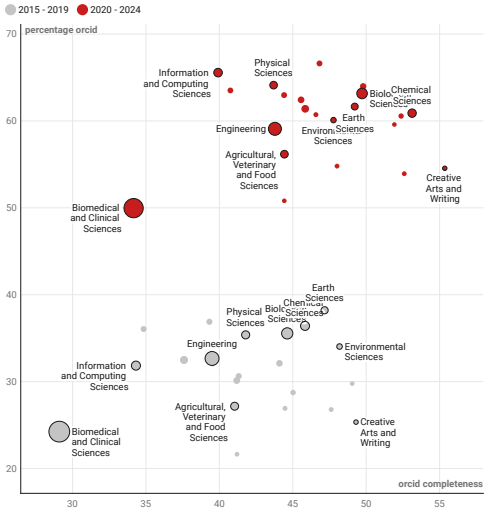


Figure 48: ORCID adoption and completeness by Field of Research 2015-2019 vs 2020-2024

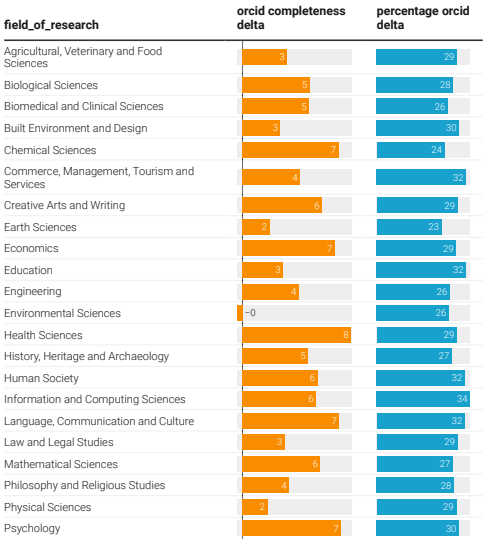


Figure 49: Change delta in ORCID adoption and completeness between time periods

Figure 50: ORCID adoption and completeness by field of research. (Click on the figure and table for interactive versions)

field of research	percentage orcid global	percentage orcid australia	adoption diff	orcid completeness global	orcid completeness australia	completeness diff
Law and Legal Studies	51%	78%	28%	44%	59%	14%
Creative Arts and Writing	55%	80%	26%	55%	63%	8%
History, Heritage and Archaeology	54%	78%	24%	53%	63%	10%
Language, Communication and Culture	61%	84%	23%	52%	62%	10%
Built Environment and Design	60%	82%	22%	52%	58%	6%
Philosophy and Religious Studies	55%	74%	19%	48%	57%	9%
Education	63%	82%	19%	44%	60%	16%
Biomedical and Clinical Sciences	50%	67%	17%	34%	34%	−0%
Engineering	59%	76%	17%	44%	49%	6%
Health Sciences	61%	79%	17%	46%	51%	5%
Chemical Sciences	61%	78%	17%	53%	58%	5%
Human Society	64%	81%	17%	50%	59%	9%
Environmental Sciences	60%	76%	16%	48%	55%	7%
Commerce, Management, Tourism and Services	62%	78%	15%	46%	52%	7%
Physical Sciences	64%	79%	15%	44%	45%	1%
Earth Sciences	62%	76%	14%	49%	52%	3%
Biological Sciences	63%	77%	14%	50%	56%	6%
Information and Computing Sciences	66%	80%	14%	40%	44%	4%
Economics	61%	75%	14%	47%	48%	1%
Psychology	67%	80%	13%	47%	49%	2%
Agricultural, Veterinary and Food Sciences	56%	68%	12%	44%	48%	4%
Mathematical Sciences	64%	75%	12%	41%	42%	2%

Figure 51: ORCID adoption and completeness comparisons to global mean by Field of Research 2000-2024. (Click on the table for an interactive version)

3.1.5 ORCID by Institution

After funding interventions, perhaps the largest impact on the acceptance and use of ORCID by researchers is institutional advocacy and outreach. As indicated in figure 52, a number of Australian institutions have very strong adoption and completion rates including the University of South Australia, Macquarie University, Queensland University of Technology. Whilst some of the differences between institutions can be explained by different discipline compositions, significant differences can still be observed between like institutions.

Further information on the nature of institutional interventions can be gathered by looking at the source of assertions that have been made in the ORCID registry (figure 53). The University of Queensland appears to have encouraged researchers to populate their ORCID using Scopus, whereas the University of Western Australia, Macquarie University and the University of Technology Sydney have promoted the use of their institutional research information systems as the primary source.

The source of ORCID assertions is important, particularly in the context facilitating national reporting. Self assertions are useful, however assertions that are also recorded in an institutional research information system have an additional layer of trust. Assertions that have come directly from the publisher have an even higher level of trust as they come directly via publication publishing workflows. Here, it should be noted that the University of Queensland has also been particularly successful in encouraging its researchers to turn on the Crossref integration in their ORCID records.

Overtime, in a system where researchers are encouraged to supply their ORCID always, we would expect to see a shift away from self asserted records, as a greater percentage of records can be directly asserted via Crossref provided researchers leverage the Crossref integration in their ORCID record. Completing the circle, these records should also be communicated via an institutional connection.

Figure 52: ORCID adoption and completeness by Australian Research Institution
. (Click on the figure for an interactive version)

Recommendation 3.1.4: Australian researchers are encouraged to turn on both Crossref and DataCite integrations to ensure that the population of ORCID records is as automatic as possible..

Recommendation 3.1.5: Australian Institutions require ORCID registration within their institu-

name	Scopus - Elsevier	Crossref	Web of Science Researcher Profile Sync	ResearcherID	Europe PubMed Central	Crossref Metadata Search	institution
University of Western Australia	44,517	15,840	5,447	8,593	2,742	3,097	87,401
Macquarie University	38,012	12,037	1,658	2,609	1,269	2,753	72,172
University of Technology Sydney	32,243	14,159	2,712	3,099	965	4,200	49,312
Monash University	102,830	39,258	8,024	9,250	8,630	7,705	40,609
Griffith University	48,066	17,159	5,510	7,693	3,060	5,605	25,818
University of Melbourne	117,533	41,054	8,707	12,054	13,053	25,653	24,514
University of Queensland	**277,585**	**62,620**	**189,148**	**32,084**	**9,355**	**10,654**	**23,552**
UNSW Sydney	109,229	36,300	7,879	8,002	7,368	6,734	22,823
Charles Sturt University	7,087	4,305	1,572	1,095	236	770	17,454
La Trobe University	22,314	10,490	1,529	1,731	1,557	2,001	13,710
Flinders University	22,639	9,287	3,219	2,394	1,712	1,718	13,365
University of Tasmania	24,560	11,302	1,970	4,748	858	2,497	9,836
University of Canberra	7,222	3,157	207	491	376	623	8,251
Bond University	4,554	2,021	363	231	213	573	6,890
Southern Cross University	8,016	3,088	1,115	592	225	993	5,526
Charles Darwin University	6,002	2,202	159	800	346	652	2,704
Torrens University Australia	1,251	803	509	7	82	56	1,092
Victoria University	8,740	3,002	604	276	173	873	344
University of Southern Queensland	12,748	4,325	1,369	283	86	784	125
The University of Sydney	120,608	38,493	19,720	11,690	8,576	8,482	98

Figure 53: Publication assertions in ORCID records by source.

tional research information management systems.

3.1.6 ORCID 2030?

As the earlier benchmarking sections for research inputs and outputs indicate, the path forward for ORCID is not simply to improve adoption and engagement rates in publishing. The challenge now is to leverage the shared belief in the importance of ORCID across the sector, and extend that comprehensive commitment across the rest of the research graph - be that grants, research activities, instruments, dissertations, research reports, other nontraditional research outputs, datasets, or samples. Although these objectives are ambitious, perhaps the strongest indicator that these changes will be achieved is that a similar coalition of interests that came together to create the vision 2020. Provided there remains a commitment towards the objectives outlined in the Australian National Persistent Identifier Strategy, the structures and networks exist to make the required changes happen.

References

[1] Brown, Jones, Josh and, Meadows, Phill and, Murphy, Alice and, and Fiona. Incentives to invest in identifiers: A cost- benefit analysis of persistent identifiers in australian research systems. URL https://doi.org/10.5281/zenodo.7100578.

[2] Australian ORCID Governance Committee. VISION 2020 AUSTRALIA's ORCID ASPIRATIONS, . URL https://static.aaf.edu.au/media/2017/ORCID/ORCID_Aust_Vision_FIN_LR_20170706.pdf.

[3] ORCID Consortium Governance Committee. ORCID consortium governance committee, . URL https://static.aaf.edu.au/media/2018/orcid/ORCIDGovernanceCommittee_TOR_FIN_Sept2018.pdf.

[4] Australian Research Data Commons. Australian national persistent identifier (PID) strategy 2024. URL https://doi.org/10.5281/zenodo.10656276.

[5] Crossref. Metadata retrieval, 9 2025. URL https://www.crossref.org/documentation/retrieve-metadata/.

[6] Joan E. Damerow, Natalie H. Raia, Val Stanley, Saebyul Choe, Mikayla A. Borton, Neil Byers, Ellen R. Cassidy, Shreyas Cholia, Rorie Edmunds, Brieanne Forbes, Kathleen Forrest, Amy E. Goldman, John Kunze, Sara Lafia, Kerstin Lehnert, Marcella McIntyre-Redden, Richard Naples, Dylan O'Ryan, Charles Parker, Esther Plomp, Beck Powers-McCormack, Sarah Ramdeen, Stephen Richard, Anne Thessen, Cody W. Thompson, Dave Vieglais, Kristina Vrouwenvelder, Elisha M. Wood-Charlson, Lesley Wyborn, T. B. K. Reddy, and Andrea K. Thomer. Opening doors to physical sample tracking and attribution in earth and environmental sciences. *Scientific Data*, 12(1):1047, 2025. .

[7] DataCite. DataCite public data file 2024, 1 2025. URL https://datafiles.datacite.org/datafiles/public-2024.

[8] Laure Haak. Reports and ORCID recommendations from ORBIT funder working group. URL https://info.orcid.org/reports-and-orcid-recommendations-from-orbit-funder-working-group/.

[9] Daniel W. Hook, Simon J. Porter, and Christian Herzog. Dimensions: Building context for search and evaluation. *Frontiers in Research Metrics and Analytics*, 3:23, 2018. .

[10] Patrice Lopez. Research and advanced technology for digital libraries, 13th european conference, ECDL 2009, corfu, greece, september 27 - october 2, 2009. proceedings. *Lecture Notes in Computer Science*, pages 473–474, 2009. ISSN 0302-9743. .

[11] Alice Meadows and Laurel Haak. ORCID open letter - one year on report. 2017. . URL https://orcid.figshare.com/articles/journal_contribution/ORCID_Open_Letter_-_One_Year_On_Report/4828312/1.

[12] ORCID. ORCID in publications. 2016. URL https://info.orcid.org/requiring-orcid-in-publications/.

[13] Julie Petro. ORCID partners with digital science to make openness even more accessible, 3 2025. URL https://info.orcid.org/orcid-partners-with-digital-science-to-make-openness-even-more-accessible/.

[14] Simon J. Porter. Measuring research information citizenship across ORCID practice. *Frontiers in Research Metrics and Analytics*, 7:779097, 2022. .

[15] Research Organization Registry. ROR data, 2025. URL https://doi.org/10.5281/zenodo.16950727.

[16] Digital Science. GRID passes the torch to ROR – FAQs. URL https://www.digital-science.c

om/grid-passes-the-torch-to-ror-faqs/.

[17] Natasha Simons. ORCID joint statements launched in australia, . URL https://info.orcid.org/orcid-joint-statements-launched-in-australia/.

[18] Natasha Simons. Australian ORCID consortium model released for comment, . URL https://info.orcid.org/australian-orcid-consortium-model-released-for-comment/.

[19] David Zeitlyn and Daniel W. Hook. Perception, prestige and PageRank. *PLoS ONE*, 14(5): e0216783, 2019. .

4 Benchmarking Recipes

4.1 Benchmarks for Research Inputs

Research Input 1: Research Partners (ROR)

toolkit query 4.1.1: Number of Australian Insititutions with a ROR id. Query Explanation

- This query calculates the potential number of Australian research organizations that could be represented in ROR by identifying all Australian organizations that have authored publications between 2020 and 2024. It uses the Dimensions dataset as a comprehensive source of publication and affiliation data to establish this baseline.

Tables Used

- `australian_national_pid_benchmarking.ror_data_schema_v2_2025_08_07` (Open)
- `dimensions-ai.data_analytics.publications` (Licensed)
- `dimensions-ai.data_analytics.organizations` (Licensed)

Justification for Licensed Tables

- The Dimensions tables are used to provide a comprehensive list of publications and their associated organizational affiliations, which serves as a baseline to measure the coverage of ROR against real-world research output.

Notebook

- executable notebook

toolkit query 4.1.2: Directly asserted affiliations by year for Australian institutions. Query Explanation

- This query measures the direct adoption of ROR identifiers by publishers for Australian affiliations. It counts the number of times a ROR ID for an Australian institution has been explicitly included in the metadata of a publication submitted to Crossref, tracking the trend on a yearly basis.

Tables Used

- `australian_national_pid_benchmarking.crossref_metadata` (Open)
- `australian_national_pid_benchmarking.ror_data_schema_v2_2025_08_07` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

toolkit query 4.1.3: Support for ROR ids is concentrated in specific publishers. . Query Explanation

- This query identifies which publishers are actively asserting ROR IDs for Australian affiliations in their publication metadata. It counts the number of ROR-linked affiliations per publisher, helping to pinpoint key partners and highlight opportunities for advocacy.

Tables Used

- `australian_national_pid_benchmarking.crossref_metadata` (Open)

- `australian_national_pid_benchmarking.ror_data_schema_v2_2025_08_07` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

toolkit query 4.1.4: Asserted australian affiliations by field of research. .

Query Explanation

- This query analyzes the uptake of ROR assertions across different fields of research. It compares the percentage of publications with Australian ROR assertions (from Crossref) to the total percentage of Australian publications in that field (from Dimensions), highlighting disciplines where ROR adoption is stronger or weaker.

Tables Used

- `australian_national_pid_benchmarking.crossref_metadata` (Open)
- `australian_national_pid_benchmarking.ror_data_schema_v2_2025_08_07` (Open)
- `dimensions-ai.data_analytics.publications` (Licensed)

Justification for Licensed Tables

- The Dimensions 'publications' table is used to establish a baseline of Australia's total research output per field, which is necessary to contextualize the ROR assertion rates found in Crossref.

Notebook

- executable notebook

toolkit query 4.1.5: ROR Assertions in DataCite records 2020-2024. .

Query Explanation

- This query assesses the adoption of ROR identifiers in non-publication outputs registered in DataCite (such as datasets, software, etc.). It calculates the total number of works with ROR affiliations and the percentage of those affiliations that are Australian, broken down by the type of research output.

Tables Used

- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)
- `australian_national_pid_benchmarking.ror_data_schema_v2_2025_08_07` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

toolkit query 4.1.6: Compare ROR-based affiliations (Australian affiliations vs. global assertions. Query Explanation

- This query identifies which DataCite repositories are likely associated with specific Australian institutions. It works by analyzing the ROR affiliations within each repository's records, assign-

ing the repository to the most frequently occurring institution, and calculating the proportion of works this affiliation represents.

Tables Used

- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)
- `australian_national_pid_benchmarking.ror_data_schema_v2_2025_08_07` (Open)
- `australian_national_pid_benchmarking.DataCite_clients` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

Research Input 2: Research Projects (RAID)

toolkit query 4.1.7: Number of RAiDs that have been created as at Janurary 2025. . Query Explanation

- This query counts the number of registered Research Activity IDs (RAiDs) to measure the initial uptake of the identifier. It works by identifying the dedicated RAiD registration agencies (clients) within the DataCite infrastructure and summing the records they have created.

Tables Used

- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)
- `australian_national_pid_benchmarking.DataCite_clients` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

Research Input 3: Grants

toolkit query 4.1.8: . Query Explanation

- This query assesses the current state of grant registration in Crossref. It counts the number of grant DOIs registered by each funder ("publisher") and analyzes the richness of their metadata, including whether they are linked to researcher ORCIDs and institutional RORs.

Tables Used

- `australian_national_pid_benchmarking.crossref_metadata` (Open)
- `australian_national_pid_benchmarking.ror_data_schema_v2_2025_08_07` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

toolkit query 4.1.9: . Query Explanation

- This query estimates the total pool of active Australian grants to provide a baseline for PID adoption. It uses the Dimensions grants database to count grants from Australian funders that are active in the current year and cross-references them with publication data to see how many are already being acknowledged in the literature.

Tables Used

- `dimensions-ai.data_analytics.grants` (Licensed)
- `dimensions-ai.data_analytics.publications` (Licensed)
- `dimensions-ai.data_analytics.grid` (Licensed)

Justification for Licensed Tables

- The Dimensions grants and publications tables are essential for providing a comprehensive view of the grant landscape and its connection to research outputs, which isn't available in open datasets at this scale.

Notebook

- executable notebook

toolkit query 4.1.10: Calculate % of publications with valid grant IDs in funding metadata. Compare rates in Crossref to rates in Dimensions. Query Explanation

- This query compares two methods of linking publications to grants for Australian research output. It measures the percentage of publications that contain grant information in structured Crossref metadata versus the percentage identified through text mining of acknowledgements in the Dimensions dataset. This highlights the gap between informal practice and structured, machine-readable data.

Tables Used

- `australian_national_pid_benchmarking.crossref_metadata` (Open)
- `australian_national_pid_benchmarking.ror_data_schema_v2_2025_08_07` (Open)
- `dimensions-ai.data_analytics.publications` (Licensed)
- `dimensions-ai.data_analytics.grid` (Licensed)

Justification for Licensed Tables

- The Dimensions 'publications' table is used for its comprehensive grant linkage data, derived from full-text analysis, which is then compared against the structured data available in the open Crossref dataset.

Notebook

- executable notebook

Research Input 4: Research Facilities

toolkit query 4.1.11: . Query Explanation

- This query provides a simple count of how many Australian research facilities are registered in the Research Organization Registry (ROR), establishing a baseline for findability.

Tables Used

- `australian_national_pid_benchmarking.ror_data_schema_v2_2025_08_07` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

toolkit query 4.1.12: Facility Assertions 2020-2024. Query Explanation

- This query measures the research output of Australian facilities by algorithmically linking publications to them based on author affiliations. It uses the Dimensions dataset to identify authors affiliated with an organization typed as a 'Facility' and counts their associated publications.

Tables Used

- `dimensions-ai.data_analytics.publications` (Licensed)
- `dimensions-ai.data_analytics.organizations` (Licensed)

Justification for Licensed Tables

- Dimensions is used for its comprehensive, algorithmically-disambiguated affiliation data and its classification of organizations, which allows for the identification of facilities and their associated publications.

Notebook

- executable notebook

toolkit query 4.1.13: . Query Explanation

- This query compares the number of publications linked to facilities via direct ROR assertions in Crossref metadata against the number linked algorithmically in Dimensions. This highlights the difference between potential, comprehensive reporting (algorithmic) and trusted, but less common, reporting (direct assertions).

Tables Used

- `australian_national_pid_benchmarking.crossref_metadata` (Open)
- `australian_national_pid_benchmarking.ror_data_schema_v2_2025_08_07` (Open)
- `dimensions-ai.data_analytics.publications` (Licensed)
- `dimensions-ai.data_analytics.organizations` (Licensed)

Justification for Licensed Tables

- The Dimensions tables are used to provide the algorithmically-derived benchmark against which the direct assertions from the open Crossref dataset are measured.

Notebook

- executable notebook

Research Input 5: Research Data

toolkit query 4.1.14: . Query Explanation

- This comprehensive query maps the Australian repository landscape within DataCite. It identifies Australian repositories using a combination of ROR affiliations, provider country codes, and domain name analysis. It then calculates Australia's share of global research outputs for

various types (e.g., datasets, software, dissertations) registered in DataCite.

Tables Used

- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)
- `australian_national_pid_benchmarking.DataCite_clients` (Open)
- `australian_national_pid_benchmarking.ror_data_schema_v2_2025_08_07` (Open)
- `australian_national_pid_benchmarking.DataCite_providers` (Open)
- `australian_national_pid_benchmarking.DataCiteTypesRemap` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

toolkit query 4.1.15: Review of PID systems in use as registered re3data. Query Explanation

- This query inspects a snapshot of data from the Registry of Research Data Repositories (re3data.org) to identify which persistent identifier systems (e.g., DOI, Handle) are reportedly used by Australian repositories.

Tables Used

- `australian_national_pid_benchmarking.repositories_re3data` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

toolkit query 4.1.16: Repository outputs by country 2020-2024. Query Explanation

- This query aggregates DataCite repository outputs by country to rank national contributions. Using a multi-faceted approach to determine the country of each repository (based on ROR, provider info, and domain names), it counts the number of works and repositories per country, distinguishing between multi-type (generalist) and single-type repositories.

Tables Used

- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)
- `australian_national_pid_benchmarking.DataCite_clients` (Open)
- `australian_national_pid_benchmarking.ror_data_schema_v2_2025_08_07` (Open)
- `australian_national_pid_benchmarking.DataCite_providers` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

toolkit query 4.1.17: repository details. Query Explanation

- This query provides a detailed profile for each repository in DataCite. It determines the repository's country and likely institutional owner, then calculates key metrics like the total number of works, the number of works with an institutional affiliation, and the number of works containing researcher ORCIDs. This helps assess the maturity of PID adoption at the repository level.

Tables Used

- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)
- `australian_national_pid_benchmarking.DataCite_clients` (Open)
- `australian_national_pid_benchmarking.ror_data_schema_v2_2025_08_07` (Open)
- `australian_national_pid_benchmarking.DataCite_providers` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

toolkit query 4.1.18: Global data citation rates 2024. Query Explanation

- This query measures the rate of dataset citation in scholarly articles. It joins publication data with a dataset of identified data citations (derived from full-text analysis) to calculate the percentage of articles in each field of research that cite a dataset DOI.

Tables Used

- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)
- `dimensions-ai.data_analytics.publications` (Licensed)
- `australian_national_pid_benchmarking.DataCite_sentence_matches_confirmed_remapped` (Licensed)

Justification for Licensed Tables

- The 'dimensions-ai.publications' table provides the denominator (all research articles), while the 'DataCite_sentence_matches...' table provides the numerator (articles with data citations). The latter is licensed because it is the output of a computationally intensive full-text mining process across a large corpus.

Notebook

- executable notebook

toolkit query 4.1.19: Data citation percentage by country. Query Explanation

- This query breaks down the dataset citation rate by both country and field of research. It allows for international comparison of data reuse practices within specific disciplines by calculating the percentage of publications from a country that contain a data citation.

Tables Used

- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)
- `dimensions-ai.data_analytics.publications` (Licensed)

- `australian_national_pid_benchmarking.DataCite_sentence_matches_confirmed_remapped` (Licensed)

Justification for Licensed Tables

- Licensed tables are required for the same reasons as the global citation rate query: to provide a comprehensive count of publications and the derived data citation links from full-text analysis.

Notebook

- executable notebook

toolkit query 4.1.20: Data citation comparative engagement. Query Explanation

- This query calculates a comparative engagement ratio for data citation. It compares a country's share of global data citations to its share of global publications within a specific field. A ratio greater than 1 indicates that the country's researchers are more engaged in data citation than the global average for that field.

Tables Used

- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)
- `dimensions-ai.data_analytics.publications` (Licensed)
- `australian_national_pid_benchmarking.DataCite_sentence_matches_confirmed_remapped` (Licensed)

Justification for Licensed Tables

- Licensed tables are essential for providing the comprehensive global publication and data citation numbers needed to calculate the comparative ratio accurately.

Notebook

- executable notebook

Research Input 6: Samples & Observations

toolkit query 4.1.21: . Query Explanation

- This query measures Australia's contribution to the global registration of physical samples using the International Generic Sample Number (IGSN). It counts the total IGSNs registered each year and calculates the proportion originating from known Australian IGSN catalogs.

Tables Used

- `australian_national_pid_benchmarking.DataCite_clients` (Open)
- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

toolkit query 4.1.22: Comparison of affiliation and person linkages between Australian IGSN records and the rest of the world. Query Explanation

- This query assesses the metadata richness of Australian IGSN records compared to those from

the rest of the world. It measures the presence of links to people (name identifiers like ORCID) and organizations (affiliation identifiers like ROR) to gauge how well Australian samples are integrated into the wider PID graph.

Tables Used

- `australian_national_pid_benchmarking.DataCite_clients` (Open)
- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

toolkit query 4.1.23: Comparison of object relationships between Australian IGSN records and the rest of the world . Query Explanation

- This query analyzes how well IGSN sample records are linked to other research objects. It counts the number and type of relationships (e.g., "IsCitedBy", "IsSupplementTo") in the metadata of Australian vs. non-Australian IGSN records to measure their integration and contextualization.

Tables Used

- `australian_national_pid_benchmarking.DataCite_clients` (Open)
- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)

Justification for Licensed Tables

- executable notebook

Notebook

- A link to the executable notebook can be provided here.

toolkit query 4.1.24: . Query Explanation

- This query assesses the metadata richness of Australian IGSN records compared to those from the rest of the world. It measures the presence of links to people (name identifiers like ORCID) and organizations (affiliation identifiers like ROR) to gauge how well Australian samples are integrated into the wider PID graph.

Tables Used

- `australian_national_pid_benchmarking.DataCite_clients` (Open)
- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)

Justification for Licensed Tables

- executable notebook

Notebook

- A link to the executable notebook can be provided here.

Research Input 7: Instruments

toolkit query 4.1.25: . Query Explanation

- This query identifies and counts the research instruments that have been registered with a persistent identifier in DataCite. It groups the instruments by the registering repository and country to provide an overview of instrument registration practices.

Tables Used

- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)
- `australian_national_pid_benchmarking.DataCite_clients` (Open)
- `australian_national_pid_benchmarking.ror_data_schema_v2_2025_08_07` (Open)
- `australian_national_pid_benchmarking.DataCite_providers` (Open)
- `australian_national_pid_benchmarking.DataCiteTypesRemap` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

toolkit query 4.1.26: Links between instruments and affiliations. Query Explanation

- This query examines the metadata of registered instruments from key Australian repositories to see how well they are connected to the broader PID graph. It counts the number and type of 'relatedIdentifier' links, which could point to datasets, publications, or other instruments.

Tables Used

- `australian_national_pid_benchmarking.DataCite_clients` (Open)
- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

toolkit query 4.1.27: Links between instruments and DataCite records. . Query Explanation

- This query measures the reuse or citation of registered instruments within the DataCite ecosystem. It searches for DataCite records that link *to* an instrument record (i.e., inbound links), providing evidence that the instrument is being referenced by other research outputs like datasets.

Tables Used

- `australian_national_pid_benchmarking.DataCite_clients` (Open)
- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

toolkit query 4.1.28: Instrument Citations. Query Explanation

- This query identifies when registered instruments are cited in the full text of scholarly publications. It joins a list of registered instruments from DataCite with the data citation table to find explicit mentions, providing a direct measure of instrument reuse and impact.

Tables Used

- `australian_national_pid_benchmarking.DataCite_clients` (Open)
- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)
- `australian_national_pid_benchmarking.DataCite_sentence_matches_confirmed_remapped` (Licensed)

Justification for Licensed Tables

- The 'DataCite_sentence_matches...' table is essential as it contains the results of a full-text search for instrument DOIs across a large corpus of publications, which is necessary to find these citations.

Notebook

- executable notebook

4.2 Benchmarks for Research Outputs

Research Output 1: Research Reports

toolkit query 4.2.1: Reports by repository. Query Explanation

- This query identifies and ranks repositories based on the number of research reports they have registered in DataCite. It uses a sophisticated method to determine the country and institution for each repository, allowing for a global and national overview of report publishing.

Tables Used

- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)
- `australian_national_pid_benchmarking.DataCite_clients` (Open)
- `australian_national_pid_benchmarking.ror_data_schema_v2_2025_08_07` (Open)
- `australian_national_pid_benchmarking.DataCite_providers` (Open)
- `australian_national_pid_benchmarking.DataCiteTypesRemap` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

toolkit query 4.2.2: Reports interoperability with ORCID and ROR. . Query Explanation

- This query assesses the metadata quality of research reports from Australian repositories. It counts the number of reports and calculates the proportion that include links to researchers via ORCID and to affiliations via ROR, indicating how well they are connected to the PID ecosystem.

Tables Used

- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)
- `australian_national_pid_benchmarking.DataCite_clients` (Open)
- `australian_national_pid_benchmarking.DataCiteTypesRemap` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

toolkit query 4.2.3: . Query Explanation

- This query measures the impact and reach of reports from Australian repositories. It counts formal citations found in scholarly literature (via the data citation table) and also queries Altmetric data to track mentions in policy documents, social media, and Wikipedia.

Tables Used

- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)
- `australian_national_pid_benchmarking.DataCite_clients` (Open)
- `altmetric.data.research_outputs` (Licensed)
- `australian_national_pid_benchmarking.DataCite_sentence_matches_confirmed_remapped` (Licensed)

Justification for Licensed Tables

- The licensed data citation table is needed to find formal citations in the full text of publications. The licensed Altmetric table is used to track broader impact and attention outside of traditional academia.

Notebook

- executable notebook

Research Output 2: Research Methods

toolkit query 4.2.4: Articles published in Protocol journals by year. Query Explanation

- This query tracks the volume of protocol-focused articles published over time. It uses the Dimensions dataset to search for articles published in journals whose titles contain the word "protocol," providing a measure of this publication type's growth.

Tables Used

- `dimensions-ai.data_analytics.publications` (Licensed)

Justification for Licensed Tables

- Dimensions is used for its comprehensive journal and publication metadata, which allows for a broad search across the scholarly literature to identify these specific articles.

Notebook

- executable notebook

toolkit query 4.2.5: . Query Explanation

- This query identifies the primary venues for publishing research protocols. It counts the number of protocol articles per journal to rank them and identify key publishers in this space.

Tables Used

- `dimensions-ai.data_analytics.publications` (Licensed)

Justification for Licensed Tables

- Dimensions is used for its comprehensive metadata, enabling the identification and ranking of journals based on their focus on publishing protocols.

Notebook

- executable notebook

toolkit query 4.2.6: . Query Explanation

- This query searches for protocols registered in DataCite. It identifies records where "protocol" appears in the title and categorizes them by repository and resource type to understand how and where protocols are being described outside of traditional journals.

Tables Used

- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)
- `australian_national_pid_benchmarking.DataCite_clients` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

Research Output 3: Non-Traditional Research Outputs (NTROs)

toolkit query 4.2.7: Non Traditional Research Outputs in DataCite. Query Explanation

- This query measures Australia's output of various Non-Traditional Research Outputs (NTROs) as registered in DataCite. It identifies Australian repositories and then counts works across numerous categories (like 'Image', 'Audiovisual', 'Software') and also measures their citation rates in scholarly literature.

Tables Used

- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)
- `australian_national_pid_benchmarking.DataCite_clients` (Open)
- `australian_national_pid_benchmarking.ror_data_schema_v2_2025_08_07` (Open)
- `australian_national_pid_benchmarking.DataCite_providers` (Open)
- `australian_national_pid_benchmarking.DataCiteTypesRemap` (Open)
- `australian_national_pid_benchmarking.DataCite_sentence_matches_confirmed_remapped` (Licensed)

Justification for Licensed Tables

- The licensed data citation table is used to measure the reuse and citation of these NTROs in the scholarly literature, providing a key indicator of their impact.

Notebook

- executable notebook

toolkit query 4.2.8: . Query Explanation

- This query identifies Software works based on ORCID assertions linked to researchers currently associated with Australian institutions (as determined by Dimensions

Tables Used

- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)
- `ds-open-datasets.orcid.summaries_2024` (Open)
- `dimensions-ai.data_analytics.researchers` (Licensed)
- `dimensions-ai.data_analytics.organizations` (Licensed)

Justification for Licensed Tables

- The licensed tables provide an estimate of researchers that are currently associated with Australian institutions, allowing for the relevant ORCIDs to be identified.

Notebook

- executable notebook

toolkit query 4.2.9: . Query Explanation

- This query identifies Local Contexts labels and rights associated with works in dataCite

Tables Used

- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)

Notebook

- executable notebook

Research Output 4: Dissertations

toolkit query 4.2.10: Dissertations by repository and country. Query Explanation

- This query provides a comprehensive count of dissertations with PIDs by combining data from both DataCite and Crossref. It identifies the top repositories and countries for dissertation publishing, determining repository location using a multi-faceted approach including affiliation analysis and publisher metadata.

Tables Used

- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)
- `australian_national_pid_benchmarking.DataCite_clients` (Open)
- `australian_national_pid_benchmarking.ror_data_schema_v2_2025_08_07` (Open)
- `australian_national_pid_benchmarking.DataCite_providers` (Open)
- `australian_national_pid_benchmarking.crossref_metadata` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

toolkit query 4.2.11: Australian Dissertations interoperability with ROR and ORCID. Query Explanation

- This query measures the metadata completeness of Australian dissertations registered in DataCite. It calculates the percentage of dissertation records that include ORCID identifiers for authors and ROR identifiers for their institutions, assessing their integration with the PID graph.

Tables Used

- `australian_national_pid_benchmarking.DataCitesnapshot2024` (Open)
- `australian_national_pid_benchmarking.DataCite_clients` (Open)
- `australian_national_pid_benchmarking.DataCiteTypesRemap` (Open)

Justification for Licensed Tables

- No licensed tables are required for this query.

Notebook

- executable notebook

Research Output 5: Research Publications

toolkit query 4.2.12: Global works with no pid in ORCID by year. Query Explanation

- This query identifies research outputs recorded in ORCID that do not have a standard persistent identifier (like a DOI, ISBN, etc.). It links ORCID records to researcher profiles in Dimensions to determine their field of research and then analyse the works to see what proportion are added without a PID, tracking this trend over time.

Tables Used

- `ds-open-datasets.orcid.summaries_2024` (Open)
- `dimensions-ai.data_analytics.publications` (Licensed)
- `dimensions-ai.data_analytics.researchers` (Licensed)

Justification for Licensed Tables

- The Dimensions tables are used to enrich the open ORCID data, specifically to determine the primary field of research for each researcher, which allows for disciplinary analysis.

Notebook

- executable notebook

toolkit query 4.2.13: Australian works with no pid in ORCID by year. Query Explanation

- This is the same as the previous query but is filtered to focus specifically on the ORCID records of Australian researchers. It provides a national view of the trend of recording research outputs without persistent identifiers.

Tables Used

- `ds-external-data.orcid.summaries` (Open)

- `dimensions-ai.data_analytics.publications` (Licensed)
- `dimensions-ai.data_analytics.researchers` (Licensed)
- `dimensions-ai.data_analytics.organizations` (Licensed)

Justification for Licensed Tables

- The Dimensions tables are used to identify the cohort of Australian researchers and determine their fields of research for disciplinary analysis.

Notebook

- executable notebook

toolkit query 4.2.14: . Query Explanation

- This query counts the number of conference papers by conference in 2024 and orders them by volume - also returning the number of papers that have a doi.

Tables Used

- `ds-external-data.australian_national_pid_benchmarking.dblp` (Open)

Notebook

- executable notebook

toolkit query 4.2.15: Works reported in 2024 with no persistent identifier by Field of Research. . Query Explanation

- This query provides a more granular analysis of the outputs that lack PIDs in the ORCID records of Australian researchers. It breaks down these works by both the field of research and the work type as recorded in ORCID (e.g., conference paper, journal article), helping to identify specific communities or output types where PID adoption is lagging.

Tables Used

- `ds-external-data.orcid.summaries` (Open)
- `dimensions-ai.data_analytics.publications` (Licensed)
- `dimensions-ai.data_analytics.researchers` (Licensed)
- `dimensions-ai.data_analytics.organizations` (Licensed)

Justification for Licensed Tables

- The Dimensions tables are used to identify the Australian researcher cohort and their disciplines, enabling a detailed breakdown of the results.

Notebook

- executable notebook

toolkit query 4.2.16: Open Access Ratio. Query Explanation

- This query measures the percentage of Australian research articles that are Open Access. It uses the Dimensions dataset to identify Australian output and its open access status, tracking the trend year over year.

Tables Used

- `dimensions-ai.data_analytics.publications` (Licensed)

Justification for Licensed Tables

- The Dimensions dataset includes comprehensive, aggregated Open Access status information for millions of publications, making this calculation straightforward and reliable.

Notebook

- executable notebook

5 List of Acronyms used in this report

AAF Australian Access Federation

ANDS Australian National Data Service

ANZSRC Australian and New Zealand Standard Research Classification

ARC Australian Research Council

ARDC Australian Research Data Commons

ARMS Australasian Research Management Society

CAUDIT Council of Australian University Directors of Information Technology

CAUL Council of Australian University Librarians

CRIS Current Research Information System

CSIRO Commonwealth Scientific and Industrial Research Organisation

DOI Digital Object Identifier

FOR Field of Research

GRID Global Research Identifier Database

GROBID GeneRation Of BIbliographic Data

HPC High Performance Computing

IGSN International Generic Sample Number

ISBN International Standard Book Number

JATS Journal Article Tag Suite

MDPI Multidisciplinary Digital Publishing Institute

NCRIS National Collaborative Research Infrastructure Strategy

NHMRC National Health and Medical Research Council

NTRO Non-Traditional Research Output

OA Open Access

ORCID Open Researcher and Contributor ID

PID Persistent Identifier

RAID Research Activity Identifier

ROR Research Organization Registry

SMART Specific, Measurable, Achievable, Relevant, Time-bound

TK toolkit query Reference